

COURSE: HAMILTONIAN DYNAMICS OF CONSTRAINED SYSTEMS

- HAMILTONIAN FORMALISM IS THE SYSTEMATIC FRAMEWORK FOR OBTAINING AND ANALYZING LOCAL SYMMETRIES AND DEGREES OF FREEDOM IN A THEORY
- ALTHOUGH NOT THE ONLY ONE, IT PROVIDES ONE OF THE MOST DIRECT AND OPERATIONAL PROCEDURES

KNOWN METHODS:

- ✓ DIRAC-BERGMANN ALGORITHM (VERY INTUITIVE)
- LAGRANGIAN NOETHER IDENTITY APPROACH
(LOCAL SYMMETRIES ARE RELATED TO DIFF. IDENTITIES AMONG EQUATIONS OF MOTION)
- BRST / BV FORMALISM (MOST POWERFUL, ESPECIALLY IN FIELD THEORY QUANTIZATION; ADVANCED METHOD)
- ✓ COVARIANT FORMALISM (KEEP COVARIANCE MANIFEST)

Method	Strength	Weakness
Hamiltonian / Dirac-Bergmann	Direct constraint algorithm and d.o.f. count	Less manifestly covariant
Noether identities	Fully Lagrangian and covariant	Finding all identities can be hard
BRST/BV	Most systematic algebraic encoding of gauge structure	Usually assumes gauge generators or Noether identities are known
Covariant phase space	Excellent for charges and boundary symmetries	Local d.o.f. counting less direct

PLAN OF THE COURSE

1. HAMILTONIAN DYNAMICS OF SINGULAR SYSTEMS
2. LOCAL SYMMETRIES
3. SYMPLECTIC GEOMETRY
4. GRAVITATIONAL HAMILTONIAN

[~ 3 LECTURES]

[~ 2 LECTURES]

[~ 3 LECTURES]

[~ 2 LECTURES]

METHOD OF WORK

- PRESENTED LECTURES
- EXAMPLES
- QUESTIONS TIME
- HOMEWORK (FOR PRACTICING)
- ORAL PRESENTATION / SEMINAR (EVALUATION)

PART 1 HAMILTONIAN DYNAMICS OF SINGULAR SYSTEMS

BIBLIOGRAPHY PARTS 1,2,3

- MILUTIN BLAGOJEVIĆ, "GRAVITATION AND GAUGE SYMMETRIES"
- KURT SUNDERMEYER, "CONSTRAINED DYNAMICS"
- MARC HENNEAUX, CLAUDIO TEITELBOIM "QUANTIZATION OF GAUGE SYSTEMS"

REGULAR SYSTEMS (WITHOUT LOCAL SYMMETRIES)

- WE HAVE TO KNOW FIRST HOW TO DESCRIBE REGULAR SYSTEMS IN A HAMILTONIAN WAY
- CONSIDER A NON RELATIVISTIC SYSTEM WITH A FINITE NUMBER OF DEGREES OF FREEDOM

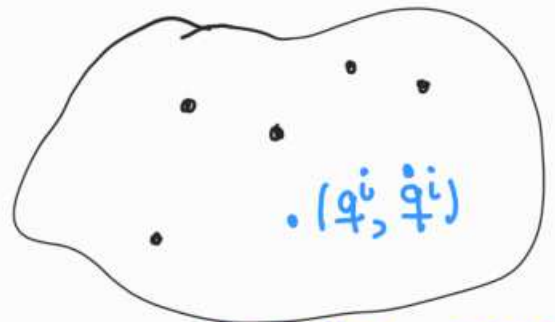
$q^i(t)$ GENERALIZED COORDINATES ($i=1, \dots, N$)

$\dot{q}^i(t) = \frac{dq^i}{dt}$... GENERALIZED VELOCITIES ($t = \text{TIME}; t_0 \leq t \leq t_1$)



CONFIGURATION
SPACE Q

(NOT ENOUGH TO DESCRIBE
DYNAMICS)



VELOCITY SPACE
OR TANGENT BUNDLE
 TQ

(IN THE LAGRANGIAN FORMALISM,
THE DYNAMICS TAKES PLACE IN TQ)

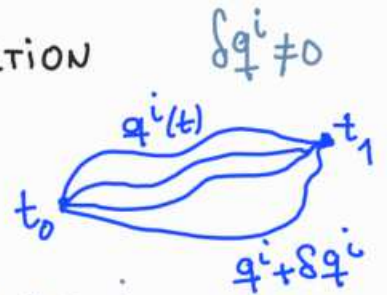
- LAGRANGIAN FORMALISM IS BASED ON THE ACTION FUNCTIONAL

$$I[q] = \int_{t_0}^{t_1} dt \quad L(q, \dot{q}, t)$$

$L(q, \dot{q}, t)$ = LAGRANGIAN
(FOR SIMPLICITY)

- DYNAMICS FOLLOWS FROM EXTREMIZING THE ACTION

$$\delta I[q] = \int_{t_0}^{t_1} dt \left(\frac{\partial L}{\partial q^i} \delta q^i + \frac{\partial L}{\partial \dot{q}^i} \delta \dot{q}^i \right)$$



$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \delta q^i \right) - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \delta q^i$$

$$\delta I = \int_{t_0}^{t_1} dt \underbrace{\left[\frac{\partial L}{\partial q^i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \right]}_{\frac{\delta I}{\delta q^i} \text{ FUNCTIONAL DERIVATIVE}} \delta q^i + \underbrace{\left(\frac{\partial L}{\partial \dot{q}^i} \delta q^i \right) \Big|_{t_0}^{t_1}}_{\substack{0, \text{ BOUNDARY TERM} \\ \delta q^i|_{t_0, t_1} = 0}}$$

COMMENT

WE OMIT WRITING
(EINSTEIN SUMMATION
CONVENTION)

$$\sum_{i=1}^N (\dots)$$

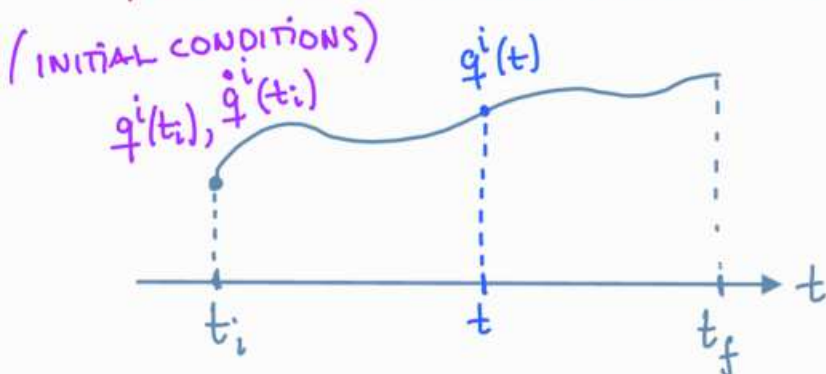
$$\sum_{i=1}^N A_i B_i \equiv A_i B_i = AB$$

- EULER-LAGRANGE EQUATIONS (EQUATIONS OF MOTION)

$$\Sigma_i = \frac{\delta I}{\delta q^i} = \frac{\partial L}{\partial q^i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) = 0$$

DESCRIBE THE DYNAMICS
OF THE SYSTEM OF
PARTICLES IN TQ

- HOW TO KNOW THAT THE TRAJECTORIES ARE FULLY DETERMINED FOR GIVEN INITIAL CONDITIONS?



IF EACH $q(t)$ SATISFIES
DIFF-EQUATION OF SECOND ORDER

$$\ddot{q}(t) + f(q, \dot{q}) = 0 \quad (2 \text{ INTEGRATION CONSTANTS})$$

ITS EVOLUTION IS DETERMINED

by 2 INITIAL CONDITIONS:

$$q_0 \equiv q(t_0), \quad \dot{q}_0 \equiv \dot{q}(t_0)$$

- THEN IT COUNTS AS 1 PROPAGATING DEGREE OF FREEDOM (DOF)

$$\boxed{1 \text{ DOF} = 2 \text{ INITIAL CONDITIONS}} \quad (\text{DETERMINE 2 INT. CONST.})$$

- THIS DEPENDS ON THE EULER-LAGRANGE EQUATIONS

$$\mathcal{E}_i = \frac{\partial L}{\partial q^i} - \underbrace{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right)}_{\frac{\partial}{\partial q^j} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \dot{q}^j + \frac{\partial}{\partial \dot{q}^j} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \ddot{q}^j} = -F_i(q, \dot{q}) - \underbrace{W_{ij}(q, \dot{q}) \ddot{q}^j}_{\text{Hessian (Matrix)}} = 0$$

HESSIAN (MATRIX) $W_{ij} = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$

$W_{ij} = W_{ji}$ (SYMMETRIC)

$$\Rightarrow \mathcal{E}_i = 0 \Leftrightarrow W_{ij} \ddot{q}^j + F_i = 0$$

$W \ddot{q} + F = 0$ (MATRIX EQUATION)

$W = [W_{ij}]$

$\ddot{q} = [\ddot{q}^i], F = [F_i]$

- IF THE INVERSE MATRIX W^{-1} EXISTS, THE EVOLUTION OF ALL $q^i(t)$ ($i=1, \dots, N$) IS DETERMINED BY $2N$ INITIAL CONDITIONS $q_0^i, \dot{q}_0^i$ THAT FIX $2N$ INTEGRATION CONSTANTS

$\Rightarrow$ THE SYSTEM HAS N DOF

W^{-1} EXISTS IF $\det(W) \neq 0 \Rightarrow$ SYSTEM IS REGULAR
 W^{-1} DOESN'T EXIST, $\det(W) = 0 \Rightarrow$ SYSTEM IS SINGULAR

- WHEN $\det(W) = 0$, SOME q HAVE EQUATIONS OF MOTION OF THE FIRST OR ZEROth ORDER AND ITS EVOLUTION REMAIN ARBITRARY

EXAMPLE $L = \frac{1}{2} \dot{q}_1^2 + \frac{\alpha}{2} \dot{q}_2^2 + \dot{q}_1 \dot{q}_2$ $N=2$ PARTICLES $q_1(t), q_2(t)$

HESSIAN $W = \begin{pmatrix} 1 & 1 \\ 1 & \alpha \end{pmatrix} \Rightarrow \det(W) = \alpha - 1$

$\alpha = 1$ SINGULAR SYSTEM

$\alpha \neq 1$ REGULAR SYSTEM

E-L EQUATIONS:

$$\mathcal{E}_1 = -\frac{d}{dt}(\dot{q}_1 + \dot{q}_2) \Rightarrow q_1 + q_2 = at + b$$

$$\mathcal{E}_2 = -\frac{d}{dt}(\dot{q}_1 + \alpha \dot{q}_2) \Rightarrow q_1 + \alpha q_2 = ct + d$$

1) $\alpha \neq 1$ DETERMINED EVOLUTION

$$q_1(t) = \frac{c - \alpha a}{1 - \alpha} t + \frac{d - \alpha b}{1 - \alpha}$$

$$q_2(t) = \frac{a - c}{1 - \alpha} t + \frac{b - d}{1 - \alpha}$$

4 INTEGRATION CONSTANTS (a, b, c, d) ARE DETERMINED:

$$a = \dot{q}_{10} + \dot{q}_{20}, \quad b = q_{10} + q_{20}$$

$$c = \dot{q}_{10} + \alpha \dot{q}_{20}, \quad d = q_{10} + \alpha q_{20}$$

2 DOF

2) $\alpha = 1$ EVOLUTION OF ONE PARTICLE REMAINS ARBITRARY

$$q_1(t) = -q_2(t) + at + b$$

2 INTEGRATION CONSTANTS

1 DOF

LOCAL SYMMETRY APPEARS:

($\mathcal{E}_1, \mathcal{E}_2$ DON'T CHANGE)

$$q_1'(t) = q_1(t) + \varepsilon(t)$$

$$q_2'(t) = q_2(t) - \varepsilon(t)$$

$$\mathcal{E}_1' = \mathcal{E}_1, \quad \mathcal{E}_2' = \mathcal{E}_2$$

THERE IS ONLY 1 PARTICLE $q \equiv q_1 + q_2$

$$L = \frac{1}{2} \dot{q}^2$$

$$\text{SYMMETRY: } q' = q, \quad L' = L, \quad I' = I$$

HAMILTONIAN FORMALISM FOR REGULAR SYSTEMS ($\det W \neq 0$)

• WE REPLACE THE VELOCITY $\dot{q}^i$ BY THE MOMENTUM p_i IN DESCRIPTION OF THE DYNAMICS

GENERALIZED CANONICAL MOMENTUM

$$p_i = \frac{\partial L}{\partial \dot{q}^i} \quad (i=1, \dots, N)$$

TQ

VELOCITY SP.
TANGENT BUNDLE



LAGRANGIAN FORMALISM
LIVES ON TQ

LAGRANGIAN $L(q, \dot{q})$

LEGENDRE
TRANSFORMATION



$T^*Q \equiv \Gamma$

PHASE SPACE,
COTANGENT
BUNDLE

HAMILTONIAN FORMALISM
LIVES ON T^*Q

HAMILTONIAN $H(q, p) = p_i \dot{q}^i - L$

- THE EFFECT OF LEGENDRE TRANSFORMATION IS TO CHANGE DYNAMICAL VARIABLES

$L \dots$ HAS DYN. VARIABLES $q, \dot{q}$

$$\delta L = \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q}$$

$H \dots$ HAS DYN. VARIABLES q, p

$$\delta H = \frac{\partial H}{\partial q} \delta q + \frac{\partial H}{\partial p} \delta p$$

PROOF

$$\begin{aligned} \delta H &= \delta (p \dot{q} - L) \\ &= \delta p \dot{q} + p \delta \dot{q} - \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) \\ &= \dot{q} \delta p - \frac{\partial L}{\partial q} \delta q \quad \checkmark \text{ Q.E.D.} \end{aligned}$$

$$\Rightarrow \delta H = \underbrace{\dot{q}}_{\frac{\partial H}{\partial p}} \delta p - \underbrace{\left(\frac{\partial L}{\partial q} \right)}_{= \frac{\partial H}{\partial q}} \delta q \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \dot{p} \quad (\text{Euler-Lagrange})$$

$\Rightarrow$

$$\begin{aligned} \dot{q} &= \frac{\partial H}{\partial p} \\ \dot{p} &= -\frac{\partial H}{\partial q} \end{aligned}$$

HAMILTON'S EQUATIONS

2N EQUATIONS OF FIRST ORDER

$\Rightarrow$ AT MOST 2N INTEGRATION CONSTANTS

THESE EQUATIONS CAN BE RE-WRITTEN IN
TERMS OF POISSON BRACKETS

POISSON BRACKET (PB) $\{, \}$

- PROVIDES THE BASIC OPERATION IN HAMILTONIAN MECHANICS
- DESCRIBE EVOLUTION AND SYMMETRY TRANSFORMATIONS
- PREPARE THE BRIDGE TO QUANTUM MECHANICS : $\{ \} \rightarrow \frac{1}{i\hbar} [,]$

DEFINITION:

$$\{A(t), B(t)\} = \frac{\partial A}{\partial q^i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q^i}$$

$A, B \dots$ FUNCTIONS ON Γ

$$\begin{aligned} A(t) &\equiv A(q(t), p(t)) \\ B(t) &\equiv B(q(t), p(t)) \end{aligned}$$

CANONICAL PBs:

$$\{q^i, q^j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q^i, p_j\} = \delta_j^i$$

• PROPERTIES: $A, B, C \in \Gamma$

ANTISYMMETRY $\{A, B\} = -\{B, A\}$

LINEARITY $\{\alpha A + \beta B, C\} = \alpha \{A, C\} + \beta \{B, C\} \quad (\alpha, \beta \in \mathbb{C})$

JACOBI IDENTITY $\{A, \{B, C\}\} + \{B, \{C, A\}\} + \{C, \{A, B\}\} = 0$

PRODUCT RULE (LEIBNIZ RULE) $\{AB, C\} = A \{B, C\} + \{A, C\} B$

• EVOLUTION: $A \in \Gamma$

$$\dot{A} = \frac{\partial A}{\partial t} + \underbrace{\frac{\partial A}{\partial q}}_{\frac{\partial H}{\partial p}} \dot{q} + \underbrace{\frac{\partial A}{\partial p}}_{-\frac{\partial H}{\partial q}} \dot{p} \Rightarrow \dot{A} = \frac{\partial A}{\partial t} + \{A, H\}$$

FIELD THEORY GENERALIZATION

CLASSICAL MECHANICS $\longrightarrow$

t
 $\sim$ TIME

GEN. COORDINATES $q^i(t)$

ACTION $I = \int dt L(q, \dot{q})$

FIELD THEORY

$$(t, x^i) = x^\mu \quad (\mu = 0, 1, 2, 3)$$

$\underbrace{\quad}_{\text{TIME+SPACE}} \quad \underbrace{\quad}_{\text{MINKOWSKI SPACE-TIME}}$

FIELDS $\phi^a(t, \vec{r}) = \phi^a(x)$

NATURAL UNITS $c = 1$

$I = \int dt L(\phi, \dot{\phi})$

- WE WORK WITH LORENTZ-COVARIANT THEORIES SUCH THAT

$$\begin{aligned} dt &\rightarrow dt d\vec{r} = d^4x \\ \dot{q} &\rightarrow \partial_\mu \phi = \left(\frac{\partial \phi}{\partial t}, \nabla_i \phi \right) \\ L(q, \dot{q}) &\rightarrow L(\phi, \dot{\phi}) = \int d\vec{r} \mathcal{L}(\phi, \partial_\mu \phi) \end{aligned}$$

$$I[\phi] = \int d^4x \mathcal{L}(\phi, \partial \phi)$$

- EULER-LAGRANGE EQUATIONS

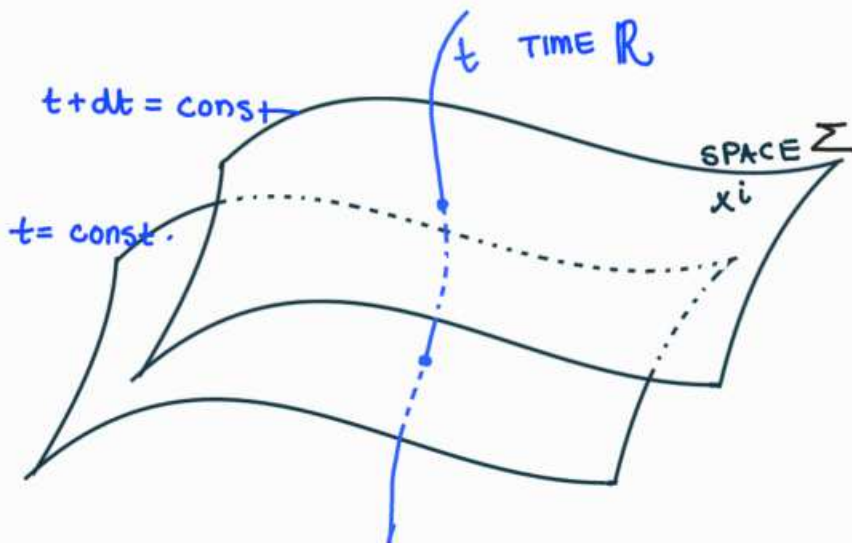
$$\mathcal{E}_a = \frac{\delta I}{\delta \phi^a} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \phi^a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^a)} = 0$$

- CANONICAL MOMENTA

$$\pi_a(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^a(x)}$$

DERIVATIVE $\frac{d}{dt}$ BREAKS EXPLICIT LORENTZ-COVARIANCE
BUT IT STILL EXISTS IMPLICITLY



IN HAMILTONIAN FORMALISM, THE SYSTEM IS DESCRIBED ON A FIXED TIME SLICE $t = \text{const.}$, AND ITS EVOLUTION FROM ONE SLICE TO ANOTHER IS GENERATED THROUGH PBs

* TIME IS TREATED AS AN EXTERNAL PARAMETER

- POISSON BRACKETS: x^i is TREATED AS A CONTINUOUS INDEX
 $\sum_{x^i} \rightarrow \int d^3x$

$A, B \in \Gamma$ MEANS $A(x) = A(\phi(x), \pi(x))$
 $B(x) = B(\phi(x), \pi(x))$

$$\underbrace{\{A(x), B(x')\}}_{\{A(t, \vec{r}), B(t, \vec{r}')\}} \Big|_{t=t'} = \int d^3x'' \left[\frac{\partial A(x)}{\partial \phi^a(x'')} \frac{\partial B(x')}{\partial \pi_a(x'')} - \frac{\partial A(x)}{\partial \pi_a(x'')} \frac{\partial B(x')}{\partial \phi^a(x'')} \right] \Big|_{t=t'=t''}$$

* WE WILL STOP WRITING $t=t'$, $t=t'=t''$, ETC., BUT IT IS ALWAYS FULFILLED: ALL EXPRESSIONS ARE VALID AT $t = \text{const.}$

• CANONICAL PBS

→ WE USE IDENTITIES $\left. \frac{\partial \phi^a(x)}{\partial \phi^b(x')} \right|_{t=t'} = \delta_b^a \delta(\vec{x} - \vec{x}')$

$$\left. \frac{\partial \pi_a(x)}{\partial \pi_b(x')} \right|_{t=t'} = \delta_b^a \delta(\vec{x} - \vec{x}')$$

$$\delta_b^a = \begin{cases} 1, & a=b \\ 0, & a \neq b \end{cases} \quad \text{KRONCKER SYMBOL}$$

$$\delta^{(3)}(\vec{x} - \vec{x}') = \begin{cases} \infty, & \vec{x} = \vec{x}' \\ 0, & \vec{x} \neq \vec{x}' \end{cases} \quad \text{DIRAC DELTA FUNCTION}$$

BASIC PROPERTIES:

$$\sum_b \delta_b^a v^b = v^a$$

$$\int d^3x' \delta(\vec{x} - \vec{x}') f(\vec{x}') = f(\vec{x})$$

$$\int d^3x' \delta(\vec{x} - \vec{x}') = 1$$

$$\int d^3x' \partial_i' \delta(\vec{x} - \vec{x}') f(\vec{x}') = -\partial_i f(\vec{x})$$

→ WE APPLY THESE PROPERTIES TO DEDUCE CANONICAL PBS

$$\begin{aligned}\{\phi^a(x), \pi_b(x')\} &= \int d^3x'' \left[\underbrace{\frac{\partial \phi^a(x)}{\partial \phi^c(x'')}}_{\delta_c^a \delta(\vec{x}-\vec{x}'')} \underbrace{\frac{\partial \pi_b(x')}{\partial \pi_c(x'')}}_{\delta_b^c \delta(\vec{x}'-\vec{x}'')} - \underbrace{\frac{\partial \phi^a(x)}{\partial \pi_c(x'')}}_0 \underbrace{\frac{\partial \pi_b(x')}{\partial \phi^c(x'')}}_0 \right] \\ &= \delta_b^a \int d^3x'' \delta(\vec{x}-\vec{x}'') \delta(\vec{x}'-\vec{x}'') = \delta_b^a \delta(\vec{x}-\vec{x}')\end{aligned}$$

$$\{\phi^a(x), \pi_b(x')\} = \delta_b^a \delta(\vec{x}-\vec{x}')$$

$$\{\phi^a(x), \phi^b(x')\} = 0$$

$$\{\pi_a(x), \pi_b(x')\} = 0$$

CANONICAL PBS

→ FROM NOW, WE WILL USE ONLY CANONICAL PBS AND PROPERTIES OF PBS TO COMPUTE ANY $\{A(x), B(x')\}$

HAMILTONIAN DYNAMICS OF SINGULAR SYSTEMS

$$q^i(t) \quad (i=1, \dots, N), \quad p_i = \frac{\partial L}{\partial \dot{q}^i}$$

HESSIAN MATRIX $W_{ij} = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$, $W = [W_{ij}]$ $N \times N$ MATRIX

$\det W = 0 \Rightarrow W \sim \left(\begin{array}{c|c} \underbrace{\begin{matrix} W_1 & & \\ & \ddots & \\ & & W_{N-P} \end{matrix}}_{N-P} & \underbrace{\begin{matrix} 0 & \dots & 0 \end{matrix}}_P \end{array} \right) \begin{matrix} \left. \vphantom{\begin{matrix} W_1 \\ \vdots \\ W_{N-P} \end{matrix}} \right\} N-P \\ \left. \vphantom{\begin{matrix} 0 \\ \vdots \\ 0 \end{matrix}} \right\} P \end{matrix}$

ASSUME THAT W HAS P INDEPENDENT ZERO MODES

W IS EQUIVALENT TO A DIAGONAL MATRIX $N \times N$

$\Rightarrow \text{rank } W = N - P$ (INVERTIBLE PART)

$\text{ker } W = \underbrace{P}_{\text{NULLITY}}$ (KERNEL, NULL SPACE) $P = \dim \text{ker } W$

$\Rightarrow$ • ONLY $N - P$ VELOCITIES CAN BE EXPRESSED IN TERMS OF MOMENTA $p = p(q, \dot{q}) \Leftrightarrow \dot{q} = \dot{q}(q, p)$

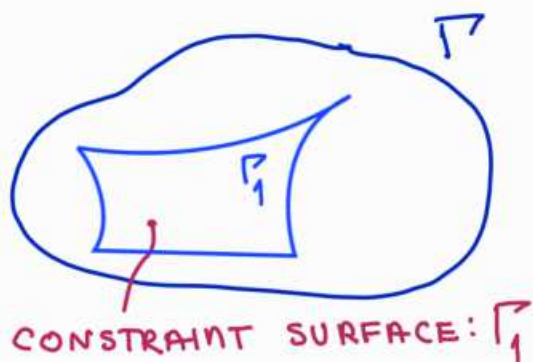
- THERE ARE P ZERO MODES OF W , SUCH THAT P MOMENTA DO NOT DEPEND ON VELOCITIES

$p = p(q) \Rightarrow$ THEY BECOME CONSTRAINTS, WITHOUT DYNAMICS

DIRAC - BERGMANN METHOD

- TREATS HAMILTONIAN SYSTEMS WITH CONSTRAINTS

$\phi_m(q, p) = 0 \quad (m=1, \dots, P)$ PRIMARY CONSTRAINTS



$$\Gamma_1 \subseteq \Gamma$$

NOT ALL (q, p) IN Γ ARE INDEPENDENT;
WHEN PRIMARY CONSTR. ARE SOLVED,
WE HAVE A SUBSPACE Γ_1

- **REGULARITY CONDITIONS:** CONSTRAINTS HAVE TO BE FUNCTIONALLY DEPENDENT, THAT IS, THE JACOBIAN

$$\left. \frac{\partial \phi_m}{\partial (q, p)} \right|_{\Gamma_1} \text{ HAS TO HAVE THE RANK } P \text{ ON } \Gamma_1$$

EXAMPLE:

$$\phi = p, \quad \tilde{\phi} = p^2 \text{ ARE NOT FUNCTIONALLY INDEPENDENT; WE HAVE TO CHOOSE ONLY } \phi$$

$$\left. \frac{\partial \phi}{\partial (q, p)} = (0 \ 1) \right|_{\Gamma_1} \quad \checkmark \quad \text{RANK } 1$$

$$\left. \frac{\partial \tilde{\phi}}{\partial (q, p)} = (0 \ 2p) \right|_{\Gamma_1} = (0 \ 0) \quad \times \quad \text{RANK } 0$$

$\Rightarrow$

$$\begin{aligned} \dim \Gamma &= 2N \\ \dim \Gamma_1 &= 2N - P \end{aligned}$$

- POISSON BRACKETS OF CONSTRAINTS CAN LEAD TO CONTRADICTION IF NOT TREATED CAREFULLY

EXAMPLE:

$$\begin{aligned} \phi = p &= 0 \quad \text{ON } \Gamma_1 \\ \phi = p &\neq 0 \quad \text{ON } \Gamma \end{aligned}$$

$$\{ \underset{0}{\phi}, q \} = -1 \neq 0 \Rightarrow \text{CONTRADICTION ON } \Gamma_1$$

- TO AVOID CONTRADICTION WE HAVE TO WORK ON Γ ONLY, AND TO INTRODUCE TWO TYPES OF EQUALITY TO TREAT Γ_1

$$\text{WEAK EQUALITY: } F(q, p) \approx 0 \Leftrightarrow F(q, p)|_{\Gamma_1} = 0$$

$$\text{STRONG EQUALITY: } F(q, p) = 0 \Leftrightarrow F, \frac{\partial F}{\partial q}, \frac{\partial F}{\partial p} \Big|_{\Gamma_1} = 0$$

* CONSTRAINTS ARE WEAK EQUALITIES

$$\phi_m(q, p) \approx 0 \quad (m=1, \dots, P)$$

- RELATION BETWEEN THESE TWO TYPES OF EQUALITIES

$$F(q,p) \approx 0 \Leftrightarrow F(q,p) = \lambda^m \phi_m(q,p)$$

HENNEAUX-
TEITELBOIM
THEOREM

MULTIPLIERS

$\Rightarrow$ IF F IS WEAKLY ZERO, IT IS A LINEAR COMBINATION OF THE CONSTRAINTS

* CANONICAL HAMILTONIAN

$$H_c = p\dot{q} - L$$

WHEN $\dot{q}(q,p)$ CANNOT BE SOLVED BECAUSE W_{ij} IS NOT INVERTIBLE, WE USE

$\phi_m \approx 0$ TO ELIMINATE $\dot{q}$

$$\Rightarrow H_c \approx H_c(q,p)$$

(IT IS ONLY WEAKLY A FUNCTION OF q,p)

* USING HENNEAUX-TEITELBOIM THEOREM

$$H_T = H_c + \lambda^m \phi_m$$

TOTAL HAMILTONIAN

ARBITRARY MULTIPLIERS, $\lambda^m(t)$

* TOTAL HAMILTONIAN DEFINES THE EVOLUTION, AND NOT THE CANONICAL ONE

SHOW: HAMILTON'S EQUATIONS DEFINE EVOLUTION OF $F(t) = F(q(t), p(t))$ AS
(HOMEWORK, THE SAME METHOD AS BEFORE)

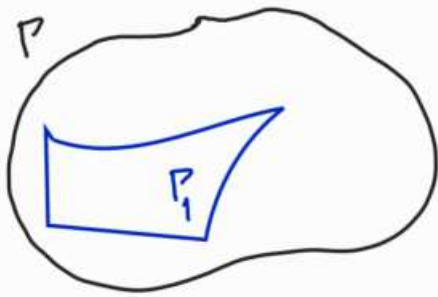
$$\dot{F} = \{F, H_c\} + \lambda^m \{F, \phi_m\} \approx \{F, H_T\}$$

$$\dot{F} \approx \{F, H_T\}$$

$\rightarrow H_T$ PRODUCE EQS. OF MOTION EQUIVALENT TO EULER-LAGRANGE EQUATION (TOGETHER WITH CONSTRAINTS)

$\rightarrow H_c$ DOES NOT GIVE EQS. OF MOTION

CONSISTENCY CONDITIONS (STABILITY CONDITIONS)



THE SURFACE P HAS TO BE CONSERVED DURING THE EVOLUTION OF THE SYSTEM

$$\phi_m \approx 0 \Rightarrow \dot{\phi}_m \approx 0$$

$$\underbrace{\{\phi_m, H_c\}}_{A_m} + u^n \underbrace{\{\phi_m, \phi_n\}}_{\Delta_{mn} = -\Delta_{nm}} \approx 0$$

(ANTISYMMETRIC MATRIX $P \times P$)

$$A_m + \Delta_{mn} u^n \approx 0$$

$$\underbrace{A}_{\text{COLUMN}} + \underbrace{\Delta}_{\text{MATRIX}} \underbrace{u}_{\text{COLUMN}} \approx 0$$

• ONE OF THE FOLLOWING SITUATIONS CAN HAPPEN:

- 1) $A \approx 0, \Delta \approx 0 \Rightarrow \dot{\phi} \approx 0$ IS AUTOMATICALLY SATISFIED, WE DO NOT GET ANYTHING NEW
- 2) $A \neq 0, \Delta \approx 0 \Rightarrow \dot{\phi} \approx A \approx 0 \Rightarrow A(q, p) \approx 0$
SECONDARY CONSTRAINT
- 3) $\Delta \neq 0 \Rightarrow \Delta^{-1}$ EXISTS
 $u = \Delta^{-1} A$ SOLVED MULTIPLIERS

• IN GENERAL, THE CONDITIONS 1)-3) ARE MIXED UP

Δ_{mn} $P \times P$ MATRIX

$\det(\Delta) \neq 0 \Rightarrow \text{rank}(\Delta) = P$, ALL MULTIPLIERS ARE SOLVED

$\det(\Delta) < 0 \Rightarrow \text{rank}(\Delta) < P$, ONLY $\text{rank}(\Delta)$ MULTIPLIERS ARE SOLVED

- SECONDARY CONSTRAINTS ALSO HAVE TO SATISFY CONSISTENCY CONDITIONS AND 1)-3) CAN HAPPEN; THEY CAN GIVE TERTIARY CONSTRAINTS, ETC.
- WE WILL CALL **SECONDARY CONSTRAINTS** ALL CONSTRAINTS THAT ARE NOT PRIMARY (SECONDARY, TERTIARY,)
- THE **NUMBER OF GENERATIONS** OF CONSTRAINTS MEANS HOW MANY SUCCESSIVE CONSISTENCY STEPS ARE NEEDED BEFORE THE DIRAC ALGORITHM CLOSES

EXAMPLE:

$$\begin{array}{ll}
 \phi^{(1)} \approx 0 & \text{(PRIMARY)} \\
 \dot{\phi}^{(1)} = \phi^{(2)} \approx 0 & \text{(SECONDARY)} \\
 \dot{\phi}^{(2)} = \phi^{(3)} \approx 0 & \text{(TERTIARY)} \\
 \dot{\phi}^{(3)} \equiv 0 &
 \end{array}
 \left. \vphantom{\begin{array}{l} \phi^{(1)} \\ \dot{\phi}^{(1)} \\ \dot{\phi}^{(2)} \end{array}} \right\} \Rightarrow 3 \text{ GENERATIONS OF CONSTRAINTS}$$

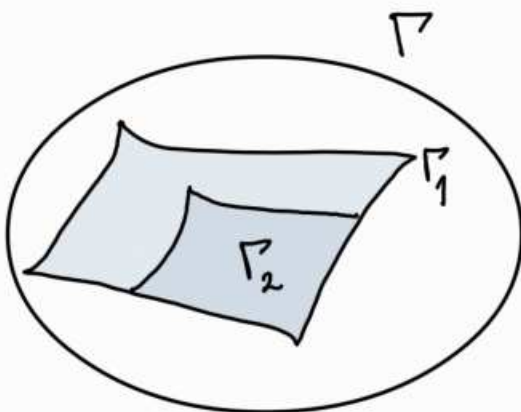
PRIMARY: $\phi^{(1)}$
 SECONDARY: $\phi^{(2)}, \phi^{(3)}$

- WE HAVE:

$$\begin{aligned}
 \phi_m \approx 0, \quad m=1, \dots, P \\
 \chi_k \approx 0, \quad k=1, \dots, S
 \end{aligned}$$

PRIMARY CONSTRAINTS
 SECONDARY CONSTRAINTS

$$\varphi_s = (\phi_m, \chi_k) \approx 0, \quad s=1, \dots, P+S \quad \text{ALL CONSTRAINTS}$$



$$\Gamma_2: \varphi_s \approx 0, \quad \dot{\varphi}_s \approx 0$$

$$\Gamma_2 \subseteq \Gamma_1 \subseteq \Gamma$$

$$\dim(\Gamma_2) = 2N - P - S$$

WHAT MULTIPLIERS HAVE BEEN DETERMINED?

$$\dot{\Psi}_S \approx \{\Psi_S, H_C\} + \mu^m \{\Psi_S, \phi_m\} \approx 0$$

ASSUME THAT N_1' MULTIPLIERS REMAIN UNSOLVED;

$$v^a(t) \quad (a=1, \dots, N_1')$$

$$\Rightarrow \mu^m = \underbrace{U^m(q, p)}_{\substack{\text{DETERMINED} \\ \text{PART} \\ \text{(PARTICULAR SOLUTION)}}} + \underbrace{v^a V_a^m(q, p)}_{\substack{\text{NOT DETERMINED} \\ \text{(HOMOGENEOUS E.G. SOLUTION)}}$$

* TOTAL HAMILTONIAN BECOMES:

$$H_T = H_C + \underbrace{\mu^m \phi_m}_{U^m + v^a V_a^m} = \underbrace{H_C + U^m \phi_m}_{H'} + \underbrace{v^a V_a^m \phi_m}_{\phi_a}$$

$$H_T = H' + v^a \phi_a \quad (a=1, \dots, N_1')$$

AFTER THE WHOLE DIRAC PROCEDURE,
WE STILL HAVE N_1' MULTIPLIERS v^a

CORRESPONDING TO CONSTRAINTS $\phi_a = V_a^m \phi_m \approx 0$

(COMBINATION OF CONSTRAINTS
IS ALSO A CONSTRAINT)

* THE EXISTENCE OF ARBITRARY $v^a(t)$ PRODUCES AN
AMBIGUITY IN THE SYSTEM'S EVOLUTION $\Rightarrow$ SYMMETRY

* HOWEVER, NOT ALL ϕ_m ARE RELATED TO THE SYMMETRY,
BUT ONLY SOME OF THEM (ϕ_a)

$\Rightarrow$ THIS MEANS THAT A CLASSIFICATION TO PRIMARY
AND SECONDARY CONSTRAINTS IS NOT VERY IMPORTANT FOR A
SYMMETRY CONCEPT; WE NEED A DIFFERENT CLASSIFICATION

THE LAST TIME WE CONCLUDED THAT ALL CONSTRAINTS SATISFY:

$$\dot{\Psi}_s \approx \{\Psi_s, H_c\} + \mu^m \{\Psi_s, \phi_m\} \approx 0 \quad (s=1, \dots, P+S; m=1, \dots, P)$$

GENERAL SOLUTION:

$$\mu^m = \underbrace{U^m(q, p)}_{\text{DETERMINED PART}} + v^a \underbrace{V_a^m(q, p)}_{\text{UNSOLVED MULTIPLIERS}}$$

DETERMINED PART

UNSOLVED MULTIPLIERS
($a=1, \dots, N_1^1; N_1^1 \leq P$)

(PARTICULAR SOLUTION)

(HOMOGENOUS EQ. SOLUTION)

* TOTAL HAMILTONIAN BECOMES:

$$H_T = H_c + \underbrace{\mu^m \phi_m}_{U^m + v^a V_a^m} = \underbrace{H_c + U^m \phi_m}_{H'} + \underbrace{v^a V_a^m \phi_m}_{\phi_a}$$

$$H_T = H' + v^a \phi_a \quad (a=1, \dots, N_1^1)$$

AFTER THE WHOLE DIRAC PROCEDURE,
WE STILL HAVE N_1^1 MULTIPLIERS v^a

CORRESPONDING TO CONSTRAINTS $\phi_a = V_a^m \phi_m \approx 0$

(COMBINATION OF CONSTRAINTS
IS ALSO A CONSTRAINT)

* THE EXISTENCE OF ARBITRARY $v^a(t)$ PRODUCES AN
AMBIGUITY IN THE SYSTEM'S EVOLUTION $\Rightarrow$ SYMMETRY

* HOWEVER, NOT ALL ϕ_m ARE RELATED TO THE SYMMETRY,
BUT ONLY SOME OF THEM (ϕ_a)

$\Rightarrow$ THIS MEANS THAT A CLASSIFICATION TO PRIMARY
AND SECONDARY CONSTRAINTS IS NOT VERY IMPORTANT FOR A
SYMMETRY CONCEPT; WE NEED A DIFFERENT CLASSIFICATION

• FIRST CLASS FUNCTION

$$\{F, \psi_s\} \approx 0 \quad (\forall s)$$

• SECOND CLASS FUNCTION

$$\{F, \psi_s\} \neq 0 \quad (\exists s)$$

* IF F IS I CLASS $\{F, \psi_s\} \approx 0 \Rightarrow \{F, \psi_s\} = F_s^r \psi_r$

HENNEAUX-
TEITELBOIM

* IF F, G ARE I CLASS

JACOBI IDENTITY: $\{\{F, G\}, \psi_s\} + \{\{G, \psi_s\}, F\} + \{\{\psi_s, F\}, G\} = 0$

$$\underbrace{\{G_s^r \psi_r, F\}}_{\approx G_s^r \{\psi_r, F\}} + \underbrace{\{-F_s^r \psi_r, G\}}_{\approx -F_s^r \{\psi_r, G\}} = -G_s^r F_r^k \psi_k \approx 0 = F_s^r G_r^k \psi_k \approx 0$$

$$\Rightarrow \{\{F, G\}, \psi_s\} \approx 0 \quad (\forall s)$$

$\{F, G\}$ IS I CLASS

WE CONCLUDE:

$$F, G \dots \text{I CLASS} \Rightarrow \{F, G\} \dots \text{I CLASS}$$

* CLAIM: $\phi_a (a=1, \dots, N_1')$ ARE PRIMARY FIRST CLASS CONSTRAINTS (PFC)

PROOF: $\phi_a = V_a^m \phi_m \approx 0$ IS PRIMARY BY CONSTRUCTION (BECAUSE ϕ_m ARE PRIMARY)

• WE ONLY HAVE TO SHOW THAT ϕ_a ARE I CLASS

$$\{\phi_a, \psi_s\} = \{V_a^m \phi_m, \psi_s\} = V_a^m \{\phi_m, \psi_s\} + \underbrace{\phi_m \{V_a^m, \psi_s\}}_{\approx 0}$$

WE KNOW: $0 = \dot{\psi}_s = \{\psi_s, H_c\} + u^m \{\psi_s, \phi_m\} \approx 0$

$$u^m = U^m + v^a V_a^m$$

$$\Rightarrow 0 = \underbrace{\{\psi_s, H_c\} + U^m \{\psi_s, \phi_m\}}_{=0 \text{ (PARTICULAR SOLUTION)}} + v^a V_a^m \{\psi_s, \phi_m\}$$

$$0 = v^a v_a^m \{ \psi_s, \phi_m \} \quad (\forall v^a)$$

$$0 = v_a^m \{ \psi_s, \phi_m \}$$

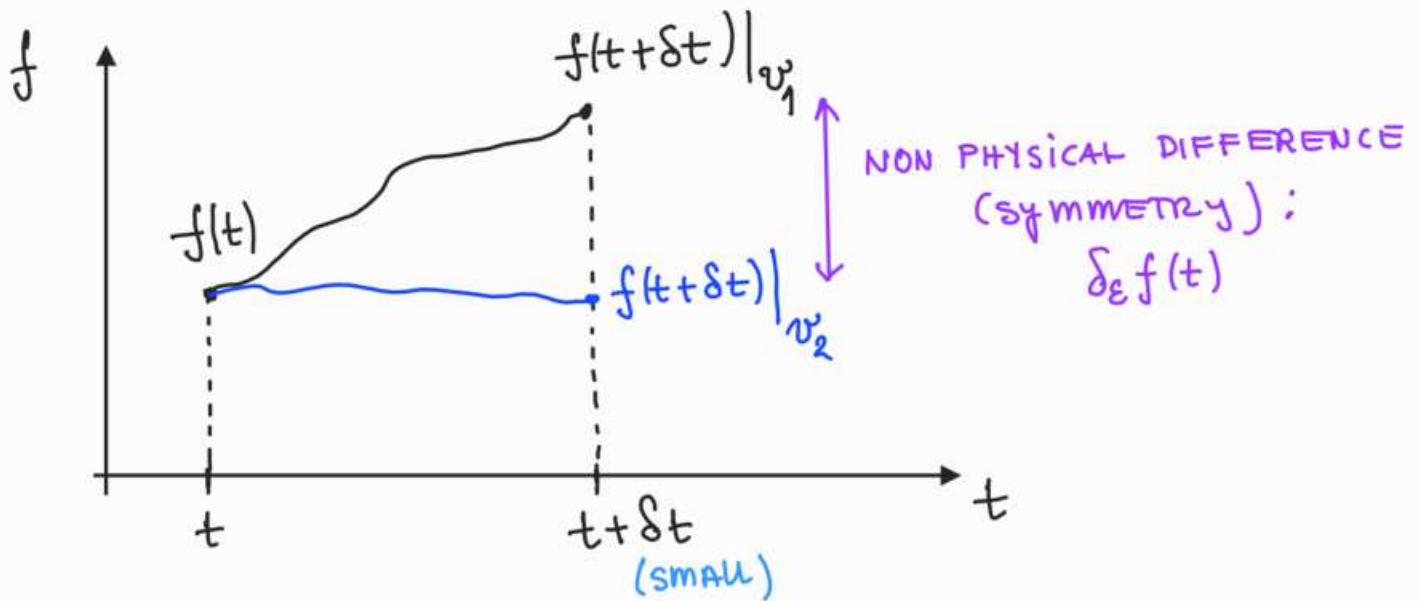
$$\Rightarrow \{ \phi_a, \psi_s \} \approx 0 \quad (\forall s) \quad \Rightarrow \phi_a \text{ ARE } \mathbb{I} \text{ CLASS}$$

* CLAIM: H' is FIRST CLASS (SHOW)

$\Rightarrow$ IF H' , ϕ_a ARE FIRST CLASS,
THUS H_T IS FIRST CLASS

CONSTRAINTS AND SYMMETRIES

- THE PRESENCE OF $v^a(t)$ INTRODUCES AN AMBIGUITY IN EVOLUTION OF $q(t), p(t)$ FOR GIVEN INITIAL CONDITIONS
- LET $f(t) = f(q(t), p(t), v(t))$ BE AN ARBITRARY FUNCTION ON Γ



$$\begin{aligned} f(t + \delta t) &\approx f(t) + \dot{f}(t) \delta t \\ &= f(t) + [\{f, H'\} + v^a \{f, \phi_a\}] \delta t \end{aligned}$$

* NON PHYSICAL DIFFERENCE IN THE RESULT DURING EVOLUTION
DUE TO THE PRESENCE OF $V^a(t)$

$$\underbrace{f(t+\delta t)|_{v_2} - f(t+\delta t)|_{v_1}}_{\delta_\epsilon f(t)} = \cancel{f(t)} + [\cancel{\{f, H'\}} + v_2^a \{f, \phi_a\}] \delta t - \cancel{f(t)} - [\cancel{\{f, H'\}} + v_1^a \{f, \phi_a\}] \delta t$$

$$= \underbrace{(v_2^a - v_1^a) \delta t}_{\epsilon^a \text{ INFINITESIMAL PARAMETER}} \{f, \phi_a\} = \epsilon^a \{f, \phi_a\}$$

$\Rightarrow$ $\delta_\epsilon f = \epsilon^a \{f, \phi_a\}$ THE CHANGE GENERATED BY PFC CONSTRAINTS WITH A LOCAL PARAMETER $\epsilon^a(t)$

- GENERATOR: $G(\epsilon) = \epsilon^a \phi_a$
- POISSON BRACKETS GENERATE A SYMMETRY TRANSFORM.
 $\delta_\epsilon(\dots) = \{\dots, G(\epsilon)\}$
- FIRST CLASS CONSTRAINTS GENERATE LOCAL/GAUGE SYMMETRY
(TRANSFORMATIONS THAT DO NOT CHANGE A PHYSICAL STATE OF THE SYSTEM)

TWO SUCCESSIVE TRANSFORMATIONS : ASSUME THERE ARE NO 2ND CLASS CONSTRAINTS

SHOW: $\underbrace{(\delta_{\epsilon_1} \delta_{\epsilon_2} - \delta_{\epsilon_2} \delta_{\epsilon_1}) f}_{\substack{2 \text{ SUCCESSIVE} \\ \text{INFINITESIMAL} \\ \text{TRANSFORMATIONS}}} = \epsilon_1^a \epsilon_1^b \{f, \{\phi_a, \phi_b\}\}$
JACOBI IDENTITY

$\Rightarrow \{\phi_a, \phi_b\} \dots$ GENERATE SUCCESSIVE TRANSFORMATIONS, THEY HAVE TO BE FIRST CLASS

$$\{\phi_a, \phi_b\} \approx 0 \Rightarrow$$

$$\{\phi_a, \phi_b\} = f_{ab}^c \phi_c + \text{SECONDARY FIRST CLASS CONSTRAINTS}$$

ALGEBRA is NOT CLOSED



WE HAVE TO INCLUDE SECONDARY FIRST CLASS CONSTRAINTS AS SYMMETRY GENERATORS

EXAMPLE

$$L = \frac{1}{2} \dot{q}_1^2 + \dot{q}_1 q_2 + \frac{1}{2} (q_1 - q_2)^2$$

(TOY MODEL, IT WILL NOT GIVE $H \geq 0$)

MOMENTA: $p_1 = \dot{q}_1 + q_2 \Rightarrow \dot{q}_1 = p_1 - q_2$

$$p_2 = 0$$

$$\Rightarrow \phi = p_2 \approx 0$$

PRIMARY CONSTRAINT

CANONICAL HAMILTONIAN:

$$H_c = p_1 \dot{q}_1 + p_2 \dot{q}_2 - L$$

$$= \underbrace{p_1 (p_1 - q_2)}_{\substack{\downarrow \\ \frac{1}{2} (p_1 - q_2)^2}} + \underbrace{p_2 \dot{q}_2}_{\substack{\text{SS} \\ 0}} - \frac{1}{2} (p_1 - q_2)^2 - (p_1 - q_2) q_2 - \frac{1}{2} (q_1 - q_2)^2$$

$$\frac{1}{2} (p_1 - q_2)^2$$

$$H_c = \frac{1}{2} (p_1 - q_2)^2 - \frac{1}{2} (q_1 - q_2)^2$$

TOTAL HAMILTONIAN:

$$H_T = H_c + \mu \phi$$

CONSISTENCY CONDITIONS:

$$\dot{\phi} \approx p_1 - q_1 \Rightarrow \chi = p_1 - q_1 \approx 0$$

SECONDARY CONSTR.

$$\dot{\chi} = -\chi \approx 0 \Rightarrow \text{THE END OF DIRAC PROCEDURE}$$

CLASS: $\{\phi, \chi\} = 0$

$$\Rightarrow \phi, \chi \dots \text{I CLASS}$$

PRIMARY FIRST CLASS (PFC)

SECONDARY FIRST CLASS

* TRANSFORMATIONS GENERATED BY ϕ

$$\delta_\epsilon q_i = \{q_i, \epsilon \phi\} = \{q_i, \epsilon p_2\} = \epsilon \delta_{i2}$$

$$\delta_\epsilon p_i = 0$$

* TRANSFORMATIONS GENERATED BY χ

$$\delta_\lambda q_i = \{q_i, \lambda \chi\} = \lambda \{q_i, p_1 - q_1\} = \lambda \delta_{i1}$$

$$\delta_\lambda p_i = \{p_i, \lambda \chi\} = \lambda \{p_i, p_1 - q_1\} = \lambda \delta_{i1}$$

EXTENDED HAMILTONIAN

$$H_T = H' + \underbrace{v_1^a}_{\text{PFC}} \phi_{1a} \Rightarrow \text{SYMMETRY GENERATED DUE TO INDEFINITE MULTIPLIER } v_1^a(t)$$

- SECONDARY FIRST CLASS CONSTRAINTS ARE MISSING IN THE HAMILTONIAN, THEY ALSO GENERATE SYMMETRIES

EXTENDED HAMILTONIAN

$$H_E = H_T + v_2^a \phi_{2a}$$

$$H_E = H' + \underbrace{v_1^a}_{\text{PFC}} \phi_{1a} + \underbrace{v_2^a}_{\text{SECONDARY FC}} \phi_{2a}$$

- ALGEBRA OF FIRST CLASS CONSTRAINTS IS CLOSED NOW:

$$\phi_a \equiv (\phi_{1a}, \phi_{2a})$$

(WE KEEP THE SAME NOTATION FOR SIMPLICITY, ALTHOUGH IT SHOULD BE, E.G., $\tilde{\phi}_A = (\phi_{1a}, \tilde{\phi}_{2a})$)

$$\{\phi_a, \phi_b\} = f_{ab}^c \phi_c$$

CONSTRAINT ALGEBRA

- HAMILTON'S EQUATION OBTAINED FROM H_E ARE NOT EQUIVALENT TO EULER-LAGRANGE EQUATIONS, BUT A DIFFERENCE IS NON-PHYSICAL

EXAMPLE (CONTINUATION OF THE PREVIOUS ONE)

EULER-LAGRANGE EQUATIONS:

$$\mathcal{E}_1 = \ddot{q}_1 - \ddot{q}_2 - \dot{\ddot{q}}_1 - \dot{\ddot{q}}_2 = 0$$

$$\mathcal{E}_2 = \dot{\ddot{q}}_1 + \ddot{q}_2 - \ddot{q}_1 = 0$$

HAMILTON'S EQUATIONS: $\phi = p_2 \approx 0, \chi = p_1 - q_1 \approx 0$

$$\textcircled{H_T} \quad \begin{aligned} \dot{\ddot{q}}_1 &= p_1 - q_2 (*) & \dot{p}_1 &= q_1 - q_2 \\ \dot{\ddot{q}}_2 &= m & \dot{p}_2 &= p_1 - q_1 \approx 0 \end{aligned}$$

$$\left. \begin{aligned} (*)' &\Rightarrow \mathcal{E}_1 = 0 \\ (*) &\Rightarrow \mathcal{E}_2 \approx 0 \end{aligned} \right\} \Rightarrow \text{EQUIVALENT TO EULER-LAGRANGE}$$

$$\textcircled{H_E} \quad H_E = H_T + \tilde{m} \chi$$

$$\begin{aligned} \dot{\ddot{q}}_1 &= p_1 - q_2 + \tilde{m} (**) & \dot{p}_1 &= q_1 - q_2 + \tilde{m} \\ \dot{\ddot{q}}_2 &= m & \dot{p}_2 &= p_1 - q_1 \approx 0 \end{aligned}$$

$$\left. \begin{aligned} (**)' &\Rightarrow \mathcal{E}_1 + \dot{\tilde{m}} + \tilde{m} = 0 \\ (**') &\Rightarrow \mathcal{E}_2 - \tilde{m} \approx 0 \end{aligned} \right\} \begin{array}{l} \text{BECOME EULER-LAGRANGE} \\ \text{ONLY WHEN } \tilde{m} = 0 \\ \text{(TOTAL HAMILTONIAN)} \end{array}$$

DIRAC BRACKETS

→ SO FAR WE HAVE INTERPRETED ONLY FIRST CLASS CONSTRAINTS AND RELATED THEM TO LOCAL SYMMETRIES

→ SECONDARY CONSTRAINTS ARE NOT SYMMETRY-RELATED, THEY JUST ELIMINATE REDUNDANT CANONICAL VARIABLES

→ THUS, WE CAN SOLVE THEM AND REDUCE THE PHASE SPACE, BUT WE HAVE TO DO THIS CONSISTENTLY

EXAMPLE:

$$q^1 \approx 0, p_1 \approx 0$$

$\overline{\text{II}}$ CLASS CONSTRAINTS,
DO NOT GENERATE ANY SYMMETRY

- HOWEVER, WE CANNOT SET $q^1=0, p_1=0$ BECAUSE $\{q^1, p_1\} = 1$

HOW TO ELIMINATE (q^1, p_1) ?

- WE DEFINE NEW BRACKETS: $\{q^1, p_1\}^* = 0$
 $\{q^i, p_j\}^* = \delta_j^i, i, j \geq 2$

$$A, B \in \Gamma, \quad \{A, B\}^* = \sum_{i \geq 2} \left(\frac{\partial A}{\partial q^i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q^i} \right)$$

↑
WE JUST EXCLUDE $i=1$ FROM THE SUM

FINAL EFFECT:

$$q^1 \approx 0, p_1 \approx 0$$
$$\{q^i, p_j\} = \delta_j^i$$
$$i, j = 1, 2, \dots$$



$$q^1 = 0, p_1 = 0$$

$$\{q^i, p_j\}^* = \begin{cases} 0, & i=1 \text{ OR } j=1 \\ \delta_j^i, & i, j \geq 2 \end{cases}$$

⇒ THANKS TO $\{, \}^*$, THE CONSTRAINTS BEHAVE AS STRONG EQUALITIES AND CAN BE ELIMINATED

GENERAL CASE

$$\phi_a \approx 0$$

$$\theta_r \approx 0$$

$$a = 1, \dots, N_1$$

$$r = 1, \dots, N_2$$

I CLASS CONSTRAINTS

$\overline{\text{II}}$ CLASS CONSTRAINTS

$$\{\phi_a, \phi_b\} \approx 0$$

$$\{\phi_a, \theta_r\} \approx 0$$

$$\{\theta_r, \theta_p\} \approx \Delta_{rp}$$

NON-SINGULAR MATRIX

$\det \Delta \neq 0 \Rightarrow \text{EXISTS } \Delta^{-1}$

- $\Delta \dots N_2 \times N_2$ MATRIX
- BECAUSE Δ IS ANTISYMMETRIC, N_2 IS AN EVEN NUMBER

DIRAC BRACKETS (CAN BE DEFINED FOR JUST A SUBSET OF $\theta_r \approx 0$)

$$\{A, B\}^* = \{A, B\} - \{A, \theta_r\} \Delta_{rp}^{-1} \{\theta_p, B\}$$

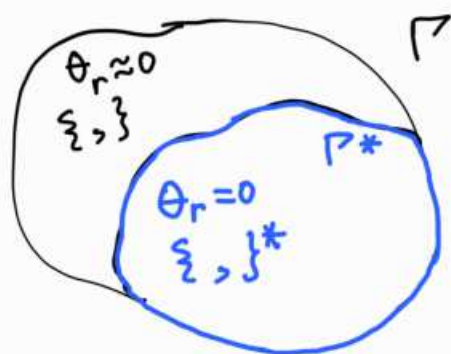
- THEY SATISFY ALL THE PROPERTIES OF POISSON BRACKETS
- THEY ARE ZERO IF A OR B IS II CLASS CONSTRAINT

$$\{A, \theta_s\}^* = \{A, \theta_s\} - \underbrace{\{A, \theta_r\} \Delta_{rp}^{-1} \underbrace{\{\theta_p, \theta_s\}}_{\Delta_{ps}}}_{\delta_{rs}} = 0$$

- THUS, USING $\xi, \}$ INSTEAD OF $\xi, \}$, WE CAN WRITE $\theta_r = 0$ STRONGLY

- EVOLUTION: $\dot{F} \approx \{F, H_T\} \approx \{F, H_T\}^*$

$\Rightarrow$ II CLASS CONSTRAINTS DISAPPEAR FROM THE PHASE SPACE



REDUCED PHASE SPACE (PARTIALLY OR TOTALLY)

GAUGE CONDITIONS IN HAMILTONIAN FORMALISM

- I CLASS CONSTRAINTS ALSO CONTAIN NON-PHYSICAL VARIABLES DUE TO THE EXISTENCE OF A SYMMETRY

$$\left. \begin{aligned} q^i &\sim q'^i = q^i + \delta_\epsilon q^i \\ p_i &\sim p'_i = p_i + \delta_\epsilon p_i \end{aligned} \right\} \text{PHYSICALLY EQUIVALENT } (\sim) \text{ VARIABLE}$$

$[q^i]_\varepsilon, [p_i]_\varepsilon \dots$ EQUIVALENCE CLASSES (ONE PER ε^a);
EACH CLASS CORRESPONDS TO ONLY
ONE PHYSICAL VARIABLE

- IN LAGRANGIAN METHOD, WE USE A GAUGE FIXING CONDITION TO SOLVE ALL $\varepsilon^a(t)$

EXAMPLE: $\delta_\varepsilon q^i = \varepsilon^i, \quad q^i \sim q'^i = q^i + \varepsilon^i$

→ WE CAN FIX $q'^i = 0$ BY CHOOSING $\varepsilon^i = -q^i$

→ THIS IS A GOOD GAUGE FIXING BECAUSE
A SOLUTION FOR ε^i EXISTS

- HOW TO FIX A GAUGE IN HAMILTONIAN METHOD?

→ WE CONVERT I CLASS CONSTRAINTS INTO II CLASS CONSTR.

$$\begin{array}{ll} \phi^a \approx 0 & \text{I CLASS CONSTRAINT} \\ \Omega_a \approx 0 & \text{GAUGE CONDITION} \end{array} \quad a = 1, \dots, N_{\text{gauge}}$$

- GOOD GAUGE CONDITIONS HAVE TO SATISFY THE FOLLOWING:

(i) THEY ARE ACCESSIBLE

THERE IS A LOCAL TRANSFORMATION $(q, p) \rightarrow (q', p')$
SUCH THAT $\Omega_a(q', p') \approx 0$

$$\varepsilon^a \quad (a=1, \dots, N_{\text{PAR}})$$

SYMMETRIES THAT WE WANT
TO FIX ($N_{\text{PAR}} \leq N_1$)

$$N_{\text{GAUGE}} \leq N_{\text{PAR}}$$

(WE CANNOT HAVE MORE GAUGE CONDITIONS THAN
THE PARAMETERS THAT HAVE TO ACCESS THESE CONDITIONS)

(ii) THEY COMPLETELY FIX THE SYMMETRY

THERE ARE NO GAUGE TR. THAT PRESERVE $\Omega_a \approx 0$

$$\boxed{N_{\text{GAUGE}} \geq N_{\text{PAR}}} \quad (\text{WE CANNOT HAVE LESS CONDITIONS THAN SYMMETRIES})$$

$$\Rightarrow N_{\text{GAUGE}} = N_{\text{PAR}} = N_1 \quad (\text{IF WE FIX ALL SYMMETRIES})$$

$$\Rightarrow \Omega_a \approx 0 \quad \text{HAVE TO DETERMINE ALL MULTIPLIERS } v^a_{\text{CH}}$$

- Ω_a ARE CONSTRAINTS, SO WE CAN APPLY CONSISTENCY CONDITIONS

$$\dot{\Omega}_a \approx \{\Omega_a, H_E\} \approx \{\Omega_a, H'\} + v^b \underbrace{\{\Omega_a, \phi_b\}}_{\text{TO BE ABLE TO SOLVE ALL } v^b, \text{ THIS MATRIX HAS TO BE INVERTIBLE}} \approx 0$$

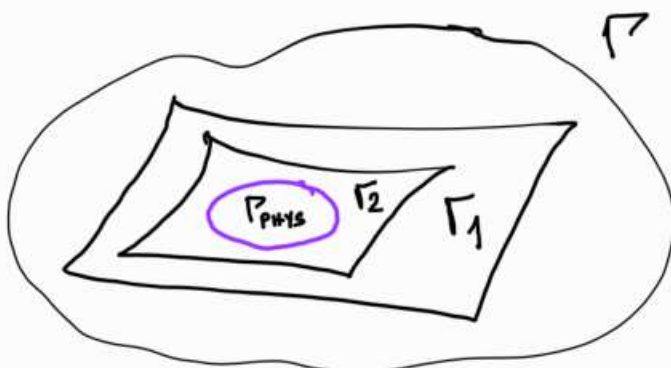
TO BE ABLE TO SOLVE ALL v^b , THIS MATRIX HAS TO BE INVERTIBLE

$$\boxed{\det \{\Omega_a, \phi_b\} \neq 0}$$

CONDITION THAT GUARANTEES **LOCALLY** THAT Ω_a IS A GOOD GAUGE CONDITION

(**GLOBALLY**, WE HAVE TO STUDY CASE BY CASE)

- WHEN WE SOLVE ALL II CLASS CONSTRAINTS AND GAUGE FIX ALL I CLASS CONSTRAINTS, WE WILL OBTAIN **PHYSICAL REDUCED PHASE SPACE**, WHICH CONTAINS ONLY PHYSICAL DEGREES OF FREEDOM



$$\Gamma_{\text{PHYS}} \subseteq \Gamma_2 \subseteq \Gamma_1 \subseteq \Gamma$$

? ?

SECONDARY PRIMARY
+ PRIMARY

$$2N_{\text{phys}} = \dim \Gamma_{\text{phys}} = 2N - N_2 - N_1 - N_1$$

(q^i, p_i) II class I class gauge conditions
 $i=1, \dots, N$ (EVEN number)

$$N_{\text{phys}} = N - N_1 - \frac{1}{2} N_2$$

DIRAC FORMULA

NUMBER OF PHYSICAL DEGREES OF FREEDOM

(PROPAGATING DEGREES OF FREEDOM)

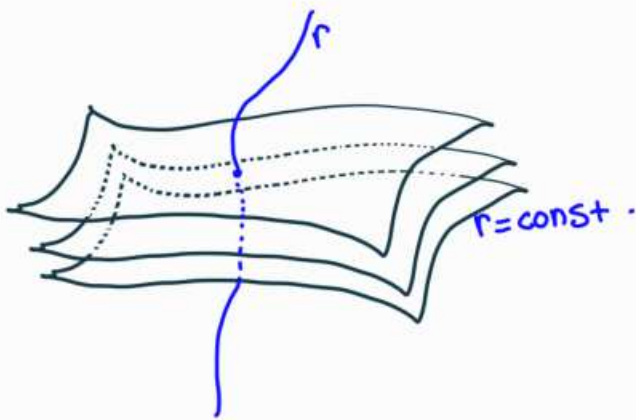
IMPORTANT COMMENTS

1) IN A FIELD THEORY - FOLIATION CHOICE

$$q^i = q^i(t, \vec{x})$$

$$p_i = p_i(t, \vec{x})$$

- WE CAN CHOOSE DIFFERENT (THAN t) EVOLUTION PARAMETER
- EXAMPLE - RADIAL FOLIATION



$$q^i(r, \underbrace{t, \vec{y}}_{\vec{x}}), p_i(r, \underbrace{t, \vec{y}}_{\vec{x}})$$

$$\bullet = \frac{d}{dr}$$

IN CFT : $r = e^t$ IN A COMPLEX PLANE

$$(t, \vec{x}) \rightarrow (r, \varphi) \quad z = e^{t+i\varphi} = r e^{i\varphi}$$

2) RESIDUAL SYMMETRY AFTER GAUGE FIXING

- LAGRANGIAN FORMALISM

$$\delta_\epsilon q(t, x) = \partial_x \epsilon(t, x)$$

$$\text{GAUGE: } q' = 0 \Rightarrow \partial_x \epsilon = -q$$

$$\epsilon(t, x) = \underbrace{\tilde{\epsilon}(t)}_{\text{RESIDUAL PARAMETER}} - \int dx q(t, x)$$

- HAMILTONIAN FORMALISM : NOT ALL $v^a(t, x)$ ARE SOLVED,
THERE IS A RESIDUAL MULTIPLIER $\tilde{v}(t)$

3) CANONICAL MOMENTUM AND BOUNDARY TERM

- BOUNDARY TERMS CHANGE MOMENTA, BUT THE DO NOT CHANGE THE DYNAMICS AND D.O.F.

EXAMPLE $L = \frac{1}{2} \dot{q}_1^2 + q_1 \dot{q}_2 + \dot{q}_1 q_2 - \frac{1}{2} (q_1 - q_2)^2$

$$\tilde{L} = \frac{1}{2} \dot{q}_1^2 - \frac{1}{2} (q_1 - q_2)^2$$

$$L = \tilde{L} + (q_1 q_2)'$$

BOUNDARY TERM IN $I[q]$,
DOES NOT CHANGE EULER-LAGRANGE EQS.

CASE L

$$p_1 = \dot{q}_1 + q_2$$

$$p_2 = q_1$$

$$\rightarrow \phi = p_2 - q_1 \approx 0$$

$$H_c = \underbrace{p_1(p_1 - q_2)} + \cancel{p_2 \dot{q}_2} - \frac{1}{2} \underbrace{(p_1 - q_2)^2} - \cancel{q_1 \dot{q}_2} - \underbrace{(p_1 - q_2) q_2} + \frac{1}{2} (q_1 - q_2)^2$$

$$H_c \approx \frac{1}{2} (p_1 - q_2)^2 + \frac{1}{2} (q_1 - q_2)^2$$

$$H_T = H_c + m\phi$$

$$\begin{aligned} \dot{\phi} &\approx \underbrace{\{p_2, \frac{1}{2} (p_1 - q_2)^2 + \frac{1}{2} (q_1 - q_2)^2\}}_{-(q_2 - p_1) - (q_2 - q_1)} - \underbrace{\{q_1, \frac{1}{2} (p_1 - q_2)^2\}}_{\cancel{p_1} - q_2} = q_1 - q_2 \\ &= \cancel{p_1} + q_1 - 2q_2 \end{aligned}$$

$$\chi = q_1 - q_2 \approx 0$$

$$\{\phi, \chi\} = 1 \quad \text{II class}$$

$$\dot{\chi} = p_1 - q_1 - m \Rightarrow m = p_1 - q_2 \Rightarrow N^* = N - \underbrace{N_1}_2 - \underbrace{\frac{1}{2} N_2}_2 = 1$$

CASE $\tilde{L}$

$$p_1 = \dot{q}_1, \quad \boxed{\phi = p_2 \approx 0}$$

$$H_C = p_1^2 + p_2 \dot{q}_2 - \frac{1}{2} p_1^2 + \frac{1}{2} (q_1 - q_2)^2 = \frac{1}{2} p_1^2 + \frac{1}{2} (q_1 - q_2)^2$$

$$H_T = H_C + u \phi$$

$$\dot{\phi} \approx \{p_2, \frac{1}{2} (q_1 - q_2)^2\} = -(q_2 - q_1) \Rightarrow \boxed{\chi = q_1 - q_2 \approx 0}$$

$$\{\phi, \chi\} = 1 \quad \dots \text{II CLASS}$$

$$\dot{\chi} = \{q_1 - q_2, \frac{1}{2} p_1^2\} - u = p_1 - u \Rightarrow \boxed{u = p_1} \Rightarrow N^* = 1$$

CONCLUSION: PHYSICAL CONTENT IS THE SAME, BUT THE WAYS TO ARRIVE TO THE RESULT ARE DIFFERENT

DIRAC BRACKETS (FOR THE LAGRANGIAN $\tilde{L}$)

$$\{F, G\}^* = \{F, G\} - \underbrace{\{F, \phi\}}_{p_2} \underbrace{\{\chi, G\}}_{q_1 - q_2}$$

$$\boxed{\begin{aligned} \{q_2, p_1\}^* &= -1 \\ \{q_1, p_1\}^* &= 1 \end{aligned}}$$

ALL OTHER BRACKETS ARE ZERO,
INCLUDING $\{q_2, p_2\}^* = 0$

$$P: (q_1, q_2, p_1, p_2)$$

$$P_{\text{PHYS}}: (q_1, p_1)$$

THERE IS ONLY 1 DEGREE OF FREEDOM.

4) HAMILTON'S EQUATIONS FROM THE VARIATIONAL PRINCIPLE

$$\text{HAMILTONIAN ACTION: } \mathcal{I}_H[q, p, u] = \int_{t_0}^{t_1} dt (p_i \dot{q}^i - H_C - u^m \phi_m)$$

→ FIRST ORDER ACTION

→ VARIATIONS: $\delta q^i, \delta p_i, \delta u^m$

→ BOUNDARY CONDITIONS: $\delta q^i|_{t_0} = 0, \delta q^i|_{t_1} = 0$

$\delta I_H = 0 \Rightarrow$ HAMILTON'S EQUATIONS + CONSTRAINTS

→ ADDING A BOUNDARY TERM CHANGES THE BOUNDARY CONDITIONS;

FOR EXAMPLE:
$$\tilde{I}_H = I_H - \int_{t_0}^{t_1} dt (Pq)'$$

$$\delta p_i|_{t_0} = 0, \quad \delta p_i|_{t_1} = 0$$

5) DIRAC CONJECTURE

→ PRIMARY FIRST CLASS CONSTRAINTS (THE ONES APPEARING IN H_T) GENERATE LOCAL SYMMETRIES

→ WE EXPECT THAT SECONDARY FIRST CLASS CONSTRAINTS (THE ONES THAT APPEAR IN H_E) ALSO GENERATE LOCAL SYMM.

CONJECTURE OF DIRAC: ALL SECONDARY FIRST CLASS CONSTR. GENERATE LOCAL SYMMETRIES

→ EXISTEN COUNTEREXAMPLES OF THE CONJECTURE

→ WE WILL SEE THE PROOF OF THE CORRECT STATEMENT (WHICH INCLUDES EXCEPTIONS)

6) THE NUMBER OF GENERATIONS OF CONSTRAINTS IS NOT LIMITED

EXAMPLE: A SYSTEM WITH n CONSTRAINT GENERATIONS

$$q^i \quad (i=1, \dots, n)$$

$$L = \sum_{i=1}^{n-1} \frac{1}{2} (q^i - \dot{q}^{i+1})^2$$

$$\text{LOCAL SYMMETRIES: } \delta_\epsilon q^1 = \frac{d^{n-1} \epsilon}{dt^{n-1}}, \quad \delta_\epsilon q^2 = \frac{d^{n-2} \epsilon}{dt^{n-2}}, \dots, \delta_\epsilon q^n = 0$$

$$\text{HAM. ANALYSIS: } H_T = \sum_{i=2}^n \left(\frac{1}{2} p_i^2 + p_i q_{i-1} \right) + n p_1$$

$$\text{CONSISTENCY COND: } \left. \begin{array}{l} \dot{p}_1 = -p_2 \approx 0 \\ \dot{p}_2 = -p_3 \approx 0 \\ \vdots \\ p_n \approx 0 \end{array} \right\} \Rightarrow p_i \approx 0 \quad (i=1, \dots, n)$$

PART 2 LOCAL SYMMETRIES IN HAMILTONIAN FORMALISM

- WE ALREADY SHOWED THAT POISSON BRACKETS OF I CLASS CONSTR. GENERATE LOCAL SYMMETRIES

* WHAT IS A DIFFERENCE IN THE GENERATOR USING H_E VS. H_T ?

1) EXTENDED HAMILTONIAN

→ NUMBER OF MULTIPLIERS $v^a(t)$ ($a=1, \dots, N_1$) IS EQUAL TO THE NUMBER OF ALL FIRST CLASS CONSTRAINTS

$$G(\epsilon) = \sum_{a=1}^{N_1} \epsilon^a \phi_a, \quad \delta_\epsilon F = \sum_{a=1}^{N_1} \epsilon^a(t) \{F, \phi_a\}$$

→ THE MULTIPLIERS TRANSFORM TOO $\delta_\epsilon v^a \neq 0$

$\delta_\epsilon I_H = 0$ (OFF-SHELL SYMMETRY OF THE THEORY)

$$H_E = H' + v^a \phi_a$$

$$(*) \quad \begin{cases} \{\phi_a, \phi_b\} = f_{ab}^c \phi_c & \dots \phi_a \text{ ARE I CLASS} \\ \{\phi_a, H'\} = f_a^b \phi_b & \dots H' \text{ IS I CLASS} \end{cases}$$

$\Rightarrow$ INVARIANCE OF I_H GIVES

$$\delta_\epsilon v^a = \dot{\epsilon}^a + \epsilon^b f_b^a - v^b \epsilon^c f_{bc}^a$$

$\delta_\epsilon v^a$ ALWAYS CONTAINS $\dot{\epsilon}^a$

COMMENT: SECOND CLASS CONSTRAINTS CANNOT APPEAR ON THE R.H.S. OF (*). IMAGINE THAT THEY APPEAR.

FOR EXAMPLE:

$$\{\phi_a, \phi_b\} = f_{ab}^c \phi_c + \underbrace{h_{ab}^r}_{\text{II class}} \theta_r$$

TAKE $\{\dots, \theta_p\}$

$$\underbrace{\{\{\phi_a, \phi_b\}, \theta_p\}}_{\approx 0 \text{ } (\{\phi, \phi\} \text{ I class})} \approx f_{ab}{}^c \underbrace{\{\phi_c, \theta_p\}}_{\approx 0 \text{ } (\phi \text{ I class})} + h_{ab}{}^r \underbrace{\{\theta_r, \theta_p\}}_{\Delta_{rp} \neq 0 \text{ INVERTIBLE } (\theta \text{ II class})}$$

$$\Rightarrow h_{ab}{}^r \Delta_{rp} \approx 0 \quad \nearrow (\Delta^{-1})_{ps}$$

$$h_{ab}{}^s \approx 0$$

$$\underbrace{h_{ab}{}^s \theta_r = 0}_{\text{SQUARE OF CONSTRAINTS}} \Rightarrow h_{ab}{}^s = 0 \quad \text{TO AVOID CONTRADICTION}$$

→ THE PROOF IS SIMILAR FOR H^I AS WELL

2) TOTAL HAMILTONIAN

→ EQUIVALENT TO A LAGRANGIAN SYSTEM

→ NUMBER OF MULTIPLIERS $U^a(t)$ ($a=1, \dots, N_1'$) IS EQUAL TO THE NUMBER OF PRIMARY FIRST CLASS CONSTRAINTS

$$N_1' < N_1 = N_1' + N_1'' \quad (\text{PRIMARY} + \text{SECONDARY})$$

$$G(\varepsilon) = \sum_{a=1}^{N_1'} \varepsilon^a \phi_a \quad ? \quad \delta_\varepsilon F = \sum_{a=1}^{N_1'} \varepsilon^a(t) \{F, \phi_a\} \quad ?$$

→ THE FORM OF THE GENERATOR IS NOT OBVIOUS ANY LONGER

→ DIRAC CONJECTURE IS NOT CLEAR

* LET US COMPUTE $G(\varepsilon)$ USING ONLY HAMILTON'S EQS. AND ALL CONSTRAINTS $\psi_\delta \approx 0$

$$\left. \begin{aligned} \Psi_S &\approx 0 \\ \dot{q}^i &= \frac{\partial H^1}{\partial p_i} + v^a \frac{\partial \phi_a}{\partial p_i} \\ -\dot{p}^i &= \frac{\partial H^1}{\partial q^i} + v^a \frac{\partial \phi_a}{\partial q^i} \end{aligned} \right\} \text{EQS. THAT DESCRIBE DYNAMICS}$$

- NOW TAKE THEIR LOCAL TRANSFORMATIONS $\delta_\epsilon(\dots)$

$$\left. \begin{aligned} \delta_\epsilon \Psi_S &= \frac{\partial \Psi_S}{\partial q^i} \delta_\epsilon q^i + \frac{\partial \Psi_S}{\partial p_i} \delta_\epsilon p_i \approx 0 \\ \delta_\epsilon \dot{q}^i &= \dots \text{ (SIMILAR BUT LONGER) } \\ -\delta_\epsilon \dot{p}^i &= \dots \end{aligned} \right\} \text{EQUATIONS } \S$$

- FIRST WE ASSUME THAT A GENERATOR HAS THE FORM $\epsilon(t) G^a(q, p)$ AND WE DEFINE:

$$\begin{aligned} \delta_\epsilon q^i &= \epsilon \{q^i, G\} = \epsilon \frac{\partial G}{\partial p_i} \\ \delta_\epsilon p_i &= \epsilon \{p_i, G\} = -\epsilon \frac{\partial G}{\partial q^i} \end{aligned}$$

↑
ASSUMED,
WITHOUT
WRITING
 $a=1, \dots, N_1$

- WE ALSO USE TAKE TIME DERIVATIVE OF THE LAST EQS:

$$\begin{aligned} (\delta_\epsilon q^i)' &\approx \dot{\epsilon} \frac{\partial G}{\partial p_i} + \epsilon \left\{ \frac{\partial G}{\partial p_i}, H_T \right\} \\ -(\delta_\epsilon p_i)' &\approx \dot{\epsilon} \frac{\partial G}{\partial q^i} + \epsilon \left\{ \frac{\partial G}{\partial q^i}, H_T \right\} \end{aligned}$$

- WE REPLACE ALL OF THAT IN $\S$ AND OBTAIN THREE EQUATIONS:

$$0 \approx \frac{\partial \Psi_S}{\partial q^i} \epsilon \frac{\partial G}{\partial p_i} + \frac{\partial \Psi_S}{\partial p_i} (-\epsilon \frac{\partial G}{\partial q^i}) = \epsilon \{ \Psi_S, G \} \rightarrow \boxed{\epsilon \{ \Psi_S, G \} \approx 0}$$

$$\boxed{\begin{aligned} \frac{\partial}{\partial p_i} [\dot{\epsilon} G + \epsilon \{G, H_T\} - \delta_\epsilon v^a \phi_a] &\approx 0 \\ \frac{\partial}{\partial q^i} [\dot{\epsilon} G + \epsilon \{G, H_T\} - \delta_\epsilon v^a \phi_a] &\approx 0 \end{aligned}}$$

$\equiv A$

- WE GOT $\frac{\partial A}{\partial p_i} \approx 0, \frac{\partial A}{\partial q^i} \approx 0 \Rightarrow \{A, F\} \approx 0 \quad \forall F$

$$A = 0$$

ZERO, CONSTANT OR
 $\varphi_s^n, \varphi_{s_1}, \dots, \varphi_{s_n} \quad (n \geq 2)$

$$\dot{\varepsilon} G + \varepsilon \{G, H_T\} - \delta_\varepsilon v^a \phi_a = 0$$

$\forall \varepsilon^a$
 $\Rightarrow$

$$\begin{aligned} G &= \alpha^a \phi_a + 0 \\ \{G, H_T\} &= \beta^a \phi_a + 0 \end{aligned}$$

$$\begin{aligned} \alpha^a &= \alpha^a(q, p) \\ \beta^a &= \beta^a(q, p) \end{aligned}$$

$C_{PFC} \stackrel{\text{def}}{=} \text{PRIMARY FIRST class constr.} + 0$

$\Rightarrow$

$$\begin{aligned} G &= C_{PFC} \\ \{G, H_T\} &= C_{PFC} \end{aligned}$$

CASTELLANI ALGORITHM

WHEN THE GENERATOR HAS THE FORM $\varepsilon^a G_a \quad (a=1, \dots, N_1')$
 (ONLY 1 GENERATION OF PFC constr.)

- THESE CONDITIONS SERVE TO COMPUTE α^a, β^a
- IF THIS IS NOT POSSIBLE (2 OR MORE GENERATIONS), THE GENERATOR DOESN'T HAVE THE FORM εG

* IF THERE ARE 2 GENERATIONS OF CONSTRAINTS:

$$G = \varepsilon(t) G_0 + \dot{\varepsilon}(t) G_1$$

$$\begin{aligned} \delta_\varepsilon q^i &= \{q^i, G\} \\ \delta_\varepsilon p_i &= \{p_i, G\} \end{aligned}$$

- REPEATING THE COMPUTATION, WE GET EQS. IN G_0, G_1 :

$$\begin{aligned} G_1 &= C_{PFC} \\ G_0 + \{G_1, H_T\} &= C_{PFC} \\ \{G_0, H_T\} &= C_{PFC} \end{aligned}$$

* IF THERE ARE $K+1$ GENERATIONS OF CONSTRAINTS:

$$G = \varepsilon^{(K)}(t) G_K + \dots + \varepsilon(t) G_0 \quad \text{WITH} \quad \delta_\varepsilon q^i = \{q^i, G\}$$

$$\varepsilon^{(K)} = \frac{d^K \varepsilon}{dt^K} \quad \delta_\varepsilon p_i = \{p_i, G\}$$

• THE EQUATIONS TO DETERMINE $G_0, \dots, G_K$ ARE:

$$\begin{aligned} G_K &= \text{CPFC} \\ G_{K-1} + \{G_K, H_T\} &= \text{CPFC} \\ &\vdots \\ \{G_0, H_T\} &= \text{CPFC} \end{aligned}$$

$G_K \dots$ PRIMARY FIRST CLASS

$G_{K-1}, \dots, G_0 \dots$ FIRST CLASS
(SECONDARY)

EXAMPLE

$$L = \frac{1}{2} \dot{q}_1^2 + \dot{q}_1 q_2 + \frac{1}{2} (q_1 - q_2)^2$$

$$\phi = p_2 \approx 0 \quad \dots \text{PRIMARY I CLASS}$$

$$\chi = p_1 - q_1 \approx 0 \quad \dots \text{SECONDARY I CLASS}$$

• THERE ARE 2 GENERATORS OF I CLASS CONSTRAINTS,
SO THE GENERATOR HAS THE FORM

$$G = \varepsilon(t) G_0 + \dot{\varepsilon}(t) G_1$$

• APPLYING THE CASTELLANI METHOD

$$G_1 = \text{CPFC} \quad \longrightarrow \quad G_1 = \phi \quad (\text{IT CAN BE } \gamma \phi, \text{ BUT WE CAN CHOOSE } \gamma = 1)$$

$$G_0 + \{G_1, H_T\} = \text{CPFC} \quad \longrightarrow \quad G_0 + \chi = \alpha \phi \quad \longrightarrow \quad G_0 = \alpha \phi - \chi$$

$$\{G_0, H_T\} = \text{CPFC} \quad \longrightarrow \quad \{\alpha \phi - \chi, H_T\} = \alpha \chi - (-\chi) = (\alpha + 1)\chi = 0$$

$$\alpha = -1$$

$$G_0 = -(\phi + \chi)$$

$$G = -\varepsilon(\phi + \chi) + \dot{\varepsilon} \phi \quad \text{GENERATOR}$$

• LOCAL TRANSF. $\delta_{\epsilon} q_1 = -\epsilon, \quad \delta_{\epsilon} p_1 = -\epsilon$
 $\delta_{\epsilon} q_2 = \dot{\epsilon} - \epsilon, \quad \delta_{\epsilon} p_2 = 0$

(THIS WAS OBTAINED FROM THE INVARIANCE OF HAMILTON'S EQUATIONS)

COMMENT:

→ GENERATOR CONTAINS ϵ^a ONLY FOR PRIMARY FC CONSTRAINTS AND THE PARAMETERS CORRESPONDING TO SECONDARY, TERTIARY, ... CONSTR. ARE $\dot{\epsilon}^a, \ddot{\epsilon}^a, \dots$

→ THIS CAN BE SEEN FROM THE EXTENDED HAMILTONIAN WHERE

$$\delta_{\epsilon} v^a = \dot{\epsilon}^a + \dots \Leftrightarrow \begin{cases} \delta_{\epsilon} v_1^a = \dot{\epsilon}_1^a + \dots & \text{PRIMARY} \\ \delta_{\epsilon} v_2^a = \dot{\epsilon}_2^a + \dots & \text{SECONDARY} \end{cases}$$

→ PASSING FROM H_E TO H_T CORRESPONDS TO

$$v_2^a = 0 \Rightarrow \delta_{\epsilon} v_2^a = 0 \Rightarrow \epsilon_2^a = \epsilon_2^a(\dot{\epsilon}_1, \epsilon_1)$$

RESOLUTION OF DIRAC CONJECTURE

$G_K \rightarrow$ PRIMARY, STARTING POINT

$G_{K-1}, \dots, G_0 \rightarrow$ SECONDARY DOES IT INCLUDES ALL SECONDARY FIRST CLASS CONSTRAINTS?

• NO, IF SOMETHING INTERRUPTS CASTELLANI'S ALGORITHM BEFORE ARRIVING TO THE END

• THE ALGORITHM IS INTERRUPTED IF $\{G_{K_0}, H_T\} = \mathcal{X}^n \equiv 0$ FOR SOME $0 < K_0 < K$ ($n \geq 2$)

⇒ THE CONSTRAINT $\mathcal{X}$ DOES NOT APPEAR IN THE GENERATOR

ALL FIRST CLASS CONSTRAINTS ENTER G , EXCEPT THE ONES THAT APPEAR IN A CONSISTENCY CONDITIONS AS $\mathcal{X}^n$ ($n \geq 2$) AND THE ONES FOLLOWING FROM $\dot{\mathcal{X}} \approx 0$

EXAMPLE OF ELECTROMAGNETISM

- WE USE COVARIANT FORMULATION OF ELECTROMAGNETISM

$$x^\mu = (t, x^i) = (t, \vec{r}) \quad \mu=0,1,2,3 \quad c=1 \text{ (NATURAL UNITS)}$$

$$A_\mu(x) = (A_0, A_i) = (\phi_e, \vec{A}) \quad \text{ELECTROMAGNETIC FIELD}$$

$\underbrace{\phi_e}_{\text{ELECTRIC POTENTIAL}} \quad \underbrace{\vec{A}}_{\text{VECTOR POTENTIAL}}$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \text{MAXWELL TENSOR, FIELD-STRENGTH}$$

$$F_{\mu\nu} = -F_{\nu\mu} \quad \dots \text{ANTISYMMETRIC}$$

$$\partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0 \quad \dots \text{BIANCHI IDENTITY}$$

$$\text{ELECTRIC FIELD: } \vec{E} = [E_i] \quad F_{0i} = -E_i$$

$$\text{MAGNETIC FIELD: } \vec{B} = [B_i] \quad F_{ij} = \epsilon_{ijk} B_k$$

$$J^\mu = (j_e, \vec{j}) \quad \text{EXTERNAL CURRENT}$$

$$\text{CONSERVED: } \partial_\mu J^\mu = 0$$

$\underbrace{j_e}_{\text{ELECTRIC CHARGE DENSITY}} \quad \underbrace{\vec{j}}_{\text{ELECTRIC CURRENT DENSITY}}$

- MAXWELL ACTION

$$I[A] = \int d^4x \mathcal{L}(A, \partial A) = \int d^4x \left(-\frac{1}{4} F^2 + A_\mu J^\mu \right)$$

$$F^2 = F^{\mu\nu} F_{\mu\nu}, \quad F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta}$$

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ & & & +1 \end{pmatrix} \quad \text{MINKOWSKI METRIC}$$

* MAXWELL EQUATIONS WITH SOURCES = EULER-LAGRANGE EQS.

* MAXWELL EQUATIONS WITHOUT SOURCES = BIANCHI IDENTITY

- FOR HAMILTONIAN ANALYSIS WE WILL TAKE $J^\mu = 0$

* VELOCITY

$$\dot{A}_\mu = \frac{\partial A_\mu}{\partial x^0}$$

* CANONICAL MOMENTUM

$$\begin{aligned} \pi^\mu &= \frac{\partial \mathcal{L}}{\partial \dot{A}_\mu} = -\frac{1}{4} \frac{\partial}{\partial \dot{A}_\mu} (F^{\alpha\beta} F_{\alpha\beta}) = -\frac{1}{2} F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial \dot{A}_\mu} \\ &= -F^{\alpha\beta} \underbrace{\frac{\partial (\partial_\alpha A_\beta)}{\partial (\partial_0 A_\mu)}}_{\delta_\alpha^0 \delta_\beta^\mu} = -F^{0\mu} \Rightarrow \boxed{\pi^\mu = -F^{0\mu}} \end{aligned}$$

$\mu=0$

$$\boxed{\phi = \pi^0 \approx 0} \quad \text{PRIMARY CONSTRAINTS}$$

$\mu=i=1,2,3$

$$\pi^i = -F^{0i} = F_{0i} = \dot{A}_i - \partial_i A_0$$

$$\boxed{\dot{A}_i = \pi_i + \partial_i A_0} \quad \text{SOLVED VELOCITIES}$$

• NOTE THAT $\pi_i = -E_i$ IS THE ELECTRIC FIELD

* CANONICAL HAMILTONIAN

$$\begin{aligned} \mathcal{H}_c &= \pi^\mu \dot{A}_\mu - \mathcal{L} = \pi^\mu \dot{A}_\mu + \frac{1}{4} F^{\mu\nu} F_{\mu\nu} \\ &\approx \pi^i \dot{A}_i + \frac{1}{2} \underbrace{F^{0i}}_{\pi_i + \partial_i A_0} \underbrace{F_{0i}}_{-\pi_i} + \frac{1}{4} F^{ij} F_{ij} \end{aligned}$$

$$\begin{aligned} &= \underbrace{\pi^i \pi_i}_{\partial_i (\pi^i A_0) - A_0 \partial_i \pi^i} + \underbrace{\pi^i \partial_i A_0}_{\partial_i (\pi^i A_0) - A_0 \partial_i \pi^i} - \frac{1}{2} \pi^i \pi_i + \frac{1}{4} F^{ij} F_{ij} \end{aligned}$$

$$\boxed{\mathcal{H}_c = \frac{1}{2} \pi^i \pi_i + \frac{1}{2} F^{ij} F_{ij} - A_0 \partial_i \pi^i + \partial_i \Lambda^i}$$

TOTAL DIVERGENCE, BECOMES A BOUNDARY TERM IN THE HAMILTONIAN $H_c = \int d^3x \mathcal{H}_c$

$$\mathcal{H}_c = \underbrace{\frac{1}{2} \vec{E}^2 + \frac{1}{2} \vec{B}^2}_{\text{ENERGY DENSITY}} - \phi_e \nabla \cdot \vec{E}$$



$$\int_{\Sigma} d^3x \partial_i \Lambda^i = \oint_{\partial \Sigma} dS_i \Lambda^i$$

(STOKES THEOREM)

* AT FIRST, WE NEGLECT ALL BOUNDARY TERMS IN HAMILTONIAN EVEN THOUGH THEY ARE NOT NECESSARILY ZERO, BUT WE WILL TREAT THEM AT THE END

* TOTAL HAMILTONIAN

$$\mathcal{H}_T = \mathcal{H}_C + m \phi$$

$$H_T = H_C + \int d^3x m(x) \phi(x)$$

$m(x) = m(x^\mu) = m(t, \vec{r}) \dots$ ARBITRARY MULTIPLIER

- THE COMPONENT A_0 ENTERS LINEARLY IN $\mathcal{H}_T$, SO IT IS ALSO AN ARBITRARY MULTIPLIER

* CANONICAL POISSON BRACKETS (PBs)

$$\{A_\mu(t, \vec{r}), \pi^\nu(t, \vec{r}')\} = \delta_\mu^\nu \delta^{(3)}(\vec{r} - \vec{r}')$$

ALWAYS $t = t'$, SO WE CAN WRITE $x^\mu = (t, \vec{r})$, $x'^\mu = (t, \vec{r}')$
 $\delta^{(3)}(x - x')$

$$\{A_\mu(x), \pi^\nu(x')\} = \delta_\mu^\nu \delta^{(3)}(x - x')$$

$$\{A_\mu(x), A_\nu(x')\} = 0$$

$$\{\pi^\mu(x), \pi^\nu(x')\} = 0$$

SHORT NOTATION:

$$\{A_\mu, \pi^{\mu'}\} = \delta_\mu^{\mu'} \delta^{(3)}$$

* CONSISTENCY CONDITIONS

$$\begin{aligned}\dot{\Phi} = \{\Phi, H_T\} &= \int d^3x' \{ \Phi, \mathcal{H}'_0 + \lambda' \Phi' \} \\ &= \int d^3x' \underbrace{\{ \pi^0, -A_0' \partial_i' \pi'^i \}}_{\partial_i' \pi'^i \delta^{(3)}} = \partial_i \pi^i\end{aligned}$$

$$\mathcal{X} = \partial_i \pi^i \approx 0$$

SECONDARY CONSTRAINT

$$\nabla \cdot \vec{E} = \rho_e = 0 \text{ (GAUSS LAW)}$$

$$\dot{\mathcal{X}} \approx \{\mathcal{X}, H_T\} = \int d^3x' \left\{ \partial_i \pi^i, \frac{1}{4} F_{jk}' F'^{jk} \right\}$$

$$\begin{aligned}\frac{1}{2} F'^{jk} \{ \partial_i \pi^i, F_{jk}' \} &= F'^{jk} \{ \partial_i \pi^i, \partial_j' A_k' \} \\ &= F'^{jk} \partial_i \partial_j' \{ \pi^i, A_k' \} \\ &= - F'^{jk} \partial_i \partial_j' \delta^{(3)} \delta_k^i\end{aligned}$$

$$= - \partial_i \int d^3x' F'^{ji} \partial_j' \delta^{(3)} = \underbrace{\partial_i \partial_j}_{\text{SYMMETRIC}} \underbrace{F^{ji}}_{\text{ANTISYMMETRIC}} \equiv 0 \Rightarrow \boxed{\dot{\mathcal{X}} \approx 0}$$

NOTHING NEW

* BOTH CONSTRAINTS ARE FIRST CLASS $\{\Phi, \mathcal{X}'\} = 0$

$\Rightarrow$ THEY GENERATE LOCAL SYMMETRIES

* HAMILTON'S EQUATIONS (SHOW THEM)

$$\dot{A}_0 \approx \lambda \text{ (NOT PHYSICAL)}$$

$$\dot{\pi}^0 \approx 0 \text{ (CONSTRAINT)}$$

$$\dot{A}_i \approx \pi_i + \partial_i A_0 \text{ (MOMENTUM DEF.)}$$

$$\dot{\pi}^i = \partial_j F^{ji} \text{ (MAXWELL EQ.)}$$

* GAUGE GENERATOR - APPLICATION OF CASTELLANI'S METHOD

2 GENERATIONS OF FIRST CLASS CONSTRAINTS: $\mathcal{G} = \dot{\epsilon}(x) G_1 + \epsilon(x) G_0$

IN FIELD THEORY - SMEARED GENERATOR; $G[\epsilon] = \int d^3x \mathcal{G}$

CASTELLANI'S ALGORITHM:

$$G_1 = \text{CPFC} \rightarrow G_1 = \Phi$$

$$G_0 + \{G_1, H_T\} = \text{CPFC} \rightarrow G_0 + \mathcal{X} = \alpha \Phi \rightarrow G_0 = \alpha \Phi - \mathcal{X}$$

$$\{G_0, H_T\} = \text{CPFC} \rightarrow \alpha \mathcal{X} = \text{CPFC} \rightarrow \alpha = 0, G_0 = -\mathcal{X}$$

$$G[\varepsilon] = \int d^3x (\dot{\varepsilon} \pi^0 - \varepsilon \partial_i \pi^i)$$

* GAUGE TRANSFORMATIONS $\delta_\varepsilon F = \{F, G[\varepsilon]\}$

$$\delta_\varepsilon A_0 = \int d^3x' \{A_0, \dot{\varepsilon}' \pi'^0\} = \int d^3x' \dot{\varepsilon}' \underbrace{\{A_0, \pi'^0\}}_{\delta^{(3)}} = \dot{\varepsilon}$$

$$\delta_\varepsilon A_i = \int d^3x' \{A_i, -\varepsilon' \partial'_j \pi'^j\} = - \int d^3x' \varepsilon' \partial'_j \delta^{(3)} \delta_i^j = \partial_i \varepsilon$$

$\Rightarrow$

$$\delta_\varepsilon A_\mu = \partial_\mu \varepsilon$$

$$\delta_\varepsilon \pi^\mu = 0$$

U(1) LOCAL SYMMETRY

* NUMBER OF DEGREES OF FREEDOM

$$A_\mu (\mu=0,1,2,3)$$

$N=4$ FIELDS

$$\phi, \chi$$

$N_1=2$ FIRST CLASS CONSTRAINTS

$N_2=0$ SECOND CLASS CONSTRAINTS

$$N_{\text{phys}} = N - N_1 - \frac{1}{2} N_2 = 2 \Rightarrow 2 \text{ PHYSICAL DEGREES OF FREEDOM}$$

HOW TO IDENTIFY THEM?

$\rightarrow$ TO IDENTIFY PHYSICAL DEGREES OF FREEDOM, WE HAVE TO GAUGE FIX THE SYMMETRY AND WHAT REMAINS IN Γ_{phys} ARE PHYSICAL FIELDS

* GAUGE CONDITIONS

$\rightarrow$ NOT UNIQUE

ONE POSSIBILITY:

$$\phi = \pi^0 \approx 0$$

$$\chi = \partial_i \pi^i \approx 0$$



TWO 1 CLASS CONSTRAINTS

$$\Omega_1 = A_0 \approx 0$$

$$\Omega_2 = \partial_i A^i \approx 0$$



TWO GAUGE CONDITIONS

RADIATIVE GAUGE

• CONSISTENCY CONDITIONS OF GAUGE CONDITIONS

$$\dot{\Omega}_1 = \dot{A}_0 \approx \mu = 0 \quad (\text{SOLVES } \mu)$$

$$\dot{\Omega}_2 = \partial_i \dot{A}^i = \partial_i (\pi^i + \partial^i A_0) = \chi + \nabla^2 \Omega_1 \approx 0 \quad (\text{SATISFIED})$$

• WE HAVE TO CHECK IF $\varphi_s = (\phi, \Omega_1, \chi, \Omega_2)$ IS INVERTIBLE

$$\{\phi, \Omega_1'\} = -\delta^{(3)}$$

$$\{\chi, \Omega_2'\} = \partial_i \partial^i \delta^{(3)} = \Delta \delta^{(3)}$$

THE ONLY NON-VANISHING PBs

$$\{\varphi_s, \varphi_r'\} = \Delta_{sr}(x, x') = \left(\begin{array}{cc|cc} 0 & -\delta^{(3)} & 0 & 0 \\ \delta^{(3)} & 0 & 0 & \Delta \delta^{(3)} \\ \hline 0 & 0 & 0 & -\Delta \delta^{(3)} \end{array} \right) \quad \text{INVERTIBLE}$$

• WE USE $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \mathbb{I}_2$ TO FIND THE INVERSE

$$\Delta_{sr}^{-1}(x, x') = \left(\begin{array}{cc|cc} 0 & \delta^{(3)} & 0 & 0 \\ -\delta^{(3)} & 0 & 0 & -\mathcal{D} \\ \hline 0 & 0 & \mathcal{D} & 0 \end{array} \right) \quad \mathcal{D}(x, x') = ?$$

$$\int d^3 x'' \Delta_{sr}(x, x'') \Delta_{rp}^{-1}(x'', x') = \delta_{sp} \delta^{(3)}(x - x')$$

DEFINITION OF INVERSE MATRIX

$$\int d^3 x'' \delta^{(3)}(x - x'') \delta^{(3)}(x'' - x') = \delta^{(3)}(x - x') \quad \checkmark$$

(THE FIRST BLOCK IS CORRECT)

⊕ ... INVERSE OF $\Delta = \partial_i \partial^i$

$$\int d^3 x'' \partial_i \partial^i \delta^{(3)}(x - x'') \mathcal{D}(x'' - x') = \delta^{(3)}(x - x')$$

$$\partial_i \partial^i \underbrace{\int d^3 x'' \delta^{(3)}(x - x'') \mathcal{D}(x'' - x')}_{\mathcal{D}(x - x')} = \delta^{(3)}(x - x')$$

$$\Delta \Phi(x) = \delta^{(3)}(x)$$

$\Rightarrow \Phi$ IS A SOLUTION OF THIS EQ.

$$\Phi(x, x') = \Phi(\vec{r} - \vec{r}')$$

$$\Phi(\vec{r}) = -\frac{1}{4\pi|\vec{r}|}$$

* DIRAC BRACKETS

$$\{A, B'\}^* = \{A, B'\} - \int d^3x'' d^3x''' \{A, \psi_s''\} \Delta_{sr}^{-1}(x'', x''') \{\psi_r''', B'\}$$

• THE ONLY NON VANISHING BRACKETS SHOULD BE THE ONES BETWEEN TRANSVERSAL COMPONENTS OF A_i AND π^i

$$\{A_i, \pi^{j'}\}^* = \underbrace{\{A_i, \pi^{j'}\}}_{\delta_i^j \delta^{(3)}(x-x')} - \int d^3x'' \int d^3x''' \underbrace{\{A_i, \partial_\kappa'' \pi^{\kappa'}\}}_{\partial_\kappa'' \delta^{(3)}(x-x'') \delta_i^\kappa} (-1) \Phi(x''-x''') \underbrace{\{\partial_\ell''' A^{\ell'}, \pi^{j'}\}}_{\partial_\ell''' \delta^{(3)}(x'''-x') \delta^{\ell j'}}$$

$$\begin{aligned} &= \delta_i^j \delta^{(3)}(x-x') + \int d^3x'' \int d^3x''' \underbrace{\partial_i'' \delta^{(3)}(x-x'') \Phi(x''-x''') \partial^{\ell'} \delta^{(3)}(x'''-x')}_{\partial_i'' \delta^{(3)}(x-x'') \int d^3x''' \Phi(x''-x''') \partial^{\ell'} \delta^{(3)}(x'''-x')} \\ &= \partial_i'' \delta^{(3)}(x-x'') (-1) \partial^{\ell'} \delta^{(3)}(x-x') \\ &= \delta_i^j \delta^{(3)}(x-x') - \int d^3x'' \partial_i'' \delta^{(3)}(x-x'') \partial^{\ell'} \delta^{(3)}(x-x') \\ &\quad - \partial_i \partial^{\ell'} \delta^{(3)}(x-x') = \partial_i \partial^{\ell'} \delta^{(3)}(x-x') \end{aligned}$$

$$\Rightarrow \boxed{\{A_i, \pi^{j'}\}^* = \delta_i^j \delta^{(3)}(x-x') - \partial_i \partial^{\ell'} \delta^{(3)}(x-x')}$$

2

IT CAN BE SEEN THAT, INDEED, ONLY TRANSVERSAL COMPONENTS ARE NON-VANISHING

TRANSVERSAL COMPONENTS:

$$A_i = A_{Ti} \quad \text{IF IT SATISFIES} \quad \partial_i A^i = 0$$

$$\pi_i = \pi_{Ti} \quad \text{IF IT SATISFIES} \quad \partial_i \pi^i = 0$$

WE KNOW THAT $\{F, \Omega_1\}^* = 0, \{F, \Omega_2\}^* = 0$
(by DEFINITION OF DIRAC BRACKETS)

$$\Rightarrow \boxed{\{A_{Ti}, \pi_T^i\}^* = \delta_i^j \delta^{(3)}(x-x') - \partial_i \partial^j \mathcal{D}(x-x')} \in \Gamma_{\text{PHYS}}$$

$$\Gamma_{\text{PHYS}} : (A_{Ti}, \pi_T^i) \Rightarrow N_{\text{PHYS}} = 2$$

2 components

CONSERVED CHARGES IN HAMILTONIAN FORMALISM

REGGE-TEITELBOIM (RT) METHOD

* T. REGGE, C. TEITELBOIM "ROLE OF SURFACE INTEGRALS IN THE HAMILTONIAN FORMULATION OF GENERAL RELATIVITY"

ANNALS PHYS. 88 (1974) 286

- UNTIL NOW, WE HAVE BEEN NEGLECTING BOUNDARY TERMS IN A FIELD THEORY, BUT NOW WE HAVE TO TREAT THEM
- IN HAMILTONIAN FORMALISM, FUNCTIONAL DERIVATIVES ARE USED TO DEFINE POISSON BRACKETS

$$F[\Psi] = \int d^3x \mathcal{F}(\Psi, \partial_i \Psi) \quad \text{FUNCTIONAL ON } \Gamma$$

$\Psi^A \dots$ ALL CANONICAL FIELDS

$\Psi^A = (A_\mu, \pi^\nu)$ IN ELECTROMAGNETISM

- DIFFERENTIABLE FUNCTIONAL VARIES LIKE THIS:

$$\delta F = \int d^3x \underbrace{\frac{\delta F}{\delta \Psi^A(x)}}_{\text{FUNCTIONAL DERIVATIVE OF } F} \delta \Psi^A(x)$$

$\Rightarrow$ A FUNCTIONAL IS DIFFERENTIABLE IF IT HAS A FUNCTIONAL DERIVATIVE WELL-DEFINED, THAT IS, WITHOUT THE PRESENCE OF BOUNDARY TERM, AFTER USING BOUNDARY CONDITIONS OF THE FIELDS

- IN A GENERIC CASE, FUNCTIONALS ARE NOT DIFFERENTIABLE DUE TO THE PRESENCE OF SPATIAL DERIVATIVES $\partial_i \Psi^A(x)$

$$\delta F[\Psi] = \int d^3x \underbrace{\left[\frac{\partial \mathcal{F}}{\partial \Psi^A} - \partial_i \left(\frac{\partial \mathcal{F}}{\partial \partial_i \Psi^A} \right) \right]}_{\frac{\delta F}{\delta \Psi^A} \text{ FUNCTIONAL DERIVATIVE}} \delta \Psi^A + \underbrace{\oint d\sigma_i \frac{\partial \mathcal{F}}{\partial \partial_i \Psi^A} \delta \Psi^A}_{\text{BOUNDARY TERM}}$$

- IF THE BOUNDARY TERM DOES NOT VANISH, THIS IS A PROBLEM BECAUSE POISSON BRACKET ARE NOT WELL-DEFINED

ELECTROMAGNETISM - DIFFERENTIABILITY OF G

$$G[\epsilon] = \int d^3x (\dot{\epsilon} \pi^0 - \epsilon \partial_i \pi^i)$$

$$\delta G[\epsilon] = \int d^3x (\dot{\epsilon} \delta \pi^0 - \underbrace{\epsilon \partial_i \delta \pi^i}_{\partial_i (\epsilon \delta \pi^i) - \partial_i \epsilon \delta \pi^i})$$

$$= \int d^3x (\underbrace{\dot{\epsilon} \delta \pi^0 + \partial_i \epsilon \delta \pi^i}_{\partial_\mu \epsilon \delta \pi^\mu = \frac{\delta G[\epsilon]}{\delta \pi^\mu} \delta \pi^\mu}) - \underbrace{\int d^3x \partial_i (\epsilon \delta \pi^i)}_{= \delta Q[\epsilon]}$$

$$\partial_\mu \epsilon \delta \pi^\mu = \frac{\delta G[\epsilon]}{\delta \pi^\mu} \delta \pi^\mu$$

DIFFERENTIABLE PART

$$= \delta Q[\epsilon]$$

NON-DIFFERENTIABLE PART

$$\delta G[\epsilon] = \text{DIFF. PART} - \delta Q[\epsilon]$$

$$\delta G_Q[\epsilon] = \delta (G[\epsilon] + Q[\epsilon]) = \text{DIFFERENTIABLE}$$

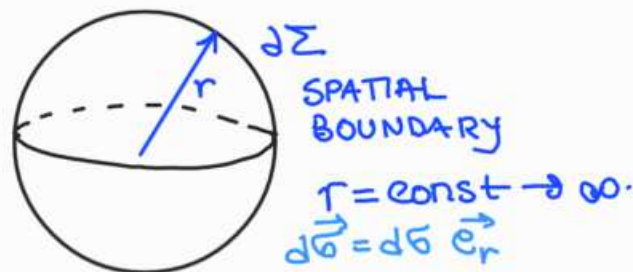
$$G_Q[\epsilon] = G[\epsilon] + Q[\epsilon] \quad \text{IMPROVED GENERATOR} =$$

$$= \text{CONSTRAINTS} + \text{BOUNDARY TERM}$$

$$\bullet \text{ WE IDENTIFY } \frac{\delta G_Q[\epsilon]}{\delta A_\mu} = 0, \quad \frac{\delta G_Q[\epsilon]}{\delta \pi^\mu} = \partial_\mu \epsilon$$

$$\delta Q[\epsilon] = \int_{\Sigma} d^3x \partial_i (\epsilon \delta \pi^i) = \oint_{\partial \Sigma} d\sigma_i \epsilon \delta \pi^i$$

STOKES THEOREM



- EL METHOD DETERMINES δQ , BUT IT IS NOT ALWAYS ENSURED THAT THE EXPRESSION CAN BE INTEGRATED AND Q OBTAINED
- IN THE MAXWELL THEORY, THIS IS POSSIBLE; ASSUMING THAT $\epsilon(x)$ DOES NOT DEPEND ON A_μ AND π^μ , IT SATISFIES $\delta \epsilon = 0$ SUCH THAT

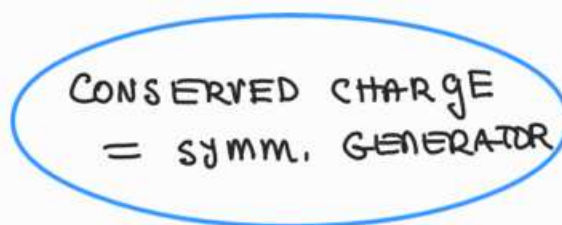
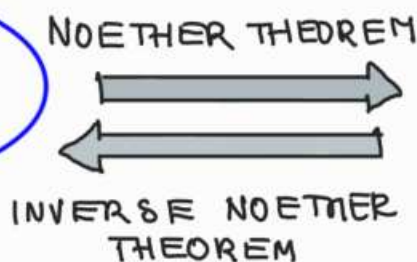
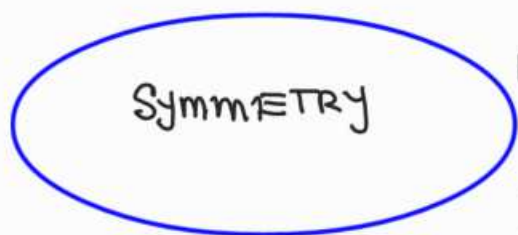
$$Q[\epsilon] = \oint d\sigma_i \epsilon \pi^i$$

- PHYSICAL DEGREES OF FREEDOM LIVE IN Γ_{phys} , WHERE WE USE $\{, \}^*$ AND WHERE $G[\epsilon]|_{\Gamma_{\text{phys}}} = 0$



$$G_Q[\epsilon] = Q[\epsilon] \text{ EN } \Gamma_{\text{phys}}$$

Q IS A SYMMETRY GENERATOR ON Γ_{phys}



(EACH CONSERVED CHARGE GENERATES A CONTINUOUS SYMMETRY)

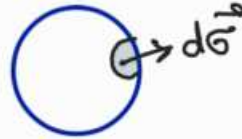


Q IS A CONSERVED CHARGE !

- CONSERVATION: $\dot{Q}[\epsilon] = \{Q[\epsilon], H_T\}^* = \{G_Q[\epsilon], H_T\}^* \stackrel{H_T \text{ is I class}}{=} 0$
- ELECTRIC CHARGE: GLOBAL SYMMETRY $\epsilon = 1$

ONE SOLUTION: $A_0 = \frac{1}{4\pi} \frac{q}{r}, \vec{A} = 0$ $\left(\begin{array}{l} \rho_e = q \delta(\vec{r}), \vec{J} = 0 \\ \text{EXTERNAL CURRENT} \\ \text{- PUNCTUAL CHARGE} \end{array} \right)$

$$\begin{aligned} \pi_i &= -F_{0i} = -\dot{A}_i + \partial_i A_0 \\ &= \frac{q}{4\pi} \partial_i \left(\frac{1}{r} \right) = \frac{q}{4\pi} \frac{x_i}{r^3} \end{aligned}$$



$$d\vec{\sigma} = \vec{e}_r r d\varphi r \sin\theta d\theta, \quad \vec{r} = (r, \theta, \varphi) \quad \text{SPHERICAL COORD.}$$

$$Q = \frac{q}{4\pi} \underbrace{\int_0^\pi \sin\theta d\theta}_{4\pi} \int_0^{2\pi} d\varphi = q \quad \Rightarrow \quad \boxed{Q = q}$$

CHARGE ALGEBRA

CONTINUOUS SYMMETRY GROUP $\Rightarrow$ ALGEBRA OF GENERATORS (INFINITESIMAL FORM)

- TRICK TO COMPUTE IT: $\delta_{\epsilon_2} F = \{F, G_Q[\epsilon_2]\}$

CHOOSE: $F = G_Q[\epsilon_1]$

$$\delta_{\epsilon_2} G_Q[\epsilon_1] = \{G_Q[\epsilon_1], G_Q[\epsilon_2]\}$$

IN ELECTROMAGNETISM: $\delta_{\epsilon_2} \pi^M = 0 \Rightarrow \delta_{\epsilon_2} G_Q[\epsilon_1] = 0$

$$\{G_Q[\epsilon_1], G_Q[\epsilon_2]\} = 0$$

ON Γ_{phys} : $\{Q[\epsilon_1], Q[\epsilon_2]\}^* = 0$

CHARGE ALGEBRA IS ABELIAN;
ASYMPTOTIC ALGEBRA
(BECAUSE Q ARE DEFINED ON THE
BOUNDARY $r \rightarrow \infty$)

GENERAL FORM OF A CHARGE ALGEBRA

$$\{Q[\epsilon_1], Q[\epsilon_2]\}^* = Q[\mathcal{E}] + C[\epsilon_1, \epsilon_2]$$

$$\mathcal{E} = \mathcal{E}(\epsilon_1, \epsilon_2)$$

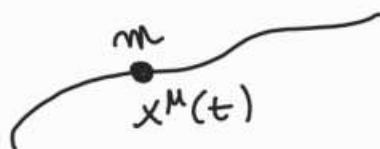
DEPENDS ON THE STRUCTURE CONSTANTS OF AN ALGEBRA

C = **CENTRAL CHARGE**, BREAKS SYMMETRY "SOFTLY", NEW GENERATOR THAT COMMUTES WITH ALL Q
(IT BEHAVES AS A NUMBER)

EXAMPLE OF RELATIVISTIC PARTICLE (WITH THE MASS $m \neq 0$)

$$\eta_{\mu\nu} = \begin{pmatrix} - & + & + & + \end{pmatrix}$$

$$\begin{aligned} ds^2 &= \eta_{\mu\nu} dx^\mu dx^\nu = \eta_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} dt^2 \\ &= \dot{x}_\mu \dot{x}^\mu dt^2 \equiv \dot{x}^2 dt^2 \end{aligned}$$



NOTE THAT $x^0 \neq t$; t IS ANY PARAMETER

• ACTION: $I[x] = -m \int ds = -m \int dt \sqrt{-\dot{x}^2} \equiv \int dt L$

$$L = -m \sqrt{-\dot{x}^2}$$

• CANONICAL MOMENTUM: $p_\mu = \frac{\partial L}{\partial \dot{x}^\mu} = \frac{m \dot{x}_\mu}{\sqrt{-\dot{x}^2}}$

• WE CANNOT INVERT $\dot{x}_\mu = \dot{x}_\mu(p)$ BECAUSE THERE ARE ONLY 3 INDEPENDENT CANONICAL MOMENTA

INDEED, HESSIAN IS: $\frac{\partial^2 L}{\partial \dot{x}^\mu \partial \dot{x}^\nu} = \frac{\partial}{\partial \dot{x}^\nu} \left(\frac{m \dot{x}_\mu}{\sqrt{-\dot{x}^2}} \right) = f(\dot{x}) (\dot{x}_\mu \dot{x}_\nu - \dot{x}^2 \eta_{\mu\nu})$

WHERE $f(\dot{x}) = m (-\dot{x}^2)^{-3/2}$

$\Rightarrow$ IT HAS EXACTLY ONE ZERO MODE $H_{\mu\nu} \dot{x}^\nu = 0$
 $\Rightarrow$ RANK OF $H_{\mu\nu} = \frac{\partial^2 L}{\partial \dot{x}^\mu \partial \dot{x}^\nu}$ IS THREE

- THERE IS ONE PRIMARY CONSTRAINT

$$p_\mu p^\mu = -m^2 \Rightarrow \boxed{\phi = p^2 + m^2 \approx 0} \text{ PRIMARY}$$

- CANONICAL HAMILTONIAN

$$H_c = p_\mu \dot{x}^\mu - L = \underbrace{\frac{m \dot{x}^2}{\sqrt{-\dot{x}^2}}}_{-m\sqrt{-\dot{x}^2}} + m\sqrt{-\dot{x}^2} = 0$$

- TOTAL HAMILTONIAN

$$\boxed{H_T = \mu \phi}$$

- POISSON BRACKETS $\{x^\mu(t), p_\nu(t)\} = \delta^\mu_\nu$

- CONSISTENCY CONDITION $\dot{\phi} = \mu \{\phi, \phi\} = 0$

$\Rightarrow$ THERE IS ONLY ONE CONSTRAINT, FIRST CLASS

- GENERATOR $G(\epsilon) = \epsilon \phi$

- LOCAL SYMMETRY $\delta_\epsilon(\dots) = \{\dots, G(\epsilon)\}$

$$\Rightarrow \begin{cases} \delta_\epsilon x^\mu = 2\epsilon p^\mu \\ \delta_\epsilon p_\mu = 0 \end{cases}$$

- NUMBER OF DEGREES OF FREEDOM:

$$N_{\text{PHYS}} = \underbrace{N}_{4} - \underbrace{N_1}_{1} - \frac{1}{2} \underbrace{N_2}_{0} = 3$$

- $x^\mu(t)$ DESCRIBES A THREE-DIMENSIONAL PARTICLE $\vec{r}(t)$
- $H_c = 0$ IS A CONSEQUENCE OF REPARAMETRIZATION SYMM.

$$\frac{dt'}{dt} > 0$$

WE ASK ONLY THAT $t'(t)$ IS MONOTONOUS, SUCH THAT WE CAN INTERPRET t' AS THE TIME COORDINATE

$$\begin{aligned} t &\rightarrow t' = t'(t) \\ x^\mu &\rightarrow x'^\mu(t') = x^\mu(t) \end{aligned}$$

- INFINITESIMAL TRANSFORMATIONS

$$\begin{aligned} t' &= t + \xi(t) \quad \Rightarrow \quad \delta_\xi t = t' - t = \xi(t) \\ x'^\mu(t + \xi) &= x^\mu(t) \quad \Rightarrow \quad \delta_\xi x^\mu = x'^\mu(t) - x^\mu(t) \\ &\simeq x'^\mu(t) + \xi \dot{x}^\mu(t) \quad \boxed{\delta_\xi x^\mu = -\xi \dot{x}^\mu} \end{aligned}$$

$\Rightarrow$ THESE ARE EXACTLY HAMILTONIAN TRANSFORMATIONS WHEN THE PARAMETERS ARE IDENTIFIED AS:

$$-\xi \dot{x}^\mu = 2\varepsilon p^\mu = 2\varepsilon \frac{m \dot{x}^\mu}{\sqrt{-\dot{x}^2}}$$

$$\boxed{\varepsilon(t) = -\frac{1}{2m} \sqrt{-\dot{x}^2} \xi(t)}$$

FIELD-DEPENDENT HAMILTONIAN PARAMETER
= LAGRANGIAN PARAMETER
(USUAL FEATURE WITH REPARAMETRIZATIONS)

- GAUGE CONDITION INTRODUCES TIME

$$\Omega = x^0 - t \simeq 0$$

$$\{\phi, \Omega\} = \{p^2 + m^2, x^0\} = -2p^0 \neq 0 \quad (m \neq 0)$$

$$\dot{\Omega} = \{\Omega, \phi\} u = -2p^0 u = 0 \Rightarrow u = 0$$

(SOLVED multiplier)

POLYAKOV ACTION

- ALTERNATIVE WAY TO DESCRIBE MASSIVE RELATIVISTIC PARTICLE
- IT IS ON-SHELL EQUIVALENT WITH THE PREVIOUS ACTION
- HAMILTONIAN ANALYSIS IS MUCH SIMPLER

$$I_P[x, \dot{x}] = \frac{m}{2} \int dt \left(\frac{1}{2} \dot{x}^2 - 2 \right)$$

alt) NON DYNAMICAL

REPARAMETRIZATION S

- ALL LAGRANGIANS INVARIANT UNDER REPARAMETRIZATIONS HAVE $H_c = 0$

REPARAMETRIZATIONS  $H_c \approx 0$

THE PROOF: $I[q] = \int dt L(q, \dot{q})$

$$\dot{q} = \frac{dq}{dt}$$

REPARAMETRIZATIONS: $t \rightarrow \tau(t)$ (NON SINGULAR TRANSFORMATIONS)

$$q' = \frac{dq}{d\tau} \quad (\text{DEFINE})$$

$$q^0 \equiv t$$

$$\dot{q} = \frac{dq}{dq^0} = \frac{dq/d\tau}{dq^0/d\tau} = \frac{q'}{q'^0}$$

$$I = \int dq^0 L\left(q, \frac{q'}{q'^0}\right) = \int d\tau \underbrace{q'^0 L\left(q, \frac{q'}{q'^0}\right)}_{L_\tau}$$

L_τ NEW LAGRANGIAN;
 (q, q^0) COORDINATES

- CANONICAL HAMILTONIAN OBTAINED FROM L_τ

$$p_0 = \frac{\partial L_\tau}{\partial q'^0} \quad , \quad p = \frac{\partial L_\tau}{\partial q'}$$

- HOMOGENOUS FUNCTION OF DEGREE n : $f(\lambda v_i) = \lambda^n f(v_i)$
- EULER THEOREM FOR HOMOGENOUS FUNCTIONS: $n f = v_i \frac{\partial f}{\partial v_i}$
- APPLICATION TO OUR CASE: L_T IS HOMOGENOUS IN VELOCITIES $v_i = (q', q'^0)$ OF DEGREE $n=1$

(EULER)

$$\Rightarrow L_T = q'^0 \frac{\partial L_T}{\partial q'^0} + q' \frac{\partial L_T}{\partial q'} = q'^0 p_0 + q' p \Rightarrow H_C = 0$$

* THIS IS THE CASE OF GRAVITY, AS WELL

* THIS CONFIRMS THAT H_C DOES NOT DETERMINE DYNAMICS, BUT H_T OR H_E

YANG-MILLS THEORY

- GENERALIZATION OF ELECTROMAGNETIC $U(1)$ THEORY (1-PARAMETER SYMMETRY) TO A NON-ABELIAN ONE (MORE-PARAMETER SYMMETRY)

$A_\mu^a(x) \dots \mu=0,1,2,3 \quad a=1,\dots,n$ GAUGE FIELD

$[T_a, T_b] = f^{ab c} T_c$ NON-ABELIAN ALGEBRA (T_a MATRICES)

$\text{TR}(T_a T_b) = g_{ab}$ CARTAN-KILLING METRIC SEMI-SIMPLE ALGEBRA ($\det g \neq 0$)

$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c$ FIELD-STRENGTH

$$I[A] = -\frac{1}{4} \int d^4x F^2 = -\frac{1}{4} \int d^4x F_{\mu\nu}^a F_{\mu\nu}^a$$

NOTE THAT NOW: $F_{\mu\nu}^a = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta}^a g_{ab}$

* HAMILTONIAN DYNAMICS

• MOMENTUM: $\pi_a^\mu = -F_a^{\mu 0} \Rightarrow \phi_a = \pi_a^0 \approx 0$

• POISSON BRACKETS:

$$\{A_\mu^a, \pi_b^{\nu'}\} = \delta_\mu^{\nu'} \delta_b^a \delta^{(3)}(x-x')$$

• CANONICAL HAMILTONIAN

$$\mathcal{H}_C = \pi_\mu^\alpha \dot{A}_\alpha^\mu - \mathcal{L} = \frac{1}{2} \pi_a^i \pi_i^a + \frac{1}{4} F_{ij}^a F_{ij}^a - A_0^a D_i \pi_a^i$$

• TOTAL HAMILTONIAN

$$\mathcal{H}_T = \mathcal{H}_C + \lambda^a \phi_a$$

• COVARIANT DERIVATIVE: $D_i V_a = \partial_i V_a + f_{ab}^c A_i^b V_c$

* ALL EXPRESSIONS ARE NON-ABELIAN GENERALIZATION OF ELECTROMAGNETISM $\partial_i \rightarrow D_i$

• CONSISTENCY CONDITIONS:

$$\dot{\phi}_a = D_i \pi_a^i \Rightarrow \boxed{\chi_a = D_i \pi_a^i \approx 0} \quad \text{SECONDARY CONSTRAINT}$$

$$\dot{\chi}_a \approx 0$$

* LET US ANALYSE THE CLASS OF CONSTRAINTS ϕ_a, χ_a

$$\{\phi_a, \phi_b^i\} = 0, \quad \{\phi_a, \chi_b^i\} = 0$$

$$\{\chi_a, \chi_b^i\} = \left\{ \underbrace{\partial_i \pi_a^i}_{(1)} + \underbrace{f_{ac}^d A_i^c \pi_d^i}_{(2)}, \underbrace{\partial_j^i \pi_b^j}_{(2)} + \underbrace{f_{be}^f A_j^e \pi_f^j}_{(3)} \right\}$$

$$\begin{aligned} (1) = \{ \partial_i \pi_a^i, f_{be}^f A_j^e \pi_f^j \} &= -f_{be}^f \pi_f^j \partial_i \delta^{(3)} \delta_j^i \delta_a^e \\ &= -f_{ba}^c \pi_c^i \partial_i \delta^{(3)} = f_{ab}^c \partial_i (\pi_c^i \delta^{(3)}) \\ &= f_{ab}^c (\partial_i \pi_c^i \delta^{(3)} + \cancel{\pi_c^i \partial_i \delta^{(3)}}) \end{aligned}$$

$$\begin{aligned} (2) = \{ f_{ac}^d A_i^c \pi_d^i, \partial_j^i \pi_b^j \} &= f_{ac}^d \pi_d^i \partial_j^i \delta^{(3)} \delta_i^j \delta_b^c \\ &= f_{ab}^d \pi_d^j \partial_j^i \delta^{(3)} = -f_{ab}^c \cancel{\pi_c^i \partial_i \delta^{(3)}} \end{aligned}$$

$$\begin{aligned}
(3) &= \{ f_{ac}^d A_i^c \pi_d^i, f_{be}^f A_j^e \pi_f^j \} \\
&= f_{ac}^d f_{be}^f \left(\delta_i^j \delta_f^c \delta^{(3)} \pi_d^i A_j^e - \delta_j^i \delta_d^e \delta^{(3)} A_i^c \pi_f^j \right) \\
&= f_{ac}^d f_{be}^e \pi_d^i A_i^e \delta^{(3)} - f_{ac}^d f_{bd}^f A_i^c \pi_f^i \delta^{(3)} \\
&= \underbrace{(f_{ae}^d f_{bc}^e - f_{ac}^e f_{be}^d)}_{f_{ab}^e f_{ec}^d \text{ (JACOBI IDENTITY)}} A_i^c \pi_d^i \delta^{(3)} = f_{ab}^e f_{ec}^d A_i^c \pi_d^i \delta^{(3)}
\end{aligned}$$

$$\Rightarrow \{ \chi_a, \chi_b^i \} = f_{ab}^c \underbrace{(D_i \pi_c^i + f_{ec}^d A_i^c \pi_d^i)}_{D_i \pi_c^i = \chi_c} \delta^{(3)}$$

$$\Rightarrow \boxed{\{ \chi_a, \chi_b^i \} = f_{ab}^c \chi_c} \quad \text{THEY SATISFY A LIE ALGEBRA}$$

CANONICAL REALIZATION OF $[T_a, T_b] = f_{ab}^c T_c$

$$\Rightarrow \{ \chi_a, \chi_b \} \approx 0, \quad \{ \phi_a, \chi_b \} = 0, \quad \{ \phi_a, \phi_b \} = 0$$

ALL CONSTRAINTS ARE 1 CLASS

• SYMMETRY GENERATOR

$$G[\epsilon] = \int d^4x (\dot{\epsilon}^a \pi_a^0 - \epsilon^a D_i \pi_a^i)$$

• LOCAL TRANSFORMATIONS

$$\begin{aligned}
\delta_\epsilon A_\mu^a &= D_\mu \epsilon^a \\
\delta_\epsilon \pi_a^0 &= 0
\end{aligned}$$

$$\delta_\epsilon \pi_a^i = \int d^4x' \left\{ \pi_a^i, -\epsilon'^b \underbrace{D_j' \pi_b^j}_{\partial_j' \pi_b^j + f_{bc}^d A_j^c \pi_d^i} \right\} = \int d^4x' \epsilon'^b f_{bc}^d \pi_d^j \delta_j^i \delta_a^c \delta^{(3)}$$

$$\delta_\epsilon \pi_a^i = -f_{ab}^c \epsilon^b \pi_c^i$$

• IMPROVED GENERATOR

$$G_Q[\epsilon] = G[\epsilon] + Q[\epsilon]$$

$$\delta G_Q[\epsilon] = \underbrace{(\dots)}_{\text{DIFFERENTIABLE PART}} - \underbrace{\int d^4x \partial_i (\epsilon^a \delta \pi_a^i)}_{=0} + \delta Q[\epsilon]$$

$$\Rightarrow \delta Q[\epsilon] = \oint d\sigma_i \epsilon^a \delta \pi_a^i$$

$$Q[\epsilon] = \oint d\sigma_i \epsilon^a \pi_a^i \quad \text{CONSERVED CHARGE}$$

* CHARGE ALGEBRA

$$\begin{aligned} \{Q[\epsilon_1], Q[\epsilon_2]\}^* &= \delta_{\epsilon_2} Q[\epsilon_1] = \oint d\sigma_i \epsilon_1^a \underbrace{\delta_{\epsilon_2} \pi_a^i}_{-f_{ab}{}^c \epsilon_2^b \pi_c^i} \\ &= -\oint d\sigma_i \underbrace{f_{ab}{}^c \epsilon_1^a \epsilon_2^b}_{[\epsilon_1, \epsilon_2]^c} \pi_c^i = Q[[\epsilon_2, \epsilon_1]] \end{aligned}$$

• LIE-ALGEBRA VALUED GAUGE PARAMETER $\mathcal{Q} = \epsilon^a T_a$
 $[\epsilon_1, \epsilon_2]^c = f_{ab}{}^c \epsilon_1^a \epsilon_2^b$

$$\Rightarrow \boxed{\{Q[\epsilon_1], Q[\epsilon_2]\}^* = Q[[\epsilon_2, \epsilon_1]]} \quad \text{NON-ABELIAN CHARGE ALGEBRA}$$

$$\Leftrightarrow \{G_Q[\epsilon_1], G_Q[\epsilon_2]\} = G_Q[[\epsilon_2, \epsilon_1]] \quad (\text{SHOW})$$

COMMENT: REDEFINING $\epsilon^a \rightarrow -\epsilon^a$ (OR $Q \rightarrow -Q$), THE ALGEBRA CAN BE WRITTEN IN A MORE NATURAL FORM

$$\{Q[\epsilon_1], Q[\epsilon_2]\}^* = Q[[\epsilon_1, \epsilon_2]]$$

THREE-DIMENSIONAL CHERN-SIMONS (CS) THEORY

- 3D CS THEORY HAS MANY APPLICATIONS
- IT IS A NON-ABELIAN GAUGE THEORY
- IT IS QUANTIZABLE
- IT DESCRIBES A QUANTUM HALL EFFECT (QUANTIZED RESISTANCE IN THE SYSTEM OF ELECTRONS IN TWO SPATIAL DIMENSIONS, AT LOW TEMPERATURES AND STRONG $\vec{B}$)
- IT CLASSIFIES KNOTS IN A KNOT THEORY
- GENERAL RELATIVITY IN $D=3$ IS A CS THEORY FOR $SL(2, \mathbb{R}) \otimes SL(2, \mathbb{R}) \cong AdS_3$, dS_3 OR POINCARÉ GAUGE GROUP

• THREE-DIMENSIONAL SPACETIME $x^\mu = (t, x^i)$ ($\mu=0,1,2$; $i=1,2$)

• GAUGE FIELD $A_\mu^a(x)$; FIELD STRENGTH $F_{\mu\nu}^a(x)$

• BIANCHI IDENTITY $\epsilon^{\mu\nu\alpha\beta} D_\nu F_{\alpha\beta}^a = 0$
 $\Leftrightarrow D_\nu F_{\alpha\beta}^a + D_\alpha F_{\beta\nu}^a + D_\beta F_{\nu\alpha}^a = 0$

• USEFUL PROPERTY $\delta F_{\mu\nu}^a = D_\mu A_\nu^a - D_\nu A_\mu^a$ (SHOW)

CONSTRUCTION OF CS ACTION

$$P = \frac{1}{4} \int_{\mathcal{M}_4} d^4x \epsilon^{MNKL} F_{MN}^a F_{KL}^a$$

PONTRYAGIN TOPOLOGICAL INVARIANT in $D=4$ DIMENSIONS

- TOPOLOGICAL INVARIANTS DEPEND ONLY ON THE TOPOLOGY OF THE MANIFOLD $\mathcal{M}_4$ (ITS BOUNDARY $\partial\mathcal{M}_4$)
- ITS VARIATION IS A BOUNDARY TERM $\Rightarrow$ DOES NOT HAVE DYNAMICS, IT IS TOPOLOGICAL

$$\delta P = \int d^4x \epsilon^{MNKL} F_{MN}^a D_K \delta A_{aL} \stackrel{\substack{\uparrow \\ \text{Bianchi id.}}}{=} \int d^4x \partial_K (\epsilon^{MNKL} F_{MN}^a \delta A_{aL})$$

$$= \oint_{\partial M_4} d\sigma_\mu \Omega^\mu$$

$$P = \int_{M_4} d^4x \mathcal{P} \leftarrow \text{PONTRYAGIN DENSITY, IT IS LOCALLY A TOTAL 4-DIVERGENCE: } \mathcal{P} = \partial_\mu \mathcal{L}_{cs}^\mu$$

$$\Rightarrow P = \int_{\partial M_4} d^3x \mathcal{L}_{cs} \leftarrow \text{CHERN-SIMONS DENSITY}$$

$$I_{cs}[A] = \int_{\mathcal{M}} d^3x \mathcal{L}_{cs}$$

$\mathcal{M} \dots$ ARBITRARY 3D MANIFOLD
(NOT NECESSARILY ∂M_4)

$\rightarrow$ THIS CONSTRUCTION GARANTIZES GAUGE SYMMETRY OF THE ACTION (BECAUSE IT COMES FROM P THAT IS GAUGE INVARIANT)

* EXPLICIT FORM OF THE ACTION

$$\mathcal{L}_{cs} = \epsilon^{\mu\nu\alpha} A_\mu^a \left(\partial_\nu A_{\alpha\alpha} + \frac{1}{3} f_{abc} A_\nu^b A_\alpha^c \right)$$

THE ACTION IS LINEAR IN $\partial_\mu A_\nu^a \Rightarrow$ LINEAR IN $\dot{A}_\mu^a$

$\Rightarrow$ IT IS **FIRST ORDER LAGRANGIAN**

$\Rightarrow I_{cs}$ IS A HAMILTONIAN ACTION

• EQUATIONS OF MOTION:

$$F_{\mu\nu}^a = 0$$

$\left\{ \begin{array}{l} \text{ONLY PURE-GAUGE FIELDS,} \\ \text{THEORY HAS 0 DEGREES OF FREEDOM (IT IS} \\ \text{TOPOLOGICAL)} \end{array} \right.$

SYMMETRIES OF CS LAGRANGIAN (by construction)

- i) $F_{MN}^a F_{aKL} \epsilon^{MNKL} \dots$ INVARIANT UNDER A GAUGE ALGEBRA
WITH STRUCTURE CONSTANT f_{ab}^c
 $\Rightarrow \mathcal{L}_{CS}$ IS ALSO INVARIANT (UP TO A BOUNDARY TERM)

GAUGE TRANSFORMATIONS $\delta_\epsilon A_\mu^a = D_\mu \epsilon^a$

- ii) $\mathcal{L}_{CS}$ CAN BE WRITTEN IN TERMS OF DIFFERENTIAL FORMS

$A(x) = A_\mu^a T_a dx^\mu \dots$ INVARIANT UNDER SPACE-TIME
Diffeomorphisms

$$x^\mu \rightarrow x'^\mu = x^\mu(x)$$

$$\left. \begin{aligned} dx'^\mu &= \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ A_\mu^a(x') &= \frac{\partial x^\alpha}{\partial x'^\mu} A_\alpha^a(x) \end{aligned} \right\} \Rightarrow A^a(x') = A^a(x)$$

CS LAGRANGIAN:
IN DIFF. FORMS $\mathcal{L}_{CS} \equiv \mathcal{L}_{CS} d^3x = \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right)$

$$\Rightarrow \mathcal{L}_{CS} \text{ IS INVARIANT UNDER SPACE-TIME diff.}$$

* SO WE CAN CONCLUDE THAT $H_C \approx 0$ FOR CS ACTION

• BUT THERE IS A SUBTLETY ...

• NAMELY, Diffeomorphisms $\mathcal{L}_\xi A_\mu^a$ ARE:

$\rightarrow$ NOT INDEPENDENT FROM GAUGE TRANSFORMATIONS

$\rightarrow$ THEY ARE JUST ON-SHELL SYMMETRY IN THE HAMILT. FRAMEWORK

THIS FOLLOWS FROM THE IDENTITY:

$$\begin{aligned} \mathcal{L}_\xi A_\mu^a &= \xi^\nu \partial_\nu A_\mu^a + A_\nu^a \partial_\mu \xi^\nu \\ &= \xi^\nu F_{\nu\mu}^a + D_\mu (\xi^\nu A_\nu^a) \stackrel{\substack{\uparrow \\ \text{ON-SHELL} \\ F_{\nu\mu}^a = 0}}{=} D_\mu \epsilon^a, \quad \epsilon^a = \xi^\nu A_\nu^a \\ &\quad \text{(FIELD-DEPENDENT)} \end{aligned}$$

HAMILTONIAN ANALYSIS OF CS ACTION

$$x^\mu = (t, x^i)$$

• MOMENTUM $\pi_a^\mu = \frac{\partial \mathcal{L}_{CS}}{\partial \dot{A}_\mu^a} = \frac{\partial}{\partial \dot{A}_\mu^a} (\epsilon^{\nu\alpha\beta} A_\nu^b \partial_\alpha A_b \partial_\beta) =$
 $= \epsilon^{\nu\alpha\beta} A_\nu^b \delta_\alpha^0 \delta_\beta^\mu g_{ab} = \epsilon^{0\mu\nu} A_{a\nu}$

$$\pi_a^0 = 0$$

$$\pi_a^i = \epsilon^{0ij} A_{aj} = \epsilon^{ij} A_{0j}$$

$$\epsilon^{0ij} = \epsilon^{ij}, \quad \epsilon_{0ij} = -\epsilon_{ij}$$

RELATION BETWEEN 3D AND 2D
LEVI-CIVITA SYMBOL

$$\phi_a = \pi_a^0 \approx 0$$

$$\phi_a^i = \pi_a^i - \epsilon^{ij} A_{aj} \approx 0$$

PRIMARY CONSTRAINTS

• CANONICAL HAMILTONIAN

$$\mathcal{H}_C = \pi_a^\mu \dot{A}_\mu^a - \mathcal{L}_{CS} = \pi_a^\mu \dot{A}_\mu^a - \epsilon^{\mu\nu\alpha} A_\mu^b (\partial_\nu A_b \partial_\alpha + \frac{1}{3} f_{abc} A_\nu^b A_\alpha^c)$$

$$\approx \pi_a^i \dot{A}_i^a - \epsilon^{ij} A_0^a (\partial_i A_{aj} + \frac{1}{3} f_{abc} A_i^b A_j^c)$$

$$+ \cancel{\epsilon^{ij} A_{aj}} + \epsilon^{ij} A_i^a (\cancel{\dot{A}_{aj}} - \partial_j A_{a0} + \frac{2}{3} f_{abc} A_0^b A_j^c)$$

$$\downarrow$$

$$\partial_j(\dots) - \epsilon^{ij} A_{0a} \partial_i A_j^a$$

(WE NEGLECT IT)

$$= -\epsilon^{ij} A_0^a (2\partial_i A_{aj} + f_{abc} A_i^b A_j^c) = -\epsilon^{ij} A_0^a F_{aij}$$

$$\partial_i A_{aj} - \partial_j A_{ai} + f_{abc} A_i^b A_j^c = F_{a ij}$$

$$\mathcal{H}_T = -\epsilon^{ij} A_0^a F_{aij} + \mu^a \phi_a + \mu_i^a \phi_a^i$$

WE NOTE THAT A_0^a IS A MULTIPLIER
(IT ENTERS LINEARLY)

- CANONICAL POISS. BRACKETS $\{A_\mu^a, \Pi_b^{\nu}\} = \delta_\mu^\nu \delta_b^a \delta^{(3)}$
- AUXILIARY BRACKETS (AMONG CONSTRAINTS) $\{\phi_a, \phi_b\} = 0, \{\phi_a, \phi_b^i\} = 0$
 $\{\phi_a^i, \phi_b^j\} = \{\pi_a^i - \epsilon^{ik} A_{ak}, \pi_b^j - \epsilon^{jl} A_{bl}\}$
 $= -2\epsilon^{ij} g_{ab} \delta^{(3)}$
- CONSISTENCY CONDITIONS

$$\dot{\phi}_a = \int d^3x' \{ \pi_a^0, -\epsilon^{ij} A_0^b F_{bij} \} = \int d^3x' \epsilon^{ij} \delta_a^b F_{bij} \delta^{(3)} = \epsilon^{ij} F_{a ij}$$

$$\chi_a = \epsilon^{ij} F_{a ij} \approx 0$$

SECONDARY CONSTRAINTS

- Obs.: IN 2 SPATIAL DIMENSIONS, F_{ij} HAS ONLY ONE IND. COMPONENT

$$\begin{cases} F_{ij} = \epsilon_{ij} F_{12} \\ F_{12} = \frac{1}{2} \epsilon^{ij} F_{ij} \end{cases}$$

$$\Rightarrow \mathcal{H}_T = -A_0^a \chi_a + m^a \phi_a + m_i^a \phi_a^i$$

- AUXILIARY BRACKETS

$$\{\phi_a, \chi_b\} = 0$$

$$\{\phi_a^i, \chi_b\} = \{\pi_a^i, \epsilon^{jk} F_{bjk}\}$$

$$= -\epsilon^{jk} (2\partial_j^i \delta^{(2)} g_{ab} \delta_k^i + 2f_{bcd} A_j^c \delta_k^i \delta_a^d \delta^{(3)})$$

$$= 2\epsilon^{ij} (-g_{ab} \partial_j \delta^{(2)} + f_{abc} A_j^c \delta^{(3)})$$

$$\{\chi_a, \chi_b\} = 0$$

$$\begin{aligned} \dot{\phi}_a^i &= \int d^3x' \{ \phi_a^i, -A_0^b \chi_b + m^b \phi_b + m_j^b \phi_b^j \} \\ &= \int d^3x' [-A_0^b 2\epsilon^{ij} (-g_{ab} \partial_j \delta^{(2)} + f_{abc} A_j^c \delta^{(3)}) - m_j^b 2\epsilon^{ij} g_{ab} \delta^{(3)}] \\ &= 2\epsilon^{ij} (D_j A_0 a - m_{aj}) \approx 0 \quad / \quad \epsilon_{j k} \quad (\text{USING } \epsilon^{ij} \epsilon_{jk} = \delta_k^i) \end{aligned}$$

$\Rightarrow$ WE SOLVE THE MULTIPLIER, WHICH MEANS THAT ϕ_a^i SHOULD BE II CLASS

$$m_i^a = D_i A_0^a$$

FINALLY:

$$\begin{aligned} \dot{\chi}_a &= \int d^2x' \{ \chi_a, \mu_i^b \phi_b^i \} = - \int d^2x' \mu_i^b 2\epsilon^{ij} (-g_{ab} \partial_j' \delta^{(3)} + f_{abc} A_j^c \delta^{(3)}) \\ &= 2\epsilon^{ij} \underbrace{(-\partial_j \mu_{ai} + f_{abc} \mu_i^b A_j^c)}_{+ \partial_i \mu_{aj}} = 2\epsilon^{ij} D_i \mu_{aj} \end{aligned}$$

$$\begin{aligned} &= 2\epsilon^{ij} D_i D_j A_{0a} \\ &\stackrel{\substack{\uparrow \\ \epsilon^{ij} \text{ IS ANTISYM.}}}{=} \epsilon^{ij} [D_i, D_j] A_{0a} \stackrel{\substack{\uparrow \\ \text{IDENTITY}}}{=} \epsilon^{ij} f_{abc} F_{ij}^b A_0^c = f_{abc} \chi^b A_0^c \approx 0 \end{aligned}$$

$[D_i, D_j] V^a = f_{bc}^a F_{ij}^b V^c$

 (SHOW)

$\Rightarrow$ THERE ARE NO MORE CONSTRAINTS

* CLASS OF THE CONSTRAINTS

$$\phi_a = \pi_a^0 \quad \text{I CLASS}$$

$$\phi_a^i = \pi_a^i - \epsilon^{ij} A_{ja} \quad \text{II CLASS} \quad (\text{SOLVES } \mu_i^a)$$

$$\{ \phi_a^i, \phi_b^j \} = -2\epsilon^{ij} g_{ab} \delta^{(3)} \neq 0$$

$$\chi_a = \epsilon^{ij} F_{aj} \quad ? \quad \text{WE DON'T KNOW}$$

$$\{ \phi_a^i, \chi_b \} = 2\epsilon^{ij} (-g_{ab} \partial_j \delta^{(3)} + f_{abc} A_j^c \delta^{(3)})$$

$\Rightarrow$ THE CONSTRAINTS $(\phi_a, \phi_a^i, \chi_a)$ ARE NOT WELL SEPARATED IN CLASSES

• THE SIMPLEST WAY TO SEPARATE THEM IS TO WRITE THEM AS $H_T = H' + \text{I CLASS CONSTR.}$ AFTER REPLACING SOLVED MULTIPLIERS

$$\mu_i^a = D_i A_0^a$$

$$\mathcal{H}_T = -A_0^a \chi_a + \mu^a \phi_a + \underbrace{D_i A_0^a \phi_a^i}_{\partial_i (A_0^a \phi_a^i) - A_0^a D_i \phi_a^i}$$

$$= -A_0^a \underbrace{(\chi_a + D_i \phi_a^i)}_{\equiv G_a} + \mu^a \phi_a + \text{TOTAL DIVERGENCE} \quad \rightarrow 0 \text{ (boundary term)}$$

• WE DEFINE $G_a = \chi_a + D_i \phi_a^i \approx 0$ (INSTEAD OF χ_a)

• NOW AGAIN WE CHECK POISSON BRACKETS AMONG $\Psi_S \equiv (\phi_a, \phi_a^i, G_a)$

$$\{\phi_a, G_b\} = 0$$

$$\{\phi_a^i, G_b\} = f^{ab^c} \phi_c^i \delta^{(3)} \approx 0$$

$$\{G_a, G_b\} = \{\chi_a + D_i \phi_a^i, \chi_b + D_j \phi_b^j\} = \dots = f^{ab^c} G_c \delta^{(3)}$$

$\Rightarrow G_a$ IS **I CLASS** !!

• LOCAL (PHYSICAL) DEGREES OF FREEDOM

$$A_\mu^a \quad (\mu=0,1,2; a=1,\dots,n)$$

$$N = 3n$$

$$\phi_a, G_a \quad (a=1,\dots,n) \quad \text{I CLASS}$$

$$N_1 = 2n$$

$$\phi_a^i \quad (i=1,2; a=1,\dots,n) \quad \text{II CLASS}$$

$$N_2 = 2n$$

$$\text{DIRAC FORMULA: } N_{\text{PHYS}} = N - N_1 - \frac{1}{2} N_2 = 3n - 2n - \frac{1}{2} \cdot 2n = 0$$

$\Rightarrow$ THERE ARE NO LOCALLY PROPAGATING DEGREES OF FREEDOM,
3D CS GAUGE THEORY IS TOPOLOGICAL

• LOCAL SYMMETRIES

$\rightarrow$ TO SIMPLIFY THE COMPUTATION, WE FIX THE GAUGE $A_0^a \approx 0$
 AND ELIMINATE (A_0^a, π_0^a) FROM THE THEORY (DEFINE
 PARTIALLY REDUCED SPACE)

$$\text{GENERATOR: } G[\varepsilon] = \int d^2x \, \varepsilon^a G_a$$

$$\Rightarrow \delta_\varepsilon A_i^a = \int d^2x' \{A_i^a, \varepsilon'^a D_i' \pi'^a\} = -D_i \varepsilon^a \quad \text{GAUGE TR.}$$

• DIFFEOMORPHISMS DO NOT APPEAR, THEY ARE DEPENDENT

PART III SYMPLECTIC GEOMETRY AND CANONICAL TRANSFORMATIONS**BIBLIOGRAPHY :**

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SYMPLECTIC GEOMETRY = GEOMETRY OF THE PHASE SPACE

* WE START WITH CLASSICAL MECHANICS AND DISCRETE SYSTEMS

- COORDINATES ON THE PHASE SPACE Γ

$$\Gamma: x^a(t) = (p_i, q^i) \quad i=1, \dots, N; \quad a=1, \dots, 2N$$

- CONVENIENT PARAMETRIZATION (THE ORDER IS IMPORTANT)

$$(x^1, \dots, x^{2N}) = (p_1, \dots, p_N, q^1, \dots, q^N)$$

SYMPLECTIC MATRIX ω^{ab}

- CANONICAL POISSON BRACKETS

$$\{q^i, p_j\} = \delta_j^i \quad \Rightarrow \quad \boxed{\{x^a, x^b\} = \omega^{ab}} = \begin{matrix} p_j & q^j \\ q^i & \end{matrix} \left(\begin{array}{c|c} 0 & -\delta_i^j \\ \hline \delta_j^i & 0 \end{array} \right)$$

- PROPERTIES OF THE SYMPLECTIC MATRIX

i) ANTISYMMETRIC: $\omega^{ab} = -\omega^{ba}$

ii) NON-SINGULAR: $\det[\omega^{ab}] \neq 0 \rightarrow$ INVERTIBILITY

- ω^{ab} DETERMINES POISSON BRACKETS $\forall A, B \in \Gamma$

$$\{A, B\} = \frac{\partial A}{\partial q^i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q^i} = \omega^{ab} \frac{\partial A}{\partial x^a} \frac{\partial B}{\partial x^b}$$

$$\frac{\partial}{\partial x^a} \equiv \partial_a \quad \Rightarrow \quad \boxed{\{A, B\} = \omega^{ab} \partial_a A \partial_b B} \quad (*)$$

• THE DEFINITION (*) OF $\{, \}$ IN TERMS OF ω^{ab} APPLIES FOR ANY COORDINATES x^a AT Γ

• IN PARTICULAR CANONICAL COORDINATES $\omega^{ab} = \begin{pmatrix} 0 & -\mathbb{I}_N \\ \mathbb{I}_N & 0 \end{pmatrix}$

iii) ω^{ab} TRANSFORMS AS A TENSOR OF RANK 2 UNDER TRANSFORMATIONS $x^a \rightarrow x'^a = x'^a(x)$

$$\boxed{\omega^{ab} \rightarrow \omega'^{ab}(x') = \frac{\partial x'^a}{\partial x^c} \frac{\partial x'^b}{\partial x^d} \omega^{cd}(x)}$$

iv) ω^{ab} SATISFIES JACOBI IDENTITY

$$\underbrace{\{A, \{B, C\}\}}_{\omega^{ab} \partial_a B \partial_b C} + \underbrace{\{B, \{C, A\}\}}_{\omega^{ab} \partial_a C \partial_b A} + \underbrace{\{C, \{A, B\}\}}_{\omega^{ab} \partial_a A \partial_b B} = 0 \quad (\text{JACOBI ID. FOR POISSON BRACKETS})$$

$$\omega^{cd} \partial_c A \partial_d (\omega^{ab} \partial_a B \partial_b C) + \omega^{cd} \partial_c B \partial_d (\omega^{ab} \partial_a C \partial_b A) + \omega^{cd} \partial_c C \partial_d (\omega^{ab} \partial_a A \partial_b B) = 0$$

$$\begin{aligned} & \omega^{cd} \partial_c A (\partial_d \omega^{ab} \partial_a B \partial_b C + \cancel{\omega^{ab} \partial_d \partial_a B \partial_b C} + \cancel{\omega^{ab} \partial_a B \partial_d \partial_b C}) \\ & + \omega^{cd} \partial_c B (\partial_d \omega^{ab} \partial_a C \partial_b A + \cancel{\omega^{ab} \partial_d \partial_a C \partial_b A} + \cancel{\omega^{ab} \partial_a C \partial_d \partial_b A}) \\ & + \omega^{cd} \partial_c C (\partial_d \omega^{ab} \partial_a A \partial_b B + \cancel{\omega^{ab} \partial_d \partial_a A \partial_b B} + \cancel{\omega^{ab} \partial_a A \partial_d \partial_b B}) = 0 \end{aligned}$$

$$\omega^{cd} \partial_d \omega^{ab} (\partial_a A \partial_b B \partial_c C + \partial_a A \partial_c B \partial_b C + \partial_a A \partial_b B \partial_c C) = 0$$

$$\underbrace{(\partial_d \omega^{ab} \omega^{cd} + \partial_d \omega^{bc} \omega^{ad} + \partial_d \omega^{ca} \omega^{bd})}_{\equiv \partial_d \omega^{ab} \omega^{cd} + \text{cyclic}(abc)} \partial_a A \partial_b B \partial_c C = 0 \quad \forall A, B, C$$

$$\boxed{\partial_d \omega^{ab} \omega^{cd} + \text{cyclic}(abc) = 0}$$

ID. DE JACOBI PARA ω^{ab}

SYMPLECTIC FORM ω_{ab}

BECAUSE THE SYMPLECTIC MATRIX ω^{ab} IS INVERTIBLE, LET US DEFINE THE INVERSE MATRIX ω_{ab}

$$\boxed{\omega^{ac} \omega_{cb} = \delta^a_b} \quad \omega^{-1} = \begin{pmatrix} 0 & -\mathbb{I}_N \\ \mathbb{I}_N & 0 \end{pmatrix}; \quad \omega = \begin{pmatrix} 0 & \mathbb{I}_N \\ -\mathbb{I}_N & 0 \end{pmatrix}$$

$$\omega = [\omega_{ab}] \quad \text{SYMPLECTIC FORM}$$

$$\omega^{-1} = [\omega^{ab}] \quad \text{SYMPLECTIC MATRIX}$$

- ω^{-1} ANTISYMMETRIC $\Rightarrow \omega$ ANTISYMMETRIC: $\omega_{ab} = -\omega_{ba}$
- ω^{-1} IS A RANK-2 TENSOR $\Rightarrow \omega$ IS A RANK-2 TENSOR

$$\boxed{\omega'^{ab}(x') = \frac{\partial x^c}{\partial x'^a} \frac{\partial x^d}{\partial x'^b} \omega_{cd}(x)}$$

$\Rightarrow \omega$ IS A 2-FORM ON $\mathbb{P}$

$$\omega = \frac{1}{2} \omega_{ab} \underbrace{dx^a \wedge dx^b}_{-dx^b \wedge dx^a} \quad \dots \text{2-FORM}$$

$\wedge \dots$ WEDGE PRODUCT, ANTISYMMETRIC by DEFINITION

$$dx'^a = \frac{\partial x'^a}{\partial x^c} dx^c \Rightarrow \boxed{\omega'(x') = \omega(x)} \quad \text{P-FORMS ARE INVARIANT UNDER REPARAMETRIZATIONS}$$

$x^a \rightarrow x'^a = x'^a(x)$

$$\omega_{ab} = \begin{pmatrix} p_i & q^j \\ q^i & -p_j \end{pmatrix} \Rightarrow \omega = \frac{1}{2} dp_i \wedge dq^i - \frac{1}{2} \underbrace{dq^i \wedge dp_i}_{-dp_i \wedge dq^i}$$

$$\omega = dp_i \wedge dq^i = dx^1 \wedge dx^{N+1} + \dots + dx^N \wedge dx^{2N}$$

Comment p-FORM is $\alpha = \frac{1}{p!} \alpha_{a_1 \dots a_p} dx^{a_1} \wedge \dots \wedge dx^{a_p}$

DARBOUX THEOREM

- IN GENERAL, LOCAL COORDINATES $x^a \in \Gamma$ CAN BE ARBITRARY, AS LONG AS $\omega^{ab}(x)$ IS ANTISYMMETRIC AND NON-SINGULAR
- DARBOUX THEOREM SAYS THAT THERE ALWAYS EXIST LOCAL COORDINATES SUCH THAT ω^{ab} IS BLOCK-DIAGONAL

$$\begin{pmatrix} (q \times q) & & \\ & \ddots & \\ & & (q \times q) \end{pmatrix}$$

OR REORDERING THE COORDINATES

THEY CAN BE PUT IN **SKEW-SYMMETRIC FORM**

$$\bar{\omega}^{-1} = \begin{pmatrix} 0 & -\mathbb{I}_N \\ \mathbb{I}_N & 0 \end{pmatrix}$$

- THESE SKEW-SYMMETRIC COORDINATES ARE EXACTLY THE CANONICAL ONES (p, q)

INTRODUCING DIFFERENTIAL STRUCTURE ON Γ

- EXTERIOR DERIVATIVE $d = dx^a \partial_a$ MAPS p -FORM TO A $(p+1)$ -FORM

$$\omega = \frac{1}{2} \omega_{ab} dx^a \wedge dx^b$$

$$d\omega = \frac{1}{2} \underbrace{\partial_c \omega_{ab}}_{\substack{\text{WE HAVE} \\ \text{TO ANTISYMMETRIZE} \\ \text{IT IN } [abc]}} \underbrace{dx^c \wedge dx^a \wedge dx^b}_{\substack{\text{ANTISYMMETRIC IN} \\ [abc]}}$$

$$d\omega = \frac{1}{3!} (\partial_a \omega_{bc} + \partial_b \omega_{ca} + \partial_c \omega_{ab}) dx^a \wedge dx^b \wedge dx^c$$

$= 0$ (CONSEQUENCE OF JACOBI IDENTITY AND INVERTIBILITY OF ω)

$d\omega = 0$

 $\iff \partial_a \omega_{bc} + \text{cycl. } (abc) = 0$

$\} \omega$ is **CLOSED**

• WE CONCLUDE THAT A SYMPLECTIC FORM IS A 2-FORM ON Γ THAT SATISFIES:

i) IT IS CLOSED, $d\omega = 0$

ii) IT IS NON SINGULAR, $(\forall v^a) \quad \omega_{ab} v^b = 0 \Rightarrow v^b = 0$
 (THERE ARE NO ZERO MODES)
 $\Leftrightarrow \det \omega \neq 0$

SYMPLECTIC MANIFOLD (Γ, ω)

(Γ WITH SYMPLECTIC STRUCTURE)

• ANALOGY WITH METRIC MANIFOLD (M, g)

MANIFOLD
WITH LOCAL
COORDINATES
OF SPACETIME
 x^μ

METRIC FIELD
 $g_{\mu\nu}(x)$;
 $g_{\mu\nu} = g_{\nu\mu}$, $\det g \neq 0$

• IN THE SAME WAY WE INTRODUCE DIFFERENTIAL GEOMETRY ON M WITH RESPECT TO THE COORDINATES x^μ , BY INTRODUCING A p -FORM BASIS $dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}$ ($p=0,1,2,\dots$),

WE CAN INTRODUCE DIFFERENTIAL GEOMETRY ON Γ WITH THE BASIS $dx^{a_1} \wedge \dots \wedge dx^{a_p}$ ($p=0,1,2,\dots$)

BASICS OF DIFFERENTIAL GEOMETRY (ON Γ)

Γ ... TARGET SPACE, SPACE WITH LOCAL COORD. $x^a \in \Gamma$

$T_x \Gamma$ TANGENT BUNDLE, TANGENT SPACE IN THE POINT x^a ;
 $x^a \rightarrow \partial_a$

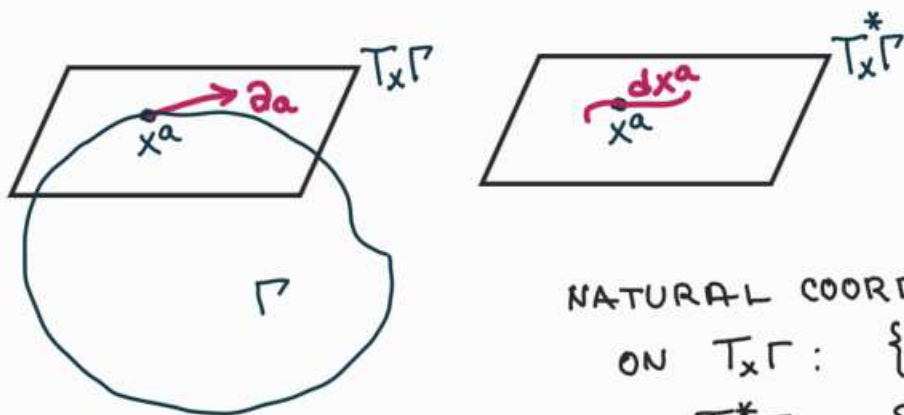
$T_x^* \Gamma$ COTANGENT BUNDLE, COTANGENT SPACE, DUAL TO $T_x \Gamma$:

DUALITY: $dx^a(\partial_b) = \delta_b^a$
 OPERATOR VECTOR NUMBER
 (OPERATOR ACTS ON A VECTOR)

$\partial_a \rightarrow dx^a$

VECTOR $X = X^a \partial_a \in T_x \Gamma$ (VECTORS LIVE ON $T_x \Gamma$) $X: \Gamma \rightarrow T_x \Gamma$

1-FORM $\alpha = \alpha_a dx^a \in T_x^* \Gamma$ (1-FORMS LIVE ON $T_x^* \Gamma$) $\alpha: \Gamma \rightarrow T_x^* \Gamma$



NATURAL COORDINATE BASIS :

ON $T_x\Gamma$: $\{\partial_a\}$

ON $T_x^*\Gamma$: $\{dx^a\}$

COMMENT : OTHER EXAMPLE OF THESE SPACES

Q ... CONFIGURATIONAL SPACE q^i (TARGET SPACE)

TQ ... TANGENT SPACE (TANGENT BUNDLE) $(q, \dot{q})$

T^*Q ... COTANGENT SPACE (q, p)

LAGRANGIAN $L: TQ \rightarrow \mathbb{R}$ $L(q, \dot{q}) \in \mathbb{R}$

HAMILTONIAN $H: T^*Q \rightarrow \mathbb{R}$ $H(q, p) \in \mathbb{R}$

USEFUL OPERATORS

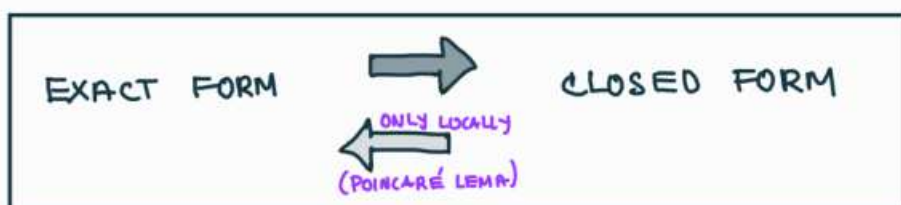
• **P-FORM** $\Omega(p) = \frac{1}{p!} \underbrace{\Omega_{a_1 \dots a_p}}_{\text{COMPLETELY ANTISYMMETRIC TENSOR}} dx^{a_1} \wedge \dots \wedge dx^{a_p}$

→ symplectic form HAS $p=2$

• **EXTERIOR DERIVATIVE** $d = dx^a \partial_a$
 $d^2 = 0$ (IT IS NILPOTENT) $dx^a \wedge dx^b \partial_a \partial_b \equiv 0$

• **CLOSED FORM** ω : $d\omega = 0$

• **EXACT FORM** β : $\beta = d\alpha$



- **INTERIOR PRODUCT** (OR CONTRACTION WITH VECTOR ξ) : i_ξ

$$p=1 \quad i_\xi \alpha = \alpha(\xi) = \xi^a \alpha_a$$

$$p=2 \quad i_\xi \omega = \omega(\xi) = \xi^a \omega_{ab} dx^b$$

$$i_\xi : p\text{-FORM} \rightarrow (p-1)\text{FORM}$$

- **GENERALIZED LEIBNIZ RULE:**

$$d(U_{(p)} \wedge V_{(q)}) = dU_{(p)} \wedge V_{(q)} + (-1)^p U_{(p)} \wedge dV_{(q)}$$

$$i_\xi (U_{(p)} \wedge V_{(q)}) = i_\xi U_{(p)} \wedge V_{(q)} + (-1)^p U_{(p)} \wedge i_\xi V_{(q)}$$

- **LIE BRACKET** $X, Y \in T_x \Gamma$

$$[X, Y] = XY - YX = (X^a \partial_a Y^b - Y^a \partial_a X^b) \partial_b \in T_x \Gamma$$

$$[,] : T_x \Gamma \times T_x \Gamma \rightarrow T_x \Gamma$$

- **LIE DERIVATIVE** IN DIRECTION X : $\mathcal{L}_X$

→ IT IS USEFUL TO DEFINE IT THROUGH THE

CARTAN IDENTITY:

$$\mathcal{L}_X = i_X d + d i_X$$

$$p\text{-FORM} \rightarrow p\text{-FORM}$$

→ SATISFIES **LEIBNIZ RULE** SO IT CAN ACT ON VECTORS TOO

$$\mathcal{L}_X (AB) = (\mathcal{L}_X A) B + A (\mathcal{L}_X B)$$

(A, B CAN BE FORMS, VECTORS, FUNCTIONS, TENSORS...)

EXAMPLES: 0-FORM

$$\mathcal{L}_X \phi = \xi^a \partial_a \phi$$

1-FORM

$$\mathcal{L}_X \alpha_a = \xi^b \partial_b \alpha_a + \partial_a \xi^b \alpha_b$$

VECTOR FIELD

$$\mathcal{L}_X X = [X, X]$$

- INTUITION: $\mathcal{L}_\xi$ DESCRIBES CHANGE OF ANY FORM OR VECTOR FIELD, $\mathcal{L}_\xi \Psi(x) = \Psi(x + \xi) - \Psi(x)$,
AS A CONSEQUENCE OF COORDINATE CHANGE
 $x^a \rightarrow x^a + \xi^a(x)$ ON Γ

SYMPLECTIC ACTION

- HAMILTONIAN ACTION

$$I_H[q] = \int dt (p_i \dot{q}^i - H)$$

$$\begin{aligned} p_i \dot{q}^i &= \frac{1}{2} p_i \dot{q}^i + \frac{1}{2} p_i \dot{q}^i \\ &= \frac{1}{2} p_i \dot{q}^i + \left(\frac{1}{2} p_i \dot{q}^i \right)' - \frac{1}{2} \dot{p}_i q^i \\ &= \frac{1}{2} \underbrace{(p_i \dot{q}^i - \dot{p}_i q^i)}_{\omega_{ab} x^a \dot{x}^b} + \cancel{\text{TOT. DER.}}^0 \text{ (NEGLECTED)} \end{aligned}$$

- SYMPLECTIC ACTION $(p_i, q^i) \rightarrow x^a$

$$I_0[x] = \int dt \left(\frac{1}{2} \omega_{ab} x^a \dot{x}^b - H \right)$$

↳ FIRST ORDER ACTION (LINEAR IN $\dot{x}^a$)

- GENERALIZATION TO ANY ACTION OF FIRST ORDER

$$I_s[x] = \int dt \left(\sigma_a(x) \dot{x}^a - V_s(x) \right)$$

$\uparrow$ SYMPLECTIC POTENTIAL

EVOLUTION OF HAMILTONIAN SYSTEM

$$\begin{aligned} \dot{F}(x) &= \{F, H\} = \omega^{ab} \partial_a F \partial_b H \\ &= \underbrace{(\omega^{ab} \partial_b H \partial_a)}_{X_H \text{ HAMILTONIAN VECTOR FIELD}} F = X_H^a \partial_a F = i_{X_H} dF \equiv X_H(F) \end{aligned}$$

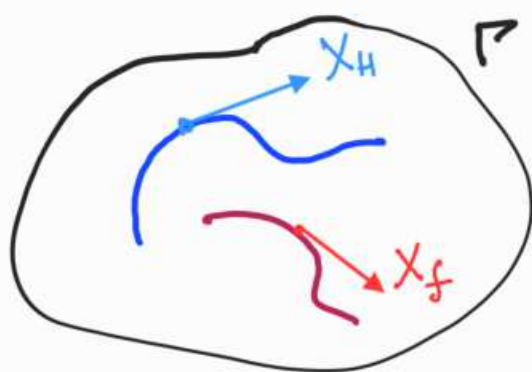
$$\dot{F}(x) = i_{X_H} dF = X_H(F)$$

$$X_H = X_H^a \partial_a = \omega^{ab} \partial_b H \partial_a$$

HAMILTONIAN VECTOR FIELD,
DETERMINES EVOLUTION
OF A SYSTEM

$\Rightarrow \frac{d}{dt}$ IS EQUIVALENT TO THE ACTION OF X_H

* ITS COMPONENTS: $\dot{x}^a = \omega^{bc} \underbrace{\partial_b x^a}_{\delta_b^a} \partial_c H = \omega^{ac} \partial_c H \Rightarrow X_H^a = \dot{x}^a$



* THE VECTOR $X_H = X_H^a \partial_a$
GENERATES A CURVE
(EVOLUTION) ON Γ

* WE CAN GENERATE ANOTHER
CURVE BY

$$X_f = X_f^a \partial_a$$

$$X_f^a = \omega^{ab} \partial_b f$$

SO, WE HAVE ON Γ :

$$\xi = \xi^a(x) \partial_a$$

GENERAL VECTOR FIELD

$$X_f = \omega^{ab} \partial_b f \partial_a$$

HAMILTONIAN VECTOR FIELD ASSOCIATED
WITH THE FUNCTION $f(x)$

- WE SAW THAT X_H GENERATES EVOLUTION, $\dot{F} = X_H(F)$.
LATER WE WILL SEE THAT X_f GENERATES A CANONICAL
TRANSFORMATION, $\delta_f F = \{F, f\} = X_f(F)$.

THIS IS THE RELEVANCE OF HAMILTONIAN VECTOR FIELDS.

- SYMPLECTIC MATRIX MAPS FUNCTIONS TO HAMILT. VECTOR FIELDS:

$$df = \partial_a f dx^a \longrightarrow X_f^a = \omega^{ab} \partial_b f$$

THREE WAYS TO WRITE POISSON BRACKETS

$$\{f, g\} = \underbrace{\omega^{ab}}_{-X_f^b} \underbrace{\partial_a f \partial_b g}_{-\omega_{ab} X_g^a} = -\omega_{ab} X_f^a X_g^b$$

i) $\{f, g\} = -\omega(X_f, X_g)$ 2-FORM AS A LINEAR OPERATOR ACTS OVER VECTORS

ii) $\{f, g\} = -i_{X_f} i_{X_g} \omega$ ← NOTE THAT $\sum_i \{ \}$ IS A DOUBLE CONTRACTION OF ω

iii) $\{f, g\} = -\mathcal{L}_{X_f} g = \mathcal{L}_{X_g} f$

* NOTE THE IDENTITY:

$$X_{\{f, g\}} = -[X_f, X_g]$$

SYMPLECTIC TRANSFORMATION = CANONICAL TRANSFORMATION ALONG A VECTOR FIELD $\xi \in T_x \Gamma$

$$\boxed{\mathcal{L}_\xi \omega = 0} \quad \text{SYMPLECTIC TRANSFORMATION}$$

$$\xi = \xi^a(x) \partial_a \quad \dots \quad \text{SYMPLECTIC VECTOR FIELD}$$

IF IT SATISFIES $\mathcal{L}_\xi \omega = 0$

→ IT IS ANALOGOUS TO ISOMETRIES $\mathcal{L}_\xi g = 0$ ON RIEMANN MANIFOLDS WHERE $\xi = \xi^\mu(x) \partial_\mu \in T_x M$ IS A KILLING VECTOR FIELD

• **CANONICAL TRANSFORMATION** $x^a = (p_i, q^i) \rightarrow x'^a = (P_i, Q^i)$
SUCH THAT $dp_i \wedge dq^i = dP_i \wedge dQ^i$

→ IN GENERAL THERE IS ω ; CANONICAL TRANSFORMATION DOES NOT CHANGE SYMPLECTIC FORM (IT IS AN ISOMETRY OF THE SYMPLECTIC FORM)

CONSEQUENCE OF HAVING AN ISOMETRY ξ^a

⇒ THERE IS A GENERATOR OF SYMPLECTIC TRANSFORMATION

• USING THE CARTAN IDENTITY:

$$\mathcal{L}_\xi \omega = \underbrace{i_\xi d\omega}_{=0 \text{ (CLOSED)}} + d i_\xi \omega = 0 \Rightarrow d(i_\xi \omega) = 0$$

POINCARÉ LEMMA: IF $i_\xi \omega$ IS CLOSED, THEN IT IS LOCALLY EXACT

$$\boxed{i_\xi \omega = dQ_\xi} \quad \xi^a \omega_{ab} = \partial_b Q_\xi$$

$Q_\xi = \text{GENERATOR OF SYMPLECTIC ISOMETRY}$

→ THE EXISTENCE OF Q_E IS GUARANTEED ONLY LOCALLY

→ TO HAVE A GLOBAL CHARGE, $i_E \omega$ MUST BE GLOBALLY EXACT, AND Q_E DEFINED ON THE WHOLE RELEVANT REGION OF PHASE SPACE

→ THE POSSIBLE OBSTRUCTION IS TOPOLOGICAL

* Q_E GENERATES ISOMETRY $\mathcal{E}$ WHEN ACTING ON $F \in T_x \Gamma$:

$$\begin{aligned}\{F, Q_E\} &= \omega^{ab} \partial_a F \partial_b Q_E = \omega^{ab} \underbrace{\partial_b Q_E}_{\mathcal{E}^c \omega_{cb}} \partial_a F = \underbrace{\omega^{ab} \omega_{cb}}_{-\delta_c^a} \mathcal{E}^c \partial_a F \\ &= -\mathcal{E}^a \partial_a F = -i_E dF = \delta_E F\end{aligned}$$



CANONICAL TRANSFORMATION

$$\delta_E F = -i_E dF = \{F, Q_E\}$$

LOCAL SYMMETRY TRANSFORMATION

$\mathcal{E} = \mathcal{E}^a(x) \partial_a$ PARAMETER OF LOCAL SYMMETRY

SATISFIES:

- 1) $\mathcal{L}_E \omega = 0$ (SYMPLECTIC ISOMETRY)
- 2) $\mathcal{L}_E H = 0$ (HAMILTONIAN ISOMETRY)

1) ⇒ THERE IS A GENERATOR Q_E DEFINED BY $i_E \omega = dQ_E$

2) ⇒ Q_E IS CONSERVED

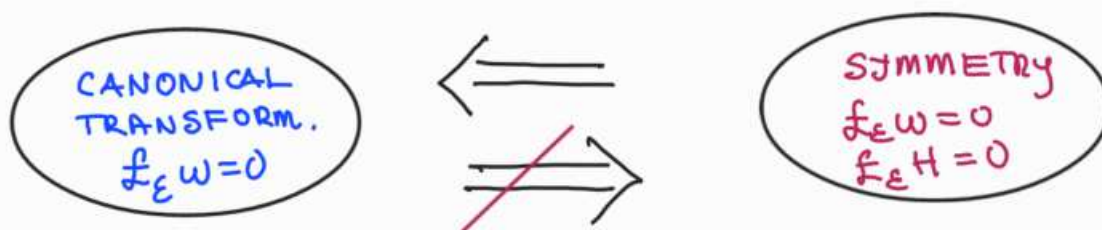
PROOF: $0 = \mathcal{L}_E H = \underbrace{d i_E H}_{\substack{\text{CARTAN} \\ \text{IDENTITY} \\ (H \text{ IS } 0\text{-FORM})}} + i_E \underbrace{dH}_{\substack{\text{CANONICAL} \\ \text{TRANSF.}}} = -\{H, Q_E\} = \dot{Q}_E$ ✓

H DOES NOT CHANGE ALONG A VECTOR FIELD, $\mathcal{L}_\epsilon H = 0$
 $\Leftrightarrow Q_\epsilon$ IS CONSERVED, $\dot{Q}_\epsilon = \{Q_\epsilon, H\} = 0$

TWO INTERPRETATION OF THE SAME FORMULA
 BECAUSE $\{Q_\epsilon, H\} = \mathcal{L}_\epsilon H = \dot{Q}_\epsilon$

COMMENTS

- ALL HAMILTONIAN SYMMETRIES ARE ALSO SYMPLECTIC SYMMETRIES



- SYMMETRY AND EVOLUTION LOOK SIMILARLY

$$\begin{array}{ll} i_\epsilon \omega = dQ_\epsilon & \text{SYMMETRY} \\ i_{X_H} \omega = -dH & \text{EVOLUTION} \end{array}$$

PROOF: $i_{X_H} \omega = X_H^a \omega_{ab} dx^b = \frac{\omega^{ac}}{\omega^{ab}} \frac{\partial H}{\partial x^c} \omega_{ab} dx^b$
 $= -\partial_b H dx^b = -dH$ $\downarrow -\delta^c_b$

- GEOMETRIC FORMULATION OF HAMILTON'S EQUATIONS

$$\begin{aligned} i_{X_H} \omega &= -dH \\ i_{X_H} (dq^i \wedge dp_i) &= \frac{\partial H}{\partial p_i} dp_i - \frac{\partial H}{\partial q^i} dq^i \\ &= i_{X_H} (dq^i) dp_i - dq^i i_{X_H} (dp_i) \\ X_H^a = \dot{x}^a &\Rightarrow \dot{q}^i dp_i - \dot{p}_i dq^i \end{aligned}$$

CHARGE ALGEBRA

LET: $X, Y \in \Gamma$ (TWO VECTOR FIELDS, ISOMETRIES)
 $\mathcal{E} = [X, Y]$ (COMPOSITION OF TWO ISOMETRIES)

$$\begin{aligned} dQ_{\mathcal{E}} = i_{\mathcal{E}}\omega &\Rightarrow \partial_a Q_{[X, Y]} = [X, Y]^b \omega_{ba} \\ &= \mathcal{L}_X Y^b \omega_{ba} = \mathcal{L}_X (Y^b \omega_{ba}) - Y^b \underbrace{\mathcal{L}_X \omega_{ba}}_{=0 \text{ (ISOMETRY)}} \\ &= \mathcal{L}_X (\partial_a Q_Y) \end{aligned}$$

$$dQ_{[X, Y]} = \mathcal{L}_X dQ_Y = (\cancel{i_X d} + d i_X) dQ_Y = d i_X dQ_Y$$

$\downarrow$
 $0 \text{ (} dd=0 \text{)}$

$$\begin{aligned} \partial_a Q_{[X, Y]} &= \partial_a (X^b \partial_b Q_Y) = \partial_a (\delta_c^b X^c \partial_b Q_Y) \\ &= \partial_a (\omega^{bd} \underbrace{\omega_{dc} X^c}_{-i_X \omega_d = -\partial_d Q_X} \partial_b Q_Y) = -\partial_a (\omega^{bd} \partial_d Q_X \partial_b Q_Y) \\ &= \partial_a (\omega^{bc} \partial_b Q_X \partial_c Q_Y) = \partial_a \{Q_X, Q_Y\} \end{aligned}$$

$$\Rightarrow \partial_a Q_{[X, Y]} = \partial_a \{Q_X, Q_Y\}$$

INTEGRATING: $Q_{[X, Y]} = \{Q_X, Q_Y\} + \text{const}$

$\{Q_X, Q_Y\} = Q_{[X, Y]} + C$

CHARGE ALGEBRA

$C = \text{CENTRAL CHARGE}$
 $\partial_a C = 0$

$$\{Q_X, C\} = 0 \text{ (COMMUTES)}$$

- ALGEBRA OF Q_X, C IS CLOSED, BUT C BREAKS SYMMETRY OF Q_X (Q_X DOESN'T CLOSE ALONE)

- C CANNOT BE ELIMINATED by A SHIFT $Q_x \rightarrow Q_x + \text{const.}$

$$[T_a, T_b] = f_{ab}^c T_c \text{ Algebra}$$

$$\text{"SHIFT": } T_a \rightarrow T_a + C_a, \quad [C_a, T_b] = 0$$

$$[T_a, T_b] = f_{ab}^c T_c + \underbrace{C_{ab}}_{f_{ab}^c C_c}, \quad C_{ab} \text{ IS NOT A CENTRAL CHARGE}$$

EXAMPLE (FROM THE FIRST PART OF THE COURSE)

$$x^a = (p_1, p_2, q_1, q_2) \quad (a=1,2,3,4)$$

$$\omega = dp_i \wedge dq_i \quad (i=1,2)$$

$$\omega_{ab} = \begin{pmatrix} 0 & \mathbb{I}_2 \\ -\mathbb{I}_2 & 0 \end{pmatrix}$$

$$L = \frac{1}{2} \dot{q}_1^2 + \dot{q}_1 q_2 + \frac{1}{2} (q_1 - q_2)^2$$

$$\text{FIRST CLASS CONSTRAINT: } p_2 = 0, \quad p_1 - q_1 = 0$$

GENERATOR Q_ε

$$i_\varepsilon \omega = \varepsilon^a \omega_{ab} dx^b = \varepsilon^1 \underbrace{\omega_{13}}_1 dx^3 + \varepsilon^2 \underbrace{\omega_{24}}_1 dx^4 + \varepsilon^3 \underbrace{\omega_{31}}_{-1} dx^1 + \varepsilon^4 \underbrace{\omega_{42}}_{-1} dx^2$$

$$dQ_\varepsilon = i_\varepsilon \omega = \varepsilon^1 dq_1 + \varepsilon^2 dq_2 - \varepsilon^3 dp_1 - \varepsilon^4 dp_2$$

- EXISTENCE OF THE GENERATOR $(d^2=0)$ (ε IS ISOMETRY)

$$ddQ_\varepsilon = 0 \Rightarrow d\varepsilon^1 \wedge dq_1 + d\varepsilon^2 \wedge dq_2 - d\varepsilon^3 \wedge dp_1 - d\varepsilon^4 \wedge dp_2 = 0$$

$$\Rightarrow d\varepsilon^a = 0 \Rightarrow \varepsilon^a \text{ ARE FIELD-INDEPENDENT}$$

$$\Rightarrow Q_\varepsilon = \varepsilon^1 q_1 + \varepsilon^2 q_2 - \varepsilon^3 p_1 - \varepsilon^4 p_2$$

- EXISTENCE OF SYMMETRY

$$\mathcal{L}_\epsilon H = \{Q_\epsilon, H\} = 0$$

$$0 = \epsilon^1 \underbrace{\dot{q}_1}_{p_1 - q_2} + \epsilon^2 \underbrace{\dot{q}_2}_u - \epsilon^3 \underbrace{\dot{p}_1}_{q_1 - q_2} - \epsilon^4 \underbrace{\dot{p}_2}_0$$

(WE USE HAMILTON'S EQS. AND CONSTRAINTS (PRIMARY))

$$\begin{cases} 0 = \epsilon^1 (p_1 - q_2) - \epsilon^3 (q_1 - q_2) + \epsilon^2 u \\ 0 = (\epsilon^1 - \epsilon^3) (q_1 - q_2) \end{cases}$$

$$\Rightarrow \epsilon^2 = 0, \epsilon^4 \rightarrow \text{ARBITRARY}$$

$$\Rightarrow \epsilon^1 = \epsilon^3$$

$$Q_\epsilon = \epsilon^1 (q_1 - p_1) - \epsilon^4 p_2$$

- SYMMETRY TRANSFORMATIONS

$$\delta_\epsilon x^a = \epsilon^a \Rightarrow \begin{aligned} \delta_\epsilon p_1 &= \epsilon^1 \equiv -\lambda \\ \delta_\epsilon p_2 &= 0 \end{aligned} \quad \begin{aligned} \delta_\epsilon q_1 &= \epsilon^1 \equiv -\lambda \\ \delta_\epsilon q_2 &= \epsilon^4 \equiv -\epsilon \end{aligned}$$

THE SAME AS BEFORE WITH $\epsilon^1 = -\lambda$, $\epsilon^4 = -\epsilon$

- IN GENERAL, TO FIND Q_ϵ , WE HAVE TO SOLVE DIFFERENTIAL EQUATIONS

$$i_\epsilon \omega = dQ_\epsilon, \quad \mathcal{L}_\epsilon H = \dot{Q}_\epsilon = 0$$

WITH GIVEN: ω, H

AND UNKNOWN; ϵ, Q_ϵ

GENERALIZATION TO FIELD THEORY

(BIBLIOGRAPHY: G. COMPERE SIMPLIFIED)

CLASSICAL MECHANICS:

FIELD THEORY:

$$x^a = (p_i, q^i) \rightarrow \phi^A(x) \in \Gamma \quad \text{CANONICAL FIELDS} \quad (a=1, \dots, 2N)$$

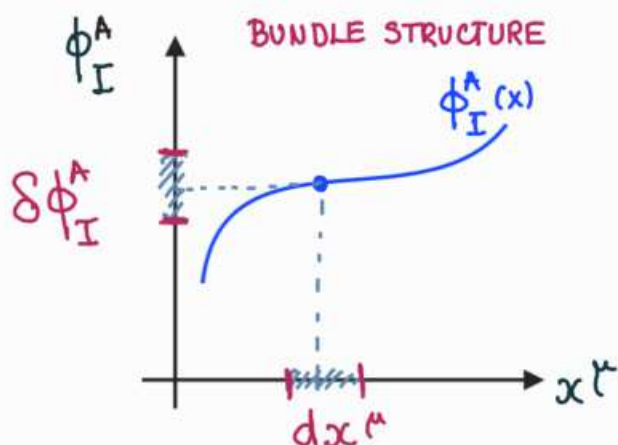
$$t \rightarrow x^\mu \in \mathcal{M} \quad \text{SPACE-TIME} \quad (\mu=0,1,2,3)$$

$$(\mathcal{M}, g_{\mu\nu}) \quad \text{METRIC MANIFOLD;} \\ \text{IN OUR CASE: } g_{\mu\nu}(x) = \eta_{\mu\nu} \\ \text{(MINKOWSKI SPACE)}$$

$$dx^a \rightarrow \begin{matrix} \delta\phi^A & 1\text{-FORM ON } \Gamma \\ dx^\mu & 1\text{-FORM ON } \mathcal{M} \end{matrix}$$

- NOTE THAT NOW WE HAVE TWO INDEPENDENT DIFFERENTIAL MANIFOLDS AND DIFFERENTIAL STRUCTURES: "HORIZONTAL" ON $\mathcal{M}$ AND "VERTICAL" ON Γ
- IN HAMILTONIAN FORMALISM, SPATIAL COORDINATES x^i ($i=1,2,3$) ARE JUST CONTINUOUS INDICES, SO WE CAN EXTEND Γ TO THE SPACE OF ALL FIELDS - **JET SPACE $\mathcal{J}$** - THAT IS x^i INDEPENDENT

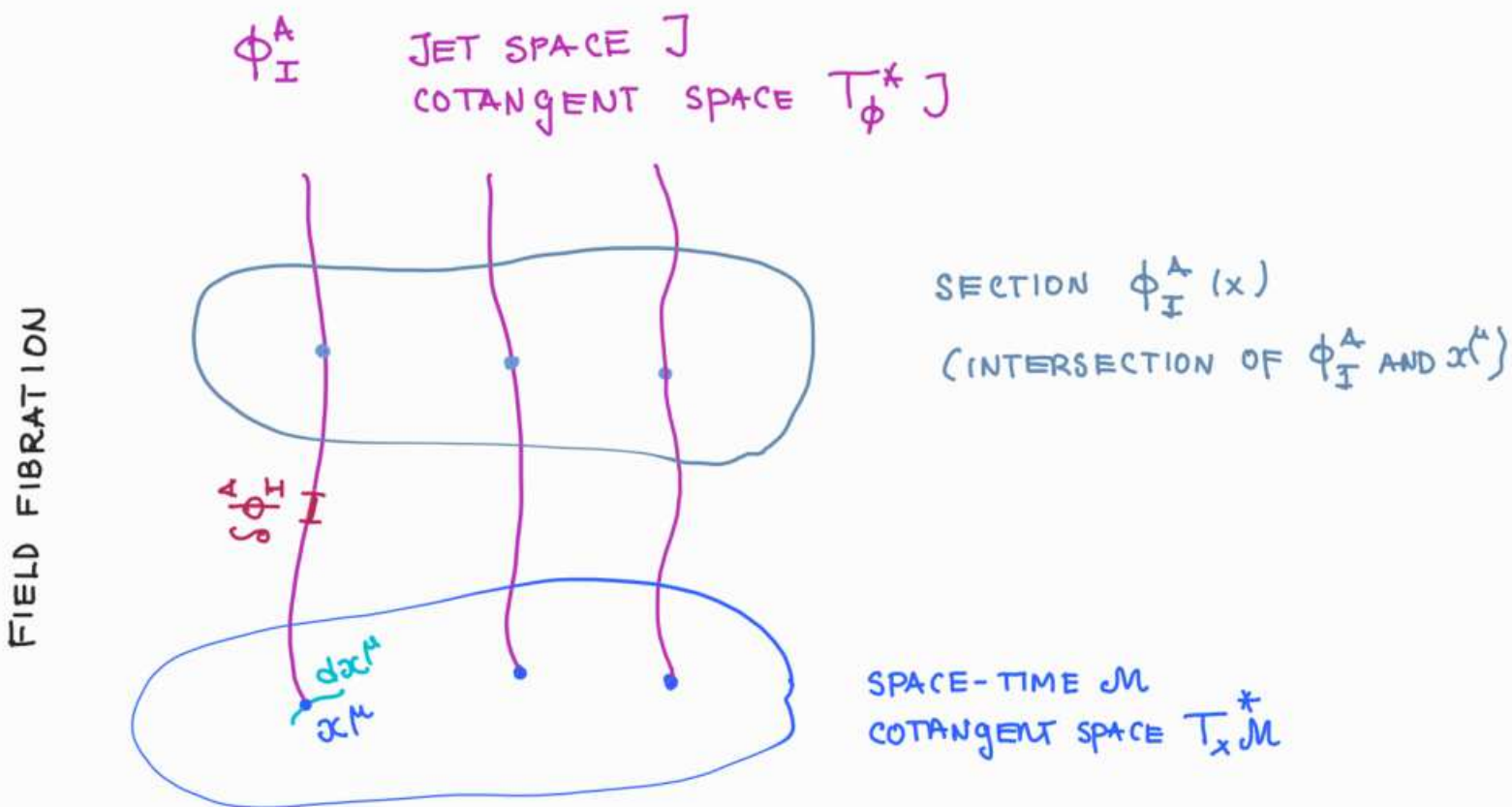
$$\mathcal{J} \ni \phi_I^A = \{ \phi^A, \partial_i \phi^A, \partial_i \partial_j \phi^A, \dots \}$$



HORIZONTAL DERIVATIVE d (ON $\mathcal{M}$)
VERTICAL DERIVATIVE δ (ON Γ)

- COTANGENT SPACE $T_x^* \mathcal{M} \rightarrow$ HAS BASIS $\{dx^\mu\}$ $x^\mu \in \mathcal{M}$
- COTANGENT SPACE $T_\phi^* \mathcal{J} \rightarrow$ HAS BASIS $\{\delta\phi_I^A\}$ $\phi_I^A \in \mathcal{J}$

• VISUALIZATION OF THE FIBRE BUNDLE STRUCTURE



RELATION BETWEEN d AND δ

$$d^2 = 0$$

$$\delta^2 = 0$$

$$d\delta = \delta d$$

- COMMENT: ONE CAN CHOOSE $d\delta = -\delta d$ AND OBTAIN A VARIATIONAL BICOMPLEX (d, δ) , AND DEFINE A (p, q) FORM

$$dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p} \delta\phi_{I_1}^{A_1} \wedge \dots \wedge \delta\phi_{I_q}^{A_q}$$

- WE WILL USE ONLY Γ FOR SIMPLICITY (AND OMIT $\mathcal{J}$)

LEAST ACTION PRINCIPLE

* HAMILTONIAN ACTION CONTAINS INFORMATION ABOUT ω_{ab}

→ SYMPLECTIC ACTION IN CLASSICAL MECHANICS

$$I_H[x] = \int dt (\sigma_a(x) \dot{x}^a - H)$$

$$\sigma_a = \omega_{ab} x^b$$

$$\omega_{ab} = \partial_a \sigma_b - \partial_b \sigma_a$$

→ SYMPLECTIC ACTION IN FIELD THEORY

(ONE EXAMPLE IS CHERN-SIMONS ACTION)

$$I[\Phi] = \int d^4x [\sigma_A(\Phi) \dot{\Phi}^A - H(\Phi)]$$

• FIELD EQUATIONS FROM THE LEAST ACTION PRINCIPLE

NOTATION: $\partial_A = \frac{\partial}{\partial \phi^A}$

$$\begin{aligned} \delta I &= \int d^4x \left(\underbrace{\partial_B \sigma_A}_{\substack{A \quad B}} \delta \phi^B \dot{\Phi}^A + \underbrace{\sigma_A \delta \dot{\Phi}^A}_{(\sigma_A \delta \phi^A)' - \dot{\sigma}_A \delta \phi^A} - \partial_A H \delta \phi^A \right) \\ &= (\sigma_A \delta \phi^A)' - \partial_B \sigma_A \dot{\Phi}^B \delta \phi^A \end{aligned}$$

$$\delta I = \int d^4x \left[\underbrace{(\partial_A \sigma_B - \partial_B \sigma_A)}_{\omega_{AB} = -\omega_{BA}} \dot{\Phi}^B - \partial_A H \right] \delta \phi^A + \oint d^3x \underbrace{\sigma_A \delta \phi^A}_{\tilde{\theta}}$$

$$\omega_{AB} = \partial_A \sigma_B - \partial_B \sigma_A$$

PRE-SYMPLECTIC FORM

$$\tilde{\theta} = \sigma_A \delta \phi^A$$

PRE-SYMPLECTIC
POTENTIAL

1-FORM ON Γ
DENSITY ON $\mathcal{M}$
(THAT'S WHY "~")

$$\delta I = \int d^4x \underbrace{(\omega_{AB} \dot{\phi}^B - \partial_A H)}_{\tilde{E}_A} \delta \phi^A + \oint d^3x \theta$$

$$\tilde{E}_A = \frac{\delta I}{\delta \phi^A} = \omega_{AB} \dot{\phi}^B - \partial_A H = 0$$

EQUATION OF MOTION
(LEAST ACTION PRINCIPLE)

$$\delta I = \int d^4x \tilde{E}_A \delta \phi^A + \oint d^3x \tilde{\theta}$$

ON Γ WE HAVE FORMS, BUT
ON $\mathcal{M}$ NOT — WE WILL
INTRODUCE THEM ON $\mathcal{M}$ AS WELL

DIFFERENTIAL FORMS ON $\mathcal{M}$

- IN $D=4$ DIMENSIONS, EVERY LAGRANGIAN IS A 4-FORM BECAUSE
4-FORMS ARE PROPORTIONAL TO VOLUME ELEMENTS

$$dx^{\mu_1} \wedge \dots \wedge dx^{\mu_4} \equiv d^4x \in \mu_1 \dots \mu_4 \quad \text{4-DIM VOLUME ELEMENT}$$

$$\Rightarrow d^4x \mathcal{L} = L \quad \text{LAGRANGIAN 4-FORM}$$

$$I_{\text{bulk}} = \int d^4x \mathcal{L} = \int L$$

• SIMILARLY: $\tilde{E}_A \delta \phi^A = E_A \delta \phi^A \quad \begin{cases} \phi^A \dots \text{p-FORM} \\ E_A \dots (4-p)\text{-FORM} \end{cases}$

- BOUNDARY TERM $\mathcal{M} \cong \mathbb{R} \times \Sigma \ni (t, x^i)$ TOPOLOGY

$$\tilde{\theta} \equiv \tilde{\theta}^0$$

$$\oint_{\Sigma} d^3x \tilde{\theta} = \int_{\mathcal{M}} d^3x dt \partial_0 \tilde{\theta}^0 = \int_{\mathcal{M}} d^4x \partial_\mu \tilde{\theta}^\mu = \int d\theta$$

LOCALIZED FIELDS:
THERE IS NO FLUX THROUGH
SPATIAL BOUNDARY $\partial \Sigma$

$$\int d^3x \partial_i \tilde{\theta}^i \equiv 0$$

3-FORM

$$\theta = \frac{1}{3!} \theta_{\mu_1 \mu_2 \mu_3} dx^{\mu_1} \wedge dx^{\mu_2} \wedge dx^{\mu_3}$$

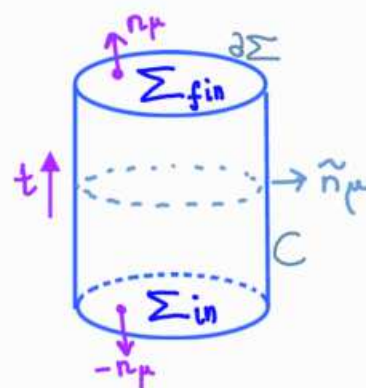
$$\tilde{\theta}^\mu = \frac{1}{3!} \epsilon^{\mu \mu_1 \mu_2 \mu_3} \theta_{\mu_1 \mu_2 \mu_3}$$

ABOUT THE BOUNDARY AND THE STOKES' THEOREM ON $\mathcal{M}$

- CHOICE OF LOCAL COORDINATES

SPACE-TIME $x^\mu = (x^0, x^i)$ $\mathcal{M} = \mathbb{R} \times \Sigma$

$\uparrow$ TIME $\uparrow$ SPACE

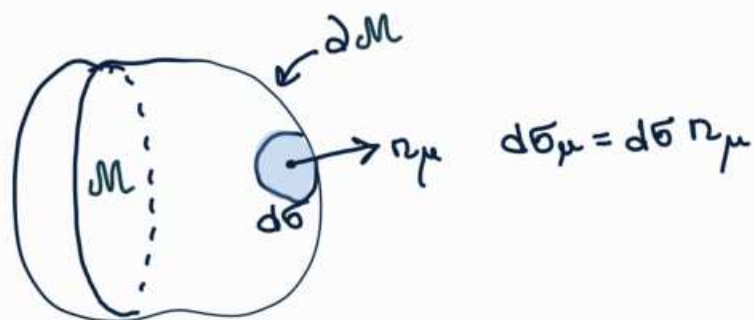


BOUNDARY $\partial\mathcal{M} = \underbrace{\partial\mathbb{R} \times \Sigma}_{\substack{t_{in}, t_{fin} \\ \text{THE CAPS OF} \\ \text{THE CYLINDER}}} + \underbrace{\mathbb{R} \times \partial\Sigma}_{\substack{\text{SIDE SURFACE} \\ C}}$

$\Sigma_{in} \cup \Sigma_{fin}$

STOKES THEOREM LET Λ BE A 3-FORM

$$\int_{\mathcal{M}} d\Lambda = \oint_{\partial\mathcal{M}} \Lambda \iff \int_{\mathcal{M}} \partial_\mu \Lambda^\mu = \oint_{\partial\mathcal{M}} d\sigma_\mu \Lambda^\mu$$



- INFINITESIMAL SURFACE ELEMENT $d\sigma_\mu$ USING DIFF. FORMS

VOLUME $d^4x \in \mu_{1\dots 4} = dx^{\mu_1} \wedge \dots \wedge dx^{\mu_4} / \epsilon_{\mu_1\dots\mu_4}$

$$d^4x = -\frac{1}{4!} \epsilon_{\mu_1\dots\mu_4} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_4}$$

- SURFACE OF CODIMENSION-1 (WITH ONE NORMAL)

$$d\sigma_\mu = -\frac{1}{1!3!} \epsilon_{\mu\mu_1\dots\mu_3} dx^{\mu_1} \wedge dx^{\mu_2} \wedge dx^{\mu_3}$$

- SURFACE OF CODIMENSION-2 (WITH A BINORMAL, TWO NORMALS)

$$d\sigma_{\mu\nu} = -\frac{1}{2!2!} \epsilon_{\mu\nu\mu_1\mu_2} dx^{\mu_1} \wedge dx^{\mu_2}$$

(ETC.)

- NOW WE ARE READY TO WRITE VARIATION OF THE ACTION $I = \int_{\mathcal{M}} L$ USING DIFFERENTIAL GEOMETRY ON Γ AND $\mathcal{M}$

$$\underbrace{\delta I}_{=} = \underbrace{\int_{\mathcal{M}} d^4x \tilde{E}_A \delta\phi^A}_{=} + \underbrace{\int_{\Sigma} d^3x \tilde{\theta}}_{=} = \int_{\mathcal{M}} \delta L = \int_{\mathcal{M}} E_A \wedge \delta\phi^A = \int_{\partial\mathcal{M}} \theta = \int_{\mathcal{M}} d\theta$$

$$\Rightarrow \boxed{\delta L = E_A \wedge \delta\phi^A + d\theta} \quad \text{ACTION VARIATION}$$

THIS IS A VERY IMPORTANT FORMULA THAT IDENTIFIES FUNCTIONAL DERIVATIVE (EULER-LAGRANGE EQUATIONS) $E_A = \frac{\delta L}{\delta\phi^A}$ AND A BOUNDARY TERM

$$\int_{\partial\mathcal{M}} d^3x \tilde{\theta} = \int_{\partial\mathcal{M}} \theta = \int_{\mathcal{M}} d\theta$$

$\tilde{\theta}, \tilde{E}_A \dots$ DENSITIES ON $\mathcal{M}$

$\theta, E_A \dots$ DIFF. FORMS ON $\mathcal{M}$

$$\int_{\mathcal{M}} d^4x \tilde{E}_A \delta\phi^A = \int_{\mathcal{M}} E_A \wedge \delta\phi^A$$

- DEFINE: $\boxed{\Theta = \int_{\partial\mathcal{M}} d^3x \tilde{\theta} = \int_{\partial\mathcal{M}} \theta = \int_{\mathcal{M}} d\theta}$

- TAKE ITS EXTERIOR DERIVATIVE ON Γ

$$\delta\Theta = \delta \int_{\partial\mathcal{M}} d^3x \tilde{\theta}_A \delta\phi^A = \int_{\partial\mathcal{M}} d^3x \delta\tilde{\theta}_A \wedge \delta\phi^A = \int_{\partial\mathcal{M}} d^3x \underbrace{\partial_B \tilde{\theta}_A}_{\delta^2=0} \delta\phi^B \wedge \delta\phi^A = \int_{\partial\mathcal{M}} d^3x \frac{1}{2} (\partial_B \tilde{\theta}_A - \partial_A \tilde{\theta}_B) \delta\phi^B \wedge \delta\phi^A$$

$\frac{1}{2} (\partial_B \tilde{\theta}_A - \partial_A \tilde{\theta}_B) = \frac{1}{2} \omega_{BA}$
ANTISIMMETRIZATION

$$\boxed{\omega = \delta\tilde{\theta}}$$

- SIMPLECTIC FORM IN FIELD THEORY

$$\boxed{\Omega \equiv \delta\Theta = \int_{\partial\mathcal{M}} d^3x \omega} \quad \omega \equiv \frac{1}{2} \omega_{AB} \delta\phi^A \wedge \delta\phi^B \dots \text{2-FORM ON } \Gamma$$

• RECEIPT TO OBTAIN THE SYMPLECTIC FORM

$$\delta L = E_A \wedge \delta \phi^A + d\theta, \quad \theta = G_A \delta \phi^A = \mathcal{G}$$

$$\delta L = E \wedge \delta \phi + \theta$$

$$\Theta = \int_{\partial \mathcal{M}} \theta \rightarrow \Omega = \delta \Theta$$

- Ω IS CLOSED ON Γ BECAUSE ω IS CLOSED ON Γ

$$\left. \begin{aligned} \omega &= \delta \mathcal{G} \\ \delta \omega &= \delta^2 \mathcal{G} = 0 \end{aligned} \right\} \rightarrow \boxed{\delta \omega = 0}$$

- Ω IS ALSO CLOSED ON $\mathcal{M}$ ON-SHELL

$$d\omega = d\delta \mathcal{G} = \delta d\mathcal{G} = \delta d(\cancel{\delta L} - \underbrace{E_A \wedge \delta \phi^A}_{\substack{0 \\ \text{ON-SHELL}}}) = 0$$

$\nearrow 0(\delta^2=0)$

- Ω HAS TO BE NON-DEGENERATE

$$i_{\xi} \Omega = 0 \Rightarrow \xi = 0$$

$$\Omega_{AB} \xi^B = 0 \Rightarrow \xi^B = 0$$

→ IN FIELD THEORY, THE DEGENERACY USUALLY REFLECTS GAUGE REDUNDANCY, AND Ω IS ALSO CALLED PRESYMPLECTIC FORM BEFORE ELIMINATING GAUGE ORBITS

EXAMPLE: ELECTROMAGNETISM $\delta_{\mathcal{E}} A_{\mu} = \partial_{\mu} \mathcal{E}$

* IF $\mathcal{E}$ IS A **PROPER GAUGE PARAMETER**, $\mathcal{E}|_{\partial \mathcal{M}} = 0$, THEN $\delta_{\mathcal{E}} A_{\mu}$ IS A **DEGENERATE DIRECTION**

* IF $\mathcal{E}$ IS **IMPROPER**, $\mathcal{E}|_{\partial \mathcal{M}} \neq 0$, WE HAVE **ASYMPTOTIC SYMMETRY**

- AMBIGUITY IN DEFINITION OF Ω

$$\delta\omega = 0 \Rightarrow \omega \sim \omega + d\alpha$$

$$d^2\alpha = 0$$

(TWO PRESIMPLECTIC FORMS ARE EQUIVALENT
GIVE THE SAME Ω)

$$\Omega = \delta\Theta \Rightarrow \Theta \sim \Theta + \delta\beta$$

(CORRESPOND TO THE SAME Ω)

- THIS AMBIGUITY CAN BE USED TO REMOVE UNWANTED FLUX THROUGH THE SPATIAL BOUNDARY $\partial\Sigma$

$\Rightarrow$ THE ONLY BOUNDARY ARE THE CUPS $\Sigma_{in} \cup \Sigma_{fin}$

SYMMETRY GENERATORS — COVARIANT FORMULATION

* IN CLASSICAL MECHANICS

- **SIMPLECTIC VECTOR FIELD** $\xi = \xi^a \partial_a$
(ISOMETRY)
- **CHARGE** $i_\xi \omega = dQ_\xi$
CONTRACTION $i_\xi = \xi^a \frac{\partial}{\partial (dx^a)}$
- **LOCAL TRANSFORMATION** $\delta_\xi F = \{F, Q_\xi\} = -i_\xi dF$ (*)

* IN FIELD THEORY

$\delta_\xi \phi^A$ KNOWN LOCAL TRANSFORMATIONS THAT
LEAVE $\bar{I}[\phi]$ INVARIANT

• SIMPLECTIC VECTOR FIELD

$$X_\varepsilon \equiv - \int d^4x \, \delta_\varepsilon \phi^A(x) \frac{\delta}{\delta \phi^A(x)}$$

COMPONENT :

$$X_\varepsilon^A \equiv - \delta_\varepsilon \phi^A$$

}

DEFINITION CONSISTENT WITH THE ONE IN CLASSICAL MECHANICS (*) BECAUSE :

$$\delta_\varepsilon \phi^A = - X_\varepsilon \cdot \delta \phi^A \quad (\text{CONTRACTION})$$

PROOF:
$$X_\varepsilon \cdot \delta \phi^A = \int d^4x' \, X_\varepsilon^B(x') \underbrace{\frac{\partial}{\partial \delta \phi^B(x)} \delta \phi^A(x)}_{\delta_B^A \delta^{(4)}(x-x')} = X_\varepsilon^A(x)$$

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2) DANIEL HARLOW, JIE-QIANG WU

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3) C. CRNKOVIC, E. WITTEN (1986)

"COVARIANT DESCRIPTION OF CANONICAL FORMALISM IN GEOMETRICAL THEORIES"

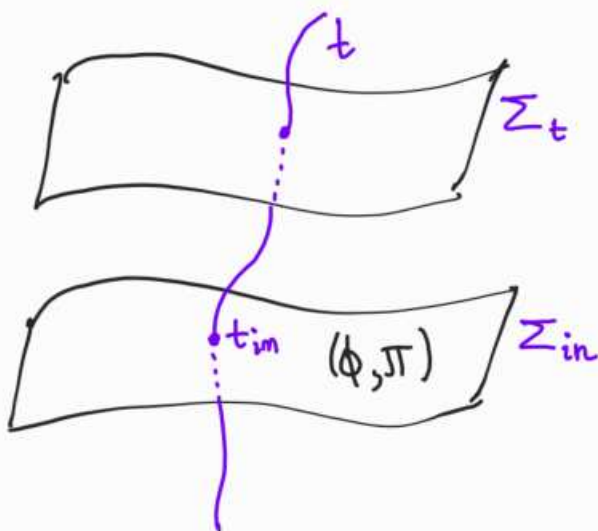
COVARIANT FORMULATION OF SINGULAR AND NON-SINGULAR SYSTEMS

- STANDARD HAMILTONIAN PHASE SPACE IS NOT EXPLICITLY COVARIANT BECAUSE IT SEPARATES TIME AND SPACE: $x^\mu = (t, x^i)$, $\mathcal{M} \sim \mathbb{R} \times \Sigma$
- IN SYMPLECTIC GEOMETRY, $\mathbb{I}$ AND $\delta\mathbb{I}$ ARE IMPORTANT, NOT DEFINITION OF $\mathbb{I}_A(x)$ THAT ENTERS IMPLICITLY IN ω_{AB}
- IT TURNS OUT, THE FORMALISM BASED ON SYMPLECTIC GEOMETRY IS VALID FOR ANY $\mathbb{I}$
- WE ONLY HAVE TO REPLACE PHASE SPACE BY JET SPACE GENERALIZED AS $\phi_I^\Delta = \{\phi^A, \partial_\mu \phi^A, \partial_\mu \partial_\nu \phi^A, \dots\}$
- USING INTEGRALS BY PARTS, ANY LAGRANGIAN L HAS VARIATION δL THAT CAN BE WRITTEN AS

$$\delta L = E \wedge \delta \phi + d\theta$$

$\phi \dots$ LAGRANGIAN FIELDS
(WITHOUT MOMENTA)

- ⊙ HAMILTONIAN PHASE SPACE IS THE SPACE OF INITIAL DATA ON CHOSEN SPATIAL HYPERSURFACE Σ_{in} : $(\phi(x), \mathbb{I}(x))$
- ⊙ COVARIANT PHASE SPACE IS THE SPACE OF SOLUTIONS OF EQUATIONS OF MOTION: $(\phi(x), E(\phi) = 0)$



HAMILTONIAN PHASE SPACE

→ IF THE INITIAL CONDITIONS PROBLEM IS WELL-DEFINED, THERE IS 1-1 RELATION BETWEEN INITIAL CONDITIONS AND SOLUTIONS

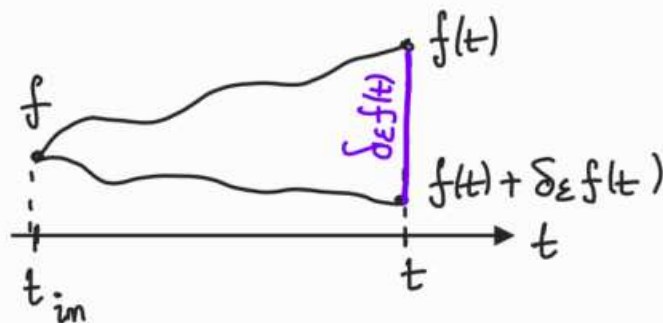
EXAMPLE MASSIVE NON RELATIVISTIC PARTICLE

$$I = \frac{1}{2} m \int dt \dot{x}^2 \Rightarrow m\ddot{x} = 0 \Rightarrow x(t) = \frac{p_0}{m} t + x_0$$

(p_0, x_0) ... COVARIANT PHASE SPACE

$(p(x), x(t))$... HAMILTONIAN PHASE SPACE

→ SYSTEMS WITH LOCAL SYMMETRIES DON'T HAVE WELL-DEFINED INITIAL CONDITIONS



→ THEREFORE, WE HAVE TO DISTINGUISH IN A COVARIANT FORMALISM PRE-PHASE SPACE $\tilde{\Gamma}_{cov}$ AND PHASE SPACE Γ_{cov}

$$\tilde{\Gamma}_{cov} = \text{ALL SOLUTIONS}$$

$$\Gamma_{cov} = \text{ALL INEQUIVALENT SOLUTIONS}$$

$$= \frac{\text{ALL SOLUTIONS}}{\text{EQUIVALENT SOLUTIONS}} \quad (\text{quotienting gauge orbits})$$

→ FIRST-CLASS CONSTRAINTS IN HAMILTONIAN FORMALISM ARE DEGENERATE DIRECTIONS OF THE COVARIANT PRESYMPECTIC FORM

→ SECOND-CLASS CONSTRAINTS ARE NOT DEGENERATE DIRECTIONS AND ARE ELIMINATED BY RESTRICTION (NOT BY QUOTIENTING)

COMPARISON

HAMILTONIAN METHOD

SPACE OF INITIAL DATA

REQUIRES THE SPLIT OF SPACE AND TIME

VARIABLES ARE (ϕ, π)

CONSTRAINTS APPEAR DIRECTLY

SYMPLECTIC FORM IS CONSTRUCTED FROM ϕ AND π

LOCAL SYMMETRIES CAN BE COMPUTED

EVOLUTION IS GENERATED BY HAMILTONIAN

COVARIANT PHASE SPACE METHOD

SPACE OF SOLUTIONS

KEEPS COVARIANCE MANIFEST

VARIABLES ARE FIELDS ϕ SOLVING EQUATIONS OF MOTION

GAUGE DIRECTIONS APPEAR AS DEGENERACIES OF Ω

SYMPLECTIC FORM IS CONSTRUCTED FROM THE ON-SHELL BOUNDARY TERM

LOCAL SYMMETRIES ARE GIVEN

EVOLUTION IS CONTAINED IN THE SOLUTION

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- R. JHA, INTRODUCTION TO HAMILTONIAN FORMULATION OF GENERAL RELATIVITY AND HOMOGENEOUS COSMOLOGIES, SciPost Phys. LECT. NOTES 73 (2023)
- E. BERTSCHINGER, HAMILTONIAN FORMULATION OF GENERAL RELATIVITY

* WE HAVE SHOWN THAT ALL THEORIES INVARIANT UNDER REPARAMETRIZATIONS HAVE $H_C = 0$

* GRAVITY IS INVARIANT UNDER REPARAMETRIZATIONS (GENERAL COORDINATE TRANSFORMATIONS), i.e.,

SPACETIME DIFFEOMORPHISMS:

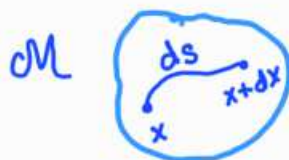
$$x^\mu \rightarrow x'^\mu = x'^\mu(x) = x^\mu + \underbrace{\xi^\mu(x)}_{\text{ARBITRARY FUNCTIONS}}$$

 $\Rightarrow$

$$H_C = 0$$

$$H_T = \text{CONSTRAINTS}$$

GRAVITATIONAL HAMILTONIAN

GENERAL RELATIVITYLINE ELEMENT: ds

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu$$

$g_{\mu\nu}(x)$ METRIC, FUNDAMENTAL FIELD

$g_{\mu\nu} = g_{\nu\mu}$ SYMMETRIC

$g = \det[g_{\mu\nu}] \neq 0$ NON-DEGENERATE; SIGNATURE $(-+++)$

$g^{\mu\nu}$... INVERSE METRIC

$$g < 0$$

- RIEMANN CURVATURE TENSOR "FIELD STRENGTH"

$$R^\mu{}_{\nu\lambda\rho} = \partial_\lambda \Gamma^\mu_{\nu\rho} - \partial_\rho \Gamma^\mu_{\nu\lambda} + \Gamma^\mu_{\sigma\lambda} \Gamma^\sigma_{\nu\rho} - \Gamma^\mu_{\sigma\rho} \Gamma^\sigma_{\nu\lambda}$$

- CHRISTOFFEL SYMBOL (SYMMETRIC: $\Gamma^\mu_{\alpha\beta} = \Gamma^\mu_{\beta\alpha}$)

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} (\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta})$$

NOT LORENTZ
TENSOR

- RICCI TENSOR (SYMMETRIC: $R_{\mu\nu} = R_{\nu\mu}$)

$$R_{\mu\nu} = R^\alpha{}_{\mu\alpha\nu}$$

- RICCI SCALAR

$$R = R^\mu{}_\mu = g^{\mu\nu} R_{\mu\nu}$$

* GRAVITATIONAL ACTION CONSTRUCTION:

→ KINETIC TERM QUADRATIC IN VELOCITIES, i.e., $(\partial g)^2$

→ LORENTZ-COVARIANT $\Gamma \sim \partial g$
 $R \sim \partial^2 g, \Gamma^2 \sim (\partial g)^2$

$\Rightarrow$ LINEAR IN R

→ LORENTZ-SCALAR $\Rightarrow$ UNIQUE POSSIBILITY $\mathcal{L}_{\text{grav}} \propto R$

- EINSTEIN'S GRAVITY : **EINSTEIN-HILBERT ACTION**

$$\mathcal{I}_{\text{EH}}[g] = \underbrace{\frac{1}{16\pi G}}_{\text{GRAVITATIONAL CONSTANT}} \int d^4x \sqrt{-g} R + \mathcal{I}_{\text{matter}}[g, \Psi]$$

G = NEWTON'S CONSTANT, FROM THE REQUIREMENT
 THAT THE FIELD EQUATIONS REPRODUCE NEWTONIAN
 GRAVITY IN THE WEAK-FIELD, SLOW-MOTION LIMIT

$$g_{00} \approx - \left(1 + \frac{2}{c^2} \Phi \right), \quad \nabla^2 \Phi = 4\pi G \underbrace{\rho_{\text{matter}}}_{\text{MASS DENSITY}}$$

* BEFORE PASSING TO HAMILTONIAN FORMALISM, WE HAVE TO DO TWO THINGS TO "PREPARE" THE ACTION

1) EH LAGRANGIAN DEPENDS ON ACCELERATIONS

$\ddot{g}_{\mu\nu} = \partial_0 \partial_0 g_{\mu\nu}$; WE HAVE TO ADD BOUNDARY TERMS TO ELIMINATE THEM, SO THAT THE LAGRANGIAN DEPENDS ONLY ON THE VELOCITIES $\dot{g}_{\mu\nu} = \partial_0 g_{\mu\nu}$

• CLASSICAL MECHANICS ANALOGUE

$$I = - \int d^4x \frac{1}{2} m \ddot{q}^2 \rightarrow \text{NOT ADAPTED TO HAMILTONIAN FORMALISM}$$

$$\begin{aligned} I_B &= I + \frac{1}{2} \oint d^3x m \dot{q}^2 \\ &= I + \frac{1}{2} \int d^4x (m \dot{q}^2)' = \int d^4x \frac{1}{2} m \dot{q}^2 \quad \checkmark \quad (\text{ADAPTED}) \end{aligned}$$

• EH ACTION: **GIBBONS-HAWKING TERM** ELIMINATES ACCELERATIONS FROM $\mathcal{L}_{\text{EH}}$

• WE WILL ASSUME:

→ PURE GRAVITY $I_{\text{matter}} = 0$

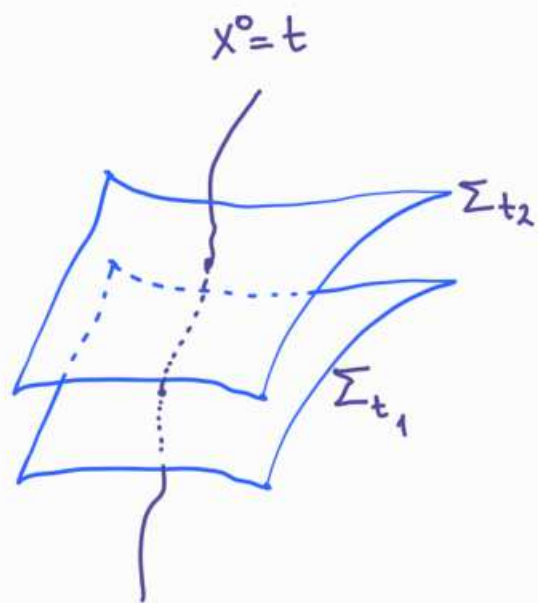
→ BOUNDARY TERMS NEGLECTED

2) **REFERENCE FRAME CHOICE**

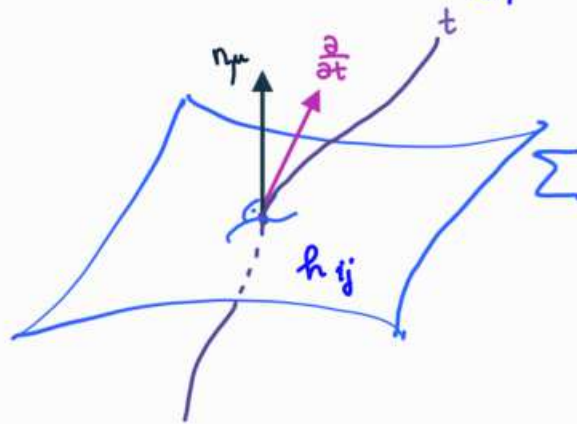
BECAUSE WE ARE IN CURVED SPACETIME

$$\mathcal{M} \sim \mathbb{R} \times \Sigma \quad x^\mu = (x^0, x^i)$$

THE TEMPORAL COORDINATE $x^0 \in \mathbb{R}$ IS NOT ORTHOGONAL TO THE SPATIAL SURFACE $\Sigma \ni x^i$



h_{ij} = INDUCED METRIC ON Σ
($g_{\mu\nu}$ with $t = \text{const.}$)



η_μ ... NORMAL VECTOR AT Σ

$\frac{\partial}{\partial t}$... TANGENT VECTOR OF THE CURVE t

η_μ AND $\frac{\partial}{\partial t}$ ARE NOT PARALLEL

HAMILTONIAN FORMALISM NEEDS
A SPACELIKE FOLIATION BY
HYPERSURFACES Σ_t OF $t = \text{const.}$

• WE HAVE TO INTRODUCE VARIABLES WHERE $\mathbb{R}$ AND Σ ARE
ORTHOGONAL $\Rightarrow$ **ADM FORMALISM (ARNOWITT, DESER, MISNER)**

• WE WILL FOCUS ON EH ACTION IN 3+1 DIMENSIONS,
BUT ALL LAGRANGIANS $\mathcal{L} = \mathcal{L}(g, \dot{g})$ HAVE THE SAME
HAMILTONIAN STRUCTURE, ONLY THE CONSTRAINTS CAN
HAVE DIFFERENT FORM

ADM FORMALISM $x^\mu = (t, x^i)$ 3+1 DIMENSIONS

• FIRST WE WILL REPLACE METRIC COMPONENTS BY ADM VARIABLES

$$ds^2 = g_{00} dt^2 + 2g_{0i} dt dx^i + g_{ij} dx^i dx^j$$

10 COMPONENTS
 g_{00}, g_{0i}, g_{ij}

$$1 + 3 + \frac{3 \cdot 4}{2} = 10$$

• **INDUCED METRIC ON Σ :**

$$ds^2|_{dt=0} = h_{ij} dx^i dx^j \Rightarrow \boxed{h_{ij} = g_{ij}}$$

• HOWEVER IF INVERSE METRICS ARE:

$$\left. \begin{array}{l} g_{\mu\nu} \rightarrow g^{\mu\nu} \\ h_{ij} \rightarrow h^{ij} \end{array} \right\}$$

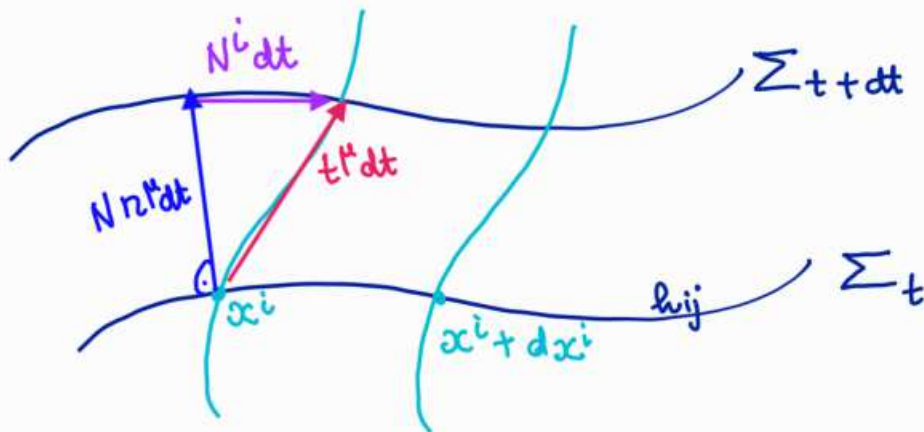
WE HAVE $h^{ij} \neq g^{ij}$
BECAUSE $g_{0i} \neq 0$

$$g_{i\mu} g^{\mu j} = \underbrace{g_{i0} g^{0j}}_{\text{WHEN } \neq 0, h^{ij} \neq g^{ij}} + g_{ik} g^{kj} = \delta_i^j$$

- WE DEFINE INVERSE INDUCED METRIC

$$h_{ik} h^{kj} = \delta_i^j$$

- THE METRIC h_{ij}, h^{ij} RISES AND LOWERS INDICES ON Σ
- FURTHERMORE, WE INTRODUCE GEOMETRIC QUANTITIES N^i, N INSTEAD OF g_{00}, g_{0i} IN THE FOLLOWING WAY:



$t^\mu \dots$ TANGENT VECTOR ON THE CURVE $t = x^0$

$n^\mu \dots$ NORMAL VECTOR

$N =$ LAPSE FUNCTION

$N^i =$ SHIFT FUNCTION

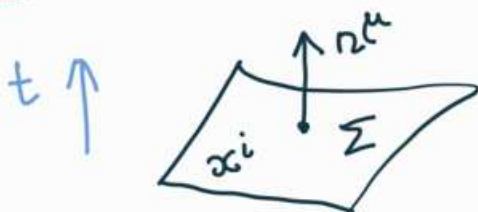
- ADM VARIABLES ARE

$$(N, N^i, h_{ij}) \text{ EQUIVALENT TO } g^{\mu\nu}$$

$$(N, N_i, h_{ij}) \text{ EQUIVALENT TO } g_{\mu\nu}$$

$$\text{HERE } N^i = h^{ij} N_j$$

* **LAPSE** IS A NORMAL VECTOR, IT DOESN'T HAVE COMPONENTS ON Σ



$$n_i = 0$$

$$n_\mu = (n_0, n_i) = (-N, 0)$$

THE SIGN ASSURES THAT n^μ POINTS IN A POSITIVE DIRECTION

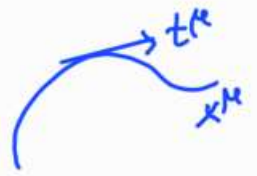
- n_μ IS AN UNIT TIMELIKE VECTOR:

$$-1 = g^{\mu\nu} n_\mu n_\nu = g^{00} N^2 \Rightarrow$$

$$N = \frac{1}{\sqrt{-g^{00}}}$$

* SHIFT

$$t^\mu = \frac{\partial x^\mu}{\partial t} \dots \text{TANGENT VECTOR}$$



• DECOMPOSITION;

$$t^\mu = N n^\mu + N^\mu$$

NORMAL
COMPONENT
 δ^μ_0

PARALLEL COMPONENT δ^μ_i ,
HAS ONLY $N^i \neq 0$

$$N^\mu = (N^0, N^i) = (0, N^i)$$

(IT IS ON Σ)

• ORTHOGONALITY: $n_\mu N^\mu = 0$

• LET US CHOOSE ORTHONORMAL BASIS ON Σ : e_a ($a=1,2,3$)

$$n_\mu e_a^\mu = 0$$

(ORTHOGONALITY)

$$e_a \cdot e_b = e_a^\mu e_{b\mu} = \delta_{ab}$$

• IN THIS BASIS: $N^\mu = e_a^\mu N^a$

$$t^\mu = N n^\mu + N^a e_a^\mu$$

• ADAPTED COORDINATES: (t, dy^a)

$$e_a^\mu = \frac{\partial x^\mu}{\partial y^a}, \quad t^\mu = \frac{\partial x^\mu}{\partial t} \Rightarrow dx^\mu = t^\mu dt + e_a^\mu dy^a$$

$$\begin{aligned} dx^\mu &= (N n^\mu + N^a e_a^\mu) dt + e_a^\mu dy^a \\ &= N n^\mu dt + (N^a dt + dy^a) e_a^\mu \end{aligned}$$

• METRIC $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$

$$= g_{\mu\nu} [N n^\mu dt + (N^a dt + dy^a) e_a^\mu] [N n^\nu dt + (N^b dt + dy^b) e_b^\nu]$$

$$= \underbrace{g_{\mu\nu} n^\mu n^\nu}_{-1} N^2 dt^2 + 2 \underbrace{(g_{\mu\nu} N n^\mu)}_{\downarrow} dt (N^b dt + dy^b) \underbrace{(e_b^\nu)}_{\downarrow}$$

$$n_\nu e_b^\nu = \frac{1}{N} e_b^0 = 0$$

$$+ \underbrace{g_{\mu\nu} e_a^\mu e_b^\nu}_{h_{ab}} (N^a dt + dy^a) (N^b dt + dy^b)$$

(PROJECTION OF $g_{\mu\nu}$ TO Σ)

$$\Rightarrow ds^2 = -N^2 dt^2 + h_{ab} (N^a dt + dy^a) (N^b dt + dy^b)$$

OR IN GENERAL COORDINATES:

$$ds^2 = -N^2 dt^2 + h_{ij} (N^i dt + dx^i) (N^j dt + dx^j)$$

$\Downarrow$

WE RECOGNIZE

$$\begin{aligned} g_{00} &= -N^2 + N_i N^i \\ g_{0i} &= N_i \\ g_{ij} &= h_{ij} \end{aligned}$$

• ADM DECOMPOSITION OF THE METRIC:

$$g_{\mu\nu} = \begin{pmatrix} N_i N^i - N^2 & N_j \\ N_i & h_{ij} \end{pmatrix}$$

• DETERMINANT

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(D) \det(A - BD^{-1}C)$$

$$\Rightarrow g = \det[g_{\mu\nu}] = \underbrace{\det[h_{ij}]}_{\equiv h} (\cancel{N_i N^i - N^2} - \cancel{N_j h^{ij} N_i}) = -N^2 h$$

$$\sqrt{-g} = N \sqrt{h}$$

$$h_{ij} \sim (+ + +)$$

• INVERSE METRIC $\delta^\mu_\nu = g^{\mu\alpha} g_{\alpha\nu}$

$$g^{00} = -\frac{1}{N^2}, \quad g^{0i} = \frac{N^i}{N^2}, \quad g^{ij} = h^{ij} - \frac{N^i N^j}{N^2}$$

$$g^{\mu\nu} = \begin{pmatrix} -1/N^2 & N^i/N^2 \\ N^i/N^2 & h^{ij} \end{pmatrix}$$

3+1 DECOMPOSITION OF THE INVERSE METRIC

$$N = \frac{1}{\sqrt{-g_{00}}}, \quad N^i = N^2 g^{0i}$$

LAPSE y SHIFT

(SE USA FRECUENTAMENTE COMO LA DEFINICIÓN)

• SHOW THAT:

$$m_{\mu} m^{\mu} = -1, \quad m_{\mu} = (-N, 0) \Rightarrow m^{\mu} = \left(\frac{1}{N}, -\frac{N^i}{N} \right)$$

EQUATIONS OF MOTION IN GENERAL RELATIVITY

• WE VARY THE EH ACTION AND USE THE ACTION PRINCIPLE

• WE USE IDENTITIES

$$1) \quad \delta \sqrt{-g} = \sqrt{-g} \frac{1}{2} g^{\mu\nu} \delta g_{\mu\nu} = -\frac{1}{2} \sqrt{-g} \delta g^{\mu\nu} g_{\mu\nu}$$

(IT FOLLOWS FROM $\det A \equiv \exp(\text{Tr} \ln A)$)

$$2) \quad \delta R^{\mu}{}_{\nu\alpha\beta} = \nabla_{\alpha} \delta \Gamma^{\mu}_{\nu\beta} - \nabla_{\beta} \delta \Gamma^{\mu}_{\nu\alpha}$$

(IT FOLLOWS FROM $\nabla_{\mu} V^{\alpha} = \partial_{\mu} V^{\alpha} + \Gamma^{\alpha}_{\mu\nu} V^{\nu}$
 $\nabla_{\mu} V_{\alpha} = \partial_{\mu} V_{\alpha} - \Gamma^{\nu}_{\mu\alpha} V_{\nu}$)

• THEN WE OBTAIN:

$$\delta I_{EH} = \frac{1}{16\pi G} \int d^4x \delta(\sqrt{-g} R)$$

$$= \frac{1}{16\pi G} \int d^4x \sqrt{-g} \underbrace{\left(-\frac{1}{2} \delta g^{\mu\nu} g_{\mu\nu} R + \delta g^{\mu\nu} R_{\mu\nu} \right)}_{\delta g^{\mu\nu} G_{\mu\nu}} + \nabla_{\alpha} \Omega^{\alpha}$$

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

EINSTEIN TENSOR

$$\Omega^{\alpha} \equiv g^{\mu\nu} \delta \Gamma^{\alpha}_{\mu\nu} - g^{\mu\alpha} \delta \Gamma^{\nu}_{\mu\nu} \quad (\text{IT WILL GIVE A BOUNDARY TERM IN THE ACTION})$$

$\Rightarrow$

$$G_{\mu\nu} = 0$$

EINSTEIN EQUATION
(WITHOUT THE MATTER)

QUANTITIES ON Σ

- WE HAVE TO DEFINE THE CURVATURE $R^{\mu\nu\alpha\beta}(g)$ IN TERMS OF QUANTITIES THAT LIVE ON Σ

$$g_{\mu\nu} \rightarrow \{N, N_i, h_{ij}\}$$

$$\partial_\mu \rightarrow \{\partial_t, \partial_i\} \quad \partial_t \text{ IS PROBLEMATIC, NOT ON } \Sigma$$

$$R^{\mu\nu\alpha\beta} \rightarrow ? \quad \text{SHOULD BE COVARIANT ON } \Sigma$$

* INTRINSIC CURVATURE ON Σ

$$\mathcal{R}^i{}_{jkl} = \mathcal{R}^i{}_{jkl}(h)$$

CURVATURE CONSTRUCTED FROM $h_{ij}, \partial_k h_{ij}, \partial_k \partial_l h_{ij}$
(IT DOESN'T CONTAIN $\frac{\partial}{\partial t}$)

* EXTRINSIC CURVATURE ON Σ

(DESCRIBES HOW A HYPERSURFACE IS CURVED INSIDE THE SURROUNDING SPACETIME)

$n_\mu \dots$ NORMAL TO Σ

$$K_{\mu\nu} = -\nabla_\mu n_\nu = -\partial_\mu n_\nu + \Gamma_{\nu\mu}^\alpha n_\alpha$$

- ONLY K_{ij} IS RELEVANT

$$K_{ij} = \frac{1}{2N} (\mathcal{L}_i N_j + \mathcal{L}_j N_i - \dot{h}_{ij})$$

$$\mathcal{L}_i K_{ij} = K_{ji} \quad (\text{SYMMETRIC})$$

$\mathcal{L}_i \dots$ COV. DERIVATIVE ON Σ

$\Rightarrow$ IN THAT WAY, $\partial_t h_{ij} = \dot{h}_{ij}$ CAN BE REPLACED BY K_{ij} (IN A COVARIANT WAY)

PROOF

$$\begin{aligned}
 K_{ij} &= -\nabla_i n_j = -\underbrace{\partial_i n_j}_{\delta} + \underbrace{\Gamma_{ij}^\mu}_{-N\delta_\mu^0} n_\mu \\
 &= -N \Gamma_{ij}^0 = -\frac{1}{2} N g^{00} (\partial_i g_{0j} + \partial_j g_{0i} - \partial_0 g_{ij}) \\
 &\quad - \frac{1}{2} N g^{0k} (\partial_i g_{kj} + \partial_j g_{ki} - \partial_k g_{ij}) \\
 &\quad \frac{N^k}{N^2} = \frac{N_e}{N^2} h^{ek}
 \end{aligned}$$

NOTE THAT

$$\boxed{\Gamma_{ij}^0 = -\frac{1}{N} K_{ij}}$$

$$\begin{aligned}
 K_{ij} &= \frac{1}{2N} (\partial_i N_j + \partial_j N_i - \dot{h}_{ij}) - \frac{N_e}{N} \underbrace{\frac{1}{2} h^{ek} (\partial_i h_{kj} + \partial_j h_{ki} - \partial_k h_{ij})}_{\gamma_{ij}^k(h)} \\
 &\quad \text{CHRISTOFFEL SYMBOL AT } \Sigma
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2N} (\partial_i N_j + \partial_j N_i - \dot{h}_{ij}) - \frac{1}{N} \underbrace{\gamma_{ij}^k N_k}_{= \frac{1}{2} (\gamma_{ij}^k N_k + \gamma_{ji}^k N_k)} \\
 &\quad \text{(DUE TO SYMMETRY)}
 \end{aligned}$$

$$\boxed{K_{ij} = \frac{1}{2N} (\mathfrak{D}_i N_j + \mathfrak{D}_j N_i - \dot{h}_{ij})}$$

$$\mathfrak{D}_i N_j = \partial_i N_j - \gamma_{ij}^k N_k$$

• WE HAVE THE FOLLOWING 3+1 DECOMPOSITION :

$$\begin{aligned}
 x^\mu &\rightarrow (x^0, x^i) \\
 g_{\mu\nu} &\rightarrow (N, N^i, h_{ij}) \\
 \partial_\alpha g_{\mu\nu} &\rightarrow (K_{ij}, R_{jke}^i(h))
 \end{aligned}$$

GAUSS - CODAZZI RELATIONS

* 3+1 DECOMPOSITION OF THE RIEMANN TENSOR $R^\mu{}_{\nu\alpha\beta}$

$$R^0{}_{ijk} = -\frac{1}{N} (\mathcal{L}_j K_{ik} - \mathcal{L}_k K_{ij})$$

$$R^i{}_{jkl} = \mathcal{R}^i{}_{jkl} - R^0{}_{jke} N^e + K^i{}_k K_{je} - K^i{}_e K_{jk}$$

$$R^0{}_{ioj} = -\frac{1}{N} \dot{K}_{ij} - K_i{}^k K_{jk} - \frac{1}{N} \mathcal{L}_i \mathcal{L}_j N + \\ + \frac{1}{N} \mathcal{L}_j (N^k K_{ik}) + \frac{1}{N} K_{jk} \mathcal{L}_i N^k$$

$$R^k{}_{ioj} = \mathcal{R}^k{}_{iej} N^e + (R^0{}_{iej} N^e - R^0{}_{ioj} N^k) \\ + N (\mathcal{L}^k K_{ij} - \mathcal{L}_i K^k{}_j)$$

ADM FORM OF THE EH ACTION

$$\frac{1}{16\pi G} \equiv 1$$

(WE CHOOSE THIS NORMALIZATION)

$$R = K_{ij} K^{ij} - K^2 + \mathcal{R} - \underbrace{\frac{2}{N\sqrt{h}} (\sqrt{h} K)^{\cdot} + \frac{2}{N\sqrt{h}} \partial_i [\sqrt{h} (KN^i - \mathcal{L}^i N)]}_{\text{BOUNDARY TERM IN } \int d^4x \sqrt{-g} (\dots) = \int d^4x N\sqrt{h} (\dots)}$$

$$\sqrt{-g} R = N\sqrt{h} (K_{ij} K^{ij} - K^2 + \mathcal{R}) + \text{TERMS THAT WILL BECOME BOUNDARY TERMS AND WE NEGLECT THEM}$$

$$K \equiv K^i{}_i = h^{ij} K_{ij} \quad (\text{TRACE})$$

$$\mathcal{I}_{\text{ADM}} [N, N_i, h_{ij}] = \int d^4x \mathcal{L}_{\text{ADM}}$$

ADM ACTION

$$\mathcal{L}_{\text{ADM}} = N\sqrt{h} (\mathcal{R} + K_{ij} K^{ij} - K^2)$$

HAMILTONIAN FORMULATION

FUNDAMENTAL FIELDS (N, N^i, h_{ij}) VELOCITIES $(\dot{N}, \dot{N}^i, \dot{h}_{ij})$
CANONICAL MOMENTA (π, π_i, π^{ij})

$$\pi = \frac{\partial \mathcal{L}_{ADM}}{\partial \dot{N}} = 0$$

$$\pi_i = \frac{\partial \mathcal{L}_{ADM}}{\partial \dot{N}^i} = 0$$

$$\pi \approx 0, \pi_i \approx 0$$

PRIMARY CONSTRAINTS

$$\pi^{ij} = \frac{\partial \mathcal{L}_{ADM}}{\partial \dot{h}_{ij}} = 2N\sqrt{h} \left(K^{mn} - K h^{mn} \right) \frac{\partial K_{mn}}{\partial \dot{h}_{ij}}$$

↑
ONLY K_{ij} DEPENDS
ON VELOCITIES

$$\frac{\partial h_{mn}}{\partial h_{ij}} = \frac{1}{2} (\delta_m^i \delta_n^j + \delta_n^i \delta_m^j)$$

WE USE THESE DEFINITIONS

$$\frac{\partial \dot{h}_{mn}}{\partial \dot{h}_{ij}} = \frac{1}{2} (\delta_m^i \delta_n^j + \delta_n^i \delta_m^j) \Rightarrow \frac{\partial K_{mn}}{\partial \dot{h}_{ij}} = -\frac{1}{4N} (\delta_m^i \delta_n^j + \delta_n^i \delta_m^j)$$

$$\pi^{ij} = \sqrt{h} (K h^{ij} - K^{ij})$$

* ALL CANONICAL MOMENTA ARE TENSOR DENSITIES OF THE WEIGHT $W=1$

TENSOR DENSITY $\sim (\sqrt{h})^W$ TENSOR

• POISSON BRACKETS $x^\mu = (t, x^i), x'^\mu = (t, x'^i)$

$$\{N, \pi'\} = \delta^{(3)}(x-x')$$

$$\{N^i, \pi_j'\} = \delta_j^i \delta^{(3)}(x-x')$$

$$\{h_{ij}, \pi'^{mn}\} = \frac{1}{2} (\delta_i^m \delta_j^n + \delta_j^m \delta_i^n) \delta^{(3)}(x-x')$$

ALL OTHERS
ARE ZERO

CANONICAL HAMILTONIAN

$$\mathcal{H}_c = \pi \dot{N} + \pi^i \dot{N}_i + \pi^{ij} \dot{h}_{ij} - \mathcal{L}_{\text{ADM}}$$

$$\approx \pi^{ij} \dot{h}_{ij} - N \sqrt{h} (\mathcal{R} + K_{ij} K^{ij} - K^2)$$

WE EXPRESS K_{ij} IN TERMS OF π^{ij} :

$$\pi^{ij} = \sqrt{h} (K h^{ij} - K^{ij}) / h^{ij}$$

$$\pi^i_i = \sqrt{h} (K \cdot 3 - K) = 2K \sqrt{h} \Rightarrow K = \frac{1}{2\sqrt{h}} \pi^i_i$$

$$\pi^{ij} = \frac{1}{2} \pi^k_k h^{ij} - \sqrt{h} K^{ij}$$

$$\Rightarrow K^{ij} = \frac{1}{\sqrt{h}} \left(\frac{1}{2} \pi^k_k h^{ij} - \pi^{ij} \right)$$

$\dot{h}_{ij}$ IS COMPUTED FROM K_{ij} :

$$\dot{h}_{ij} = \mathcal{D}_i N_j + \mathcal{D}_j N_i - 2N K_{ij}$$

$$\pi^{ij} \dot{h}_{ij} = 2 \pi^{ij} \mathcal{D}_i N_j - \frac{2N}{\sqrt{h}} \pi^{ij} \left(\frac{1}{2} \pi^k_k h_{ij} - \pi_{ij} \right)$$

$$\begin{aligned} -N \sqrt{h} K_{ij} K^{ij} &= -\frac{N}{\sqrt{h}} \left(\frac{1}{2} \pi^k_k h_{ij} - \pi_{ij} \right) \left(\frac{1}{2} \pi^m_m h^{ij} - \pi^{ij} \right) \\ &= -\frac{N}{\sqrt{h}} \left(\frac{3}{4} (\pi^k_k)^2 - (\pi^k_k)^2 + \pi_{ij} \pi^{ij} \right) \\ &= \frac{N}{\sqrt{h}} \left(\frac{1}{4} (\pi^i_i)^2 - \pi_{ij} \pi^{ij} \right) \end{aligned}$$

$$N \sqrt{h} K^2 = \frac{N}{4\sqrt{h}} (\pi^i_i)^2$$

$$\Rightarrow \mathcal{H}_c \approx \frac{N}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} (\pi^i_i)^2 \right) - N \sqrt{h} \mathcal{R} + 2 \pi^{ij} \mathcal{D}_i N_j$$

$$H_c = \int d^3x \mathcal{H}_c$$

- GENERALIZATION OF COVARIANT DERIVATIVE TO WORK WITH TENSOR DENSITIES:

$$\mathcal{D}_j T^{i\dots}_{k\dots} = \partial_j T^{i\dots}_{k\dots} + \gamma_{jl}^i T^{l\dots}_{k\dots} + \dots - \gamma_{jk}^l T^{i\dots}_{l\dots} - \dots - w \gamma_{lj}^l T^{i\dots}_{k\dots}$$

- IN THE CASE OF MOMENTA

$$\mathcal{D}_i \pi^i = \partial_i \pi^i + \cancel{\gamma_{ik}^i \pi^k} - \cancel{\gamma_{ki}^k \pi^i}$$

$$\mathcal{D}_i \pi^i = \partial_i \pi^i$$

$$\mathcal{D}_i \pi^{ij} = \partial_i \pi^{ij} + \cancel{\gamma_{ik}^i \pi^{kj}} + \cancel{\gamma_{ik}^j \pi^{kj}} - \cancel{\gamma_{ik}^k \pi^{ij}}$$

$$\mathcal{D}_i \pi^{ij} = \partial_i \pi^{ij} + \gamma_{ik}^j \pi^{ki}$$

- WE PERFORM INTEGRALS BY PARTS:

* ONLY $\int d^3x \partial_i (\dots)^i$ IS A BOUNDARY TERM

* IDENTITY THAT CHANGES $\mathcal{D}_i \rightarrow \partial_i$

$$\sqrt{h} \mathcal{D}_i V^i = \partial_i (\sqrt{h} V^i) \quad w(V) = 0$$

$$\mathcal{D}_i \pi^i = \partial_i \pi^i \quad w(\pi) = 1$$

- APPLYING TO CANONICAL HAMILTONIAN

$$2\pi^{ij} \mathcal{D}_i N_j = -2N_j \mathcal{D}_i \pi^{ij} + \mathcal{D}_i (2\pi^{ij} N_j) = -2N_j \mathcal{D}_i \pi^{ij} + \partial_i (\dots)^i$$

→ WE NEGLECT $a_i(\dots)^i$ BECAUSE IT WILL BECOME A BOUNDARY TERM IN H_c

- CANONICAL HAMILTONIAN BECOMES

$$\mathcal{H}_c = N \mathcal{H} + N_i \mathcal{H}^i$$

$$\mathcal{H} = \frac{1}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} (\pi^i_i)^2 \right) - \sqrt{h} \mathcal{R}$$

$$\mathcal{H}^i = -2 \mathcal{D}_j \pi^{ij}$$

→ $\mathcal{H}_c$ IS LINEAR IN THE LAPSE AND SHIFT FUNCTIONS, THEY ARE LAGRANGE MULTIPLIERS

TOTAL HAMILTONIAN

$$\mathcal{H}_T = \mathcal{H}_c + N \pi + N_i \pi^i$$

- CONSISTENCY CONDITIONS GIVE SECONDARY CONSTRAINTS

$$\begin{aligned} \dot{\pi} &= -\mathcal{H} \approx 0 & \Rightarrow & \boxed{\mathcal{H} \approx 0} & \text{HAMILTONIAN CONSTRAINT} \\ \dot{\pi}^i &= -\mathcal{H}^i \approx 0 & \Rightarrow & \boxed{\mathcal{H}^i \approx 0} & \text{MOMENTUM CONSTRAINT} \end{aligned}$$

$$\mathcal{H}^i \sim \mathcal{D}_i \pi^{ij} \approx 0 \quad \text{"GAUSS LAW" IN ELECTROMAGNETISM}$$

⇒ AS EXPECTED, $\mathcal{H}_c \approx 0$ (LINEAR COMBINATION OF CONSTRAINTS)

- WE WILL SHOW THAT ALL CONSTRAINTS ARE 1st CLASS BY COMPUTING THE ALGEBRA

CONSTRAINT ALGEBRA

HYPERSURFACE DEFORMATION ALGEBRA (HDA)
= DIRAC ALGEBRA

$$\begin{aligned} \mathbb{D} [\xi, \tilde{\xi}] &= \int d^3x \left(\xi^i \mathcal{H}_i + \tilde{\xi}^i \pi_i \right) \\ \mathbb{H} [\phi, \tilde{\phi}] &= \int d^3x \left(\phi \mathcal{H} + \tilde{\phi} \pi \right) \end{aligned} \left\{ \begin{array}{l} \text{FOR SIMPLICITY } \tilde{\xi}, \tilde{\phi} = 0 \\ \text{(OTHERWISE } \tilde{\xi} \sim \dot{\xi}, \tilde{\phi} \sim \dot{\phi} \\ \text{FROM CASTELLANI'S METHOD)} \end{array} \right.$$

$\mathbb{D} \dots$ GENERATES TANGENTIAL (SPATIAL) DIFFEOMORPHISMS

$\mathbb{H} \dots$ GENERATES NORMAL DIFFEOMORPHISMS

→ DIFFEOMORPHISMS ACT ON TENSORS AS LIE DERIVATIVES

$$\delta_{\xi} x^i = \xi^i(x)$$

$$\delta_{\xi} (\text{FIELD}) = -\mathcal{L}_{\xi} (\text{FIELD})$$

EXAMPLES (ON Σ)

* SCALAR FIELD $\phi'(x') = \phi(x)$

$$\phi'(x+\xi) \approx \phi'(x) + \xi^i \partial_i \phi = \phi(x)$$

$$\Rightarrow \delta_{\xi} \phi = \phi' - \phi = -\xi^i \partial_i \phi \Rightarrow \boxed{\mathcal{L}_{\xi} \phi = \xi^i \partial_i \phi}$$

* COVARIANT VECTOR $V_i'(x') = \frac{\partial x^j}{\partial x'^i} V_j(x)$

$$V_i'(x+\xi) = \frac{\partial(x^j - \xi^j)}{\partial x'^i} V_j'$$

$$V_i' + \xi^j \partial_j V_i \approx (\delta_i^j - \partial_i \xi^j) V_j' = V_i - \partial_i \xi^j V_j'$$

$$\delta_{\xi} V_i = V_i' - V_i = -\xi^j \partial_j V_i - \partial_i \xi^j V_j'$$

$$\Rightarrow \boxed{\mathcal{L}_{\xi} V_i = \xi^j \partial_j V_i + \partial_i \xi^j V_j'}$$

* CONTRAVARIANT VECTOR $V^{i'}(x') = \frac{\partial x^{i'}}{\partial x^i} V^i(x)$

$$\boxed{\mathcal{L}_{\xi} V^i = \xi^j \partial_j V^i - \partial_j \xi^i V^j}$$

- AFTER LONG COMPUTATION, THE ALGEBRA IS:

$$\{D[\xi], H[\phi]\} = \int d^3x \int d^3x' \xi_i \phi' \{g_i^i, g_i^i\}$$

$$= \int d^3x \mathcal{L}_\xi \phi \mathcal{H} = H[\mathcal{L}_\xi \phi]$$

$$\{D[\xi_1], D[\xi_2]\} = D[\mathcal{L}_{\xi_1} \xi_2]$$

$$\{H[\phi_1], H[\phi_2]\} = D[\chi], \quad \chi^i = h^{ij}(\phi_1 \partial_j \phi_2 - \phi_2 \partial_j \phi_1)$$

- ALGEBRA IS CLOSED BECAUSE

$$\{\text{GENERATOR}, \text{GENERATOR}\} = \text{GENERATOR}$$

- THE SUBALGEBRA $\{D, D\} = D$ IS A LIE ALGEBRA BECAUSE IT HAS STRUCTURE CONSTANTS

- HOWEVER $\{H, H\} = H$ IS NOT A LIE SUBALGEBRA BECAUSE THE R.H.S. HAS $h^{ij}(x)$ BY THE CONSTRAINT (STRUCTURE FUNCTIONS WHEN $h^{ij} \neq \delta^{ij}$)

- WHEN $h^{ij} = \delta^{ij}$, THEN HDA BECOMES THE POINCARÉ ALGEBRA

DEGREES OF FREEDOM COUNT

FIELDS:

$$h_{ij}, N, N^i \quad N = 6 + 1 + 3 = 10 \quad (\text{AS } g_{\mu\nu})$$

I CLASS CONSTRAINTS: $\pi, \pi_i, \mathcal{H}, \mathcal{H}_i \quad N_1 = 1 + 3 + 1 + 3 = 8$

II CLASS CONSTRAINTS: $- \quad N_2 = 0$

$$N_{\text{DOF}} = N - N_1 - \frac{1}{2} N_2 = 10 - 8 = 2$$

$\Rightarrow$ 2 DEGREES OF FREEDOM OF GRAVITON

(2 STATES OF A MASSLESS PARTICLE: SPIN $S=2$, HELICITY $\lambda_s = \pm 2$)

CONSERVED CHARGES

- WE HAVE TO OBTAIN IMPROVED GENERATORS BY ADDING BOUNDARY TERMS J AND E

$$\mathbb{D}_J[\xi] = \mathbb{D}[\xi] + J[\xi]$$

$$\mathbb{H}_E[\phi] = \mathbb{H}[\phi] + E[\phi]$$

J = ANGULAR MOMENTUM

E = ENERGY

- BOUNDARY TERMS HAVE TO BE CHOSEN SUCH THAT THE IMPROVED GENERATORS $\mathbb{D}_J$ AND $\mathbb{H}_E$ ARE DIFFERENTIABLE:

$$\delta \mathbb{D}_J[\xi] = \int d^3x \left(\frac{\delta \mathbb{D}_J[\xi]}{\delta h_{ij}} \delta h_{ij} + \frac{\delta \mathbb{D}_J[\xi]}{\delta \pi^{ij}} \delta \pi^{ij} \right)$$

$$\delta \mathbb{H}_E[\phi] = \int d^3x \left(\frac{\delta \mathbb{H}_E[\phi]}{\delta h_{ij}} \delta h_{ij} + \frac{\delta \mathbb{H}_E[\phi]}{\delta \pi^{ij}} \delta \pi^{ij} \right)$$

- THE GENERATORS WITH THE PARAMETERS $\tilde{\xi}, \tilde{\phi}$ (THAT WE DROPPED FOR SIMPLICITY) ARE ALREADY DIFFERENTIABLE AND THEY DON'T NEED BOUNDARY TERMS

$\Rightarrow$ CORRESPONDING CHARGES ARE ZERO

$\rightarrow$ THEY DESCRIBE TRIVIAL, OR PROPER TRANSFORMATIONS

TRANSFORMATIONS $\begin{cases} \text{PROPER} & (\text{CHARGE} = 0, \text{ DO NOT CHANGE BOUNDARY STATES}) \\ \text{IMPROPER} & (\text{CHARGE} \neq 0) \end{cases}$

ENERGY

- LET US ANALYZE DIFFERENTIABILITY OF $\mathbb{H}$, LOOKING ONLY AT THE TERMS CONTAINING $\partial_k \delta h_{ij}, \partial_k \delta \pi^{ij}$
- WE WILL CALL OTHER TERMS D.T. (DIFFERENTIABLE TERMS)

$$\begin{aligned}
\delta H[\phi, \tilde{\phi}] &= \int d^3x \phi \delta \mathcal{H} + \text{D.T.} \\
&= \int d^3x \phi \delta(-\sqrt{h} \mathcal{R}) + \text{D.T.} \\
&\quad \quad \quad h^{ij} \mathcal{R}^k_{ikj} \\
&= - \int d^3x \sqrt{h} \phi h^{ij} \delta \mathcal{R}^k_{ikj} + \text{D.T.} \\
&\quad \quad \quad \underbrace{\mathcal{D}_k \delta \gamma^k_{ij} - \mathcal{D}_j \delta \gamma^k_{ik}}
\end{aligned}$$

• WE HAVE TO COLLECT ALL THE BOUNDARY TERMS, THUS WE ARE INTERESTED IN TOTAL DIVERGENCES ONLY

• NOTICE THE IDENTITIES: $\mathcal{D}_k h_{ij} = 0 \Leftrightarrow \mathcal{D}_k h^{ij} = 0$

$$\begin{aligned}
\delta H[\phi, \tilde{\phi}] &= - \underbrace{\int d^3x \sqrt{h} \phi h^{ij} \mathcal{D}_k \delta \gamma^k_{ij}} \\
&\quad - \int d^3x \sqrt{h} \mathcal{D}_k (\phi h^{ij} \delta \gamma^k_{ij}) + \int d^3x \sqrt{h} \mathcal{D}_k \phi h^{ij} \delta \gamma^k_{ij} \\
&\quad + \underbrace{\int d^3x \sqrt{h} \phi h^{ij} \mathcal{D}_j \delta \gamma^k_{ik}} + \text{D.T.} \\
&\quad \quad \quad \int d^3x \sqrt{h} \mathcal{D}_j (\phi h^{ij} \delta \gamma^k_{ik}) - \int d^3x \sqrt{h} \mathcal{D}_j \phi h^{ij} \delta \gamma^k_{ik}
\end{aligned}$$

• THE TERMS $\delta \gamma^k_{ij}, \delta \gamma^k_{ik}$ ARE NOT DIFFERENTIABLE YET BECAUSE $\delta \gamma^k_{ij} = \frac{1}{2} h^{kl} (\partial_i \delta h_{lj} + \partial_j \delta h_{li} - \partial_l \delta h_{ij}) + \text{D.T.}$, BUT WE WILL LEAVE THEM FOR NOW

• THUS, WE HAVE:

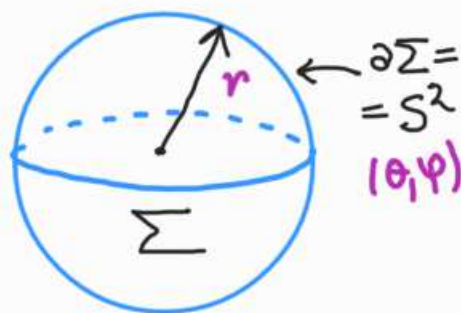
$$\begin{aligned}
\delta H[\phi, \tilde{\phi}] &= - \sum_i \int d^3x \sqrt{h} \mathcal{D}_k (\phi h^{ij} \delta \gamma^k_{ij}) + \sum_j \int d^3x \sqrt{h} \mathcal{D}_j (\phi h^{ij} \delta \gamma^k_{ik}) \\
&\quad + \mathcal{D}\phi \text{ TERMS} + \text{D.T.} \\
&= \sum_i \int d^3x \sqrt{h} \mathcal{D}_i [\phi (-h^{jk} \delta \gamma^i_{jk} + h^{ij} \delta \gamma^k_{jk})] + \mathcal{D}\phi \text{ TERMS} + \text{D.T.}
\end{aligned}$$

- APPLYING THE STOKES' THEOREM:

→ THE BOUNDARY OF Σ IS 2-SPHERE S^2 AT $r \rightarrow \infty$: S_∞^2

→ EXPRESSIONS SHOULD BE FINITE

→ WE NEED BOUNDARY CONDITIONS



$$\int_{\Sigma} d^3x \sqrt{h} \partial_i V^i = \int_{\Sigma} d^3x \partial_i (\sqrt{h} V^i) = \oint_{S_\infty^2} d\sigma_i \sqrt{h} V^i$$

$$\delta H[\phi, \tilde{\phi}] = \oint_{S_\infty^2} d\sigma_i \sqrt{h} \phi (-h^{jk} \delta x_{jk}^i + h^{ij} \delta x_{jk}^k) + \delta \phi \text{ TERMS} + \text{D.T.}$$

- NEXT, WE USE:

$$\delta x_{jk}^i = \frac{1}{2} h^{il} (\partial_j \delta h_{lk} + \partial_k \delta h_{lj} - \partial_l \delta h_{jk})$$

- REMEMBERING THAT $\partial_i h^{jk} = 0$ AND h_{ij}, h^{ij} RISE AND LOWER INDICES, WE GET:

$$\begin{aligned} -h^{jk} \delta x_{jk}^i &= -\frac{1}{2} h^{jk} h^{il} (\partial_j \delta h_{lk} + \partial_k \delta h_{lj} - \partial_l \delta h_{jk}) \\ &= -\underbrace{h^{il} \partial^k \delta h_{ek}} + \frac{1}{2} h^{jk} \partial^i \delta h_{jk} \\ &\quad \partial^k (h^{il} \delta h_{ek}) = -\partial^k (\delta h^{il} h_{ek}) = -\partial_e \delta h^{il} \\ &= \partial_j \delta h^{ij} - \frac{1}{2} \partial^i (h_{jk} \delta h^{jk}) \end{aligned}$$

$$\begin{aligned} h^{ij} \delta x_{jk}^k &= \frac{1}{2} h^{ij} h^{kl} (\partial_j \delta h_{lk} + \partial_k \delta h_{lj} - \partial_l \delta h_{jk}) \\ &= \frac{1}{2} \partial^i (h^{kl} \delta h_{kl}) = -\frac{1}{2} \partial^i (\delta h^{kl} h_{kl}) \end{aligned}$$

(SYMMETRY IN kl)

WE USED: $h_{ik} h^{kj} = \delta_i^j \Rightarrow \delta h_{ik} h^{kj} = -h_{ik} \delta h^{kj}$

$$\delta H[\phi, \tilde{\phi}] = \oint_{S_\infty^2} d\sigma_i \sqrt{h} \phi [\partial_j \delta h^{ij} - \partial^i (h_{jk} \delta h^{jk})] + \delta \phi \text{ TERMS} + \text{D.T.}$$

BOUNDARY CONDITIONS AT $\partial\Sigma = S_\infty^2$, $r \rightarrow \infty$

* T. REGGE, C. TEITELBOIM, ANN. PHYS. 88 (1974) 286

• CHOOSING ASYMPTOTICALLY FLAT COORDINATES:

$$h_{ij} \rightarrow \bar{h}_{ij} + H_{ij}$$

$$H_{ij} = \mathcal{O}(1/r)$$

$$N = 1 + \mathcal{O}(1/r)$$

$$\partial_i N = \mathcal{O}(1/r^2)$$

$$N^i = \mathcal{O}(1/r)$$

$$\partial_i N^i = \mathcal{O}(1/r^2)$$

$$\pi^{ij} = \mathcal{O}(1/r^2)$$

LOCAL PARAMETER IS: $\phi = 1 + \mathcal{O}(1/r)$
 $\partial_i \phi = \mathcal{O}(1/r^2)$

$$\Rightarrow \sqrt{h} \sim \sqrt{\bar{h}} + \mathcal{O}(1/r) \quad h^{-1} \sim \bar{h}^{-1} + \mathcal{O}(1/r)$$

$$\mathcal{D}\phi \sim \mathcal{O}(1/r^2)$$

$$x \sim \bar{x} + \mathcal{O}(1/r)$$

⊛ $\Rightarrow \mathcal{D}\phi$ TERMS ARE EITHER DIFFERENTIABLE BULK TERMS, THEN THEY GO INTO "D.T.", OR BOUNDARY TERMS, THEN THEY CAN BE NEGLECTED IN THE $r \rightarrow \infty$ LIMIT BECAUSE $\mathcal{D}_i \phi = \partial_i \phi$:

$$\sqrt{h} \mathcal{D}\phi h^{-1} \delta x|_\Sigma \sim \sqrt{\bar{h}} \partial\phi \bar{h}^{-1} (\bar{h}^{-1} \delta \bar{h}) = \mathcal{O}(1/r^2) \rightarrow 0$$

→ **FIXED BACKGROUND** IS NECESSARY TO BE ABLE TO INTEGRATE OUT THE VARIATION $\mathcal{D}\delta h^{-1}$ AS $\delta(\dots)$, THAT IS IMPOSSIBLE WITHOUT IT, SINCE $\mathcal{D}$ ALSO DEPENDS ON h

→ THANKS TO $\delta \bar{h}_{ij} = 0$, WE GET THAT THE EXPRESSION IS INTEGRABLE

$$\mathcal{D}\delta h^{-1} = \bar{\mathcal{D}}(-\delta H) = -\delta(\bar{\mathcal{D}}H)$$

BECAUSE $h = \bar{h} + H$ IMPLIES $h^{-1} = \bar{h}^{-1} - H + \mathcal{O}(1/r^2)$

→ CARTESIAN COORDINATES ARE THE SIMPLEST: $\bar{h}_{ij} = \delta_{ij}$, $\bar{x}_{ij} = 0$, $\bar{\mathcal{D}} = \partial$

→ WE CAN WRITE NOW ALL INDICES DOWN BECAUSE δ_{ij} , δ^i_i RISE AND LOWER INDICES ON Σ

• WE OBTAIN;

$$\delta H|_{\partial \Sigma} = \underbrace{-\delta \oint d\sigma_i (a_j H_{ij} - a_i H_{jj})}_{\equiv -\delta E} + \text{D.T.}$$

⇒ ENERGY IS

$$E = \oint d\sigma_i (a_j H_{ij} - a_i H_{jj})$$

→ THIS EXPRESSION FOR ENERGY WILL GIVE THE **BLACK HOLE MASS**
E=M FOR THE SOLUTION (SHOW IT!)

$$r \rightarrow \infty, \quad g_{ij} dx^i dx^j \simeq -\left(1 - \frac{2M}{r}\right) dt^2 + \left(\delta_{ij} + 2M \frac{x^i x^j}{r^3}\right) dx^i dx^j$$

→ SIMILARLY, WE CAN COMPUTE ANGULAR MOMENTUM J

→ IF WE CHANGE BOUNDARY CONDITIONS, WE CAN APPLY OBTAINED FORMULA ONLY TO THE CLASS OF SOLUTIONS BEHAVING ASYMPTOTICALLY AS THIS NEW CLASS OF ASYMPTOTIC CONDITIONS